

Teaching Cø / N*: A Graduate Instructor's Manual

Expanded lecture notes, comparative theory placement, collaborative learning, validation arguments, and course materials for the network-dynamics theory of phenomenal consciousness

Based on the Cø / N* research program of Phil Stilwell

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Preface

This manual is designed to help an instructor teach a new theory without either oversimplifying it or treating its central claims as already established. The CØ / N* research program proposes a precise candidate account of minimal phenomenal presence and then surrounds that core claim with methodological disciplines: fix the candidate system before measuring it; distinguish causal availability from report; separate scientific indeterminacy from invalid inference; audit every conjunct against simpler rivals; extend presence to phenomenal character only through an independently tested bridge; and explain what it would mean for the model to become structurally intelligible.

The intended audience is graduate students in philosophy of mind, consciousness science, cognitive neuroscience, complex systems, artificial intelligence, or philosophy of science. The register assumes intellectual maturity but not prior allegiance to a particular consciousness theory. Mathematical expressions are used to make commitments visible. Every formal claim should be translated into plain language, linked to a scientific purpose, and paired with a failure condition.

The manual supports a 14-session seminar. A shorter course can use Sessions 1, 2, 3, 5, 7, 8, 9, 10, and 13 as a nine-session core. A four-session workshop can use Sessions 1-2, 3-6 as a single component session, 7-8, and 9-14 as a synthesis session. Each full session contains:

- learning objectives;
- instructor preparation;
- a timed teaching plan;
- lecture notes written for oral delivery;
- a board or display plan;
- a slide sequence;
- a fully specified collaborative exercise;
- a structured discussion protocol;
- common misconceptions and corrective responses;
- a short assessment or exit ticket; and
- assigned and optional readings.

The manual treats the seven papers as one research constellation rather than seven competing statements. The core paper supplies the central hypothesis. The companion papers specify the bearer, one difficult component, the evidence policy, the component audit, the character extension, and the model's explanatory form.

The expanded lecture notes are an instructor resource bank, not a script that must be read in full during one meeting. The timed plan identifies the material to emphasize in a 110-minute seminar. Worked examples can be assigned before class, moved to a discussion section, or used as replacement material when students need a slower formal explanation. Protect the collaborative block: it is where students demonstrate that they can apply the distinctions rather than merely recognize them.

Instructor stance

Teach the theory in the conditional and adversarial mood. Students should be able to state what the model claims, why each element is included, what would count for and against it, and where its present evidence is incomplete. A successful course does not produce automatic agreement with the theory. It produces readers capable of evaluating it fairly and designing studies that could genuinely change its status.

Use four recurring questions:

1. What is the explanandum?

2. What is the candidate bearer?
3. What evidence would license the inference?
4. What observation would force revision?

Notation policy

The papers use both $C\emptyset$ and C_0 for minimal phenomenal presence. This manual uses **$C\emptyset$** in prose and C_0 in formulas. The three components are written N_1 , N_2 , and N_3 . The bundle is $N^* = N_1 \wedge N_2 \wedge N_3$. Viability is V . A positive classification is never treated as a direct observation of another subject's phenomenology; it is a licensed scientific attribution under a declared model and validity envelope.

Part I. Course Architecture

1. The theory in one page

The target is **minimal phenomenal presence**: the fact that there is something it is like for a candidate system to undergo a state. The target does not require verbal report, explicit attention, intelligence, self-reflection, autobiographical memory, or metacognitive ownership.

The core conjecture is:

$$H_{C_0}: V(S; R, \Delta) \wedge (N_1(S, c) \wedge N_2(S, c) \wedge N_3(S, c)) \Leftrightarrow C_0(S, c; R, \Delta).$$

In plain language: for a boundary-qualified candidate system S , content c , operating regime R , and evaluation interval Δ , minimal phenomenal presence obtains exactly when the system is independently viable and realizes integration or synergy, report-independent system-wide availability, and recurrent stability together.

The components divide explanatory labor:

Element	Role	Question answered	Characteristic false positive blocked
Boundary qualification B_S	Bearer	What system is being tested?	A score calculated over a convenient or shifting node set
V	Realized operation	Can the candidate sustain the relevant dynamics in this regime?	An inert diagram, failed preparation, or out-of-domain application
N_1	Unity	Is the content irreducibly joint rather than an aggregate of independent events?	Coactivation, common drive, or a heap of local signals
N_2	System-level availability	Can the content causally modulate functionally diverse recipients without report being required?	Passive diffusion, output behavior, or mere widespread activation
N_3	Temporal presence	Is the content maintained through bounded recurrent dynamics?	An evanescent feedforward transient or pathological fixation
C_0	Phenomenal presence	Is there something it is like for the candidate in this state?	Confusing presence with access, report, or character

The biconditional is intentionally risky. Its sufficiency direction is challenged by a viable system that passes all three conditions while strong independent evidence counts against phenomenal presence:

$$V \wedge N^* \wedge \neg C_0.$$

Its necessity direction is challenged by a well-supported conscious case in which one or more conditions are genuinely absent at the correct boundary, grain, regime, and time window:

$$C_0 \wedge \neg (V \wedge N^*).$$

The research program adds six disciplines around this hypothesis:

1. **Boundary first:** identify and freeze a sufficiently autonomous, inclusion-minimal candidate before applying the consciousness measure.
2. **Availability without report:** define N_2 through content-specific causal reach across functionally distinct recipients, not through speech, button presses, diffusion, or node count.

3. **Four evidential outcomes:** distinguish positive, negative, indeterminate, and unlicensed results.
4. **Component audit:** require every conjunct to outperform a leave-one-out rival on cases where the models actually disagree, using independent held-out anchors.
5. **Character extension:** after a positive presence gate, test a bridge from content-specific causal-trajectory geometry to independently anchored phenomenal relations.
6. **Schematic legibility:** make the theory's role structure compositional, counterfactual, fault-localizing, and teachable without confusing intelligibility with proof.

2. Placement in the consciousness-theory landscape

The most accurate one-line classification is:

CØ / N is a physicalist, information-dynamical, functional network theory of minimal phenomenal presence, cross-listed with neurobiological workspace and recurrent-processing approaches.*

In the Closer To Truth taxonomy supplied with the project, its primary home is **Materialism / Computational & Functionalism**, with a strong secondary placement under **Information** and cross-references to **Materialism / Neurobiological** and **Materialism / First-Order**. Its integration requirement creates a weak adjacency to dynamical and electromagnetic approaches, but the model does not identify consciousness with synchrony or a field.

Theory family	What CØ / N* accepts	What it rejects or tests	Relation to the model
Integrated Information Theory (IIT)	Unity, irreducible joint structure, and the need to specify a causal complex	Integration alone as sufficient; use of the final consciousness quantity to select its own bearer	N_1 inherits the integration pressure but is only one conjunct
Global Workspace / Global Neuronal Workspace (GWT/GNWT)	Broad availability to diverse processors	Report as constitutive, fixed prefrontal dominance, and observed ignition as sufficient	N_2 generalizes availability into a report-independent causal role
Recurrent Processing Theory (RPT)	Feedback and temporal stabilization	Local recurrence alone as sufficient	N_3 makes recurrence necessary but joins it to unity and availability
Predictive Processing / Free Energy	Hierarchical inference, precision, and self-organizing dynamics as possible implementations	Inference or error minimization as identical to consciousness	A potential implementation language for all three components
Higher-Order and Metacognitive theories	Self-representation can enrich consciousness and support report	A higher-order representation as necessary for minimal C_0	Placed at enriched consciousness C_1 , not the minimal gate
Attention Schema and self-model theories	Internal models can shape control, availability, and self-attribution	A self-model as an unstated minimal conjunct	Optional contributor to N_2 or C_1 unless evidence requires more
Quality-space theories	Phenomenal character has relational similarity structure	Treating structured representation as sufficient for presence	Supplies the target structure for the character extension
Representational geometry and neural manifolds	Population relations, trajectories, topology, and transitions can be compared	Renaming any latent geometry as qualia	Supplies methods for estimating G_c , subject to causal and phenomenal validation
Embodied, affective, enactive, and evolutionary views	Organismic regulation, interoception, action, value, and learning can matter	Assuming in advance that all embodiment is either constitutive or irrelevant	These factors may enter V , the system boundary, content dynamics, or a future conjunct
C-test and theory-derived indicator approaches	Nonreporting systems require validated, multi-source tests	One marker, one behavior, or one theory-conditional indicator as a universal verdict	The full program is a candidate C-test architecture

Theory family	What CØ / N* accepts	What it rejects or tests	Relation to the model
Illusionism	Reports and self-ascriptions require a functional explanation	Replacing the phenomenal target with the disposition to claim phenomenality	CØ / N* remains realist about minimal phenomenal presence
Panpsychism, fundamentalism, and universalism	Substrate independence is a live possibility	Consciousness from mere existence, complexity, or abstract isomorphism	Requires causal realization, correct grain, effective coupling, and viability
Biological autonomy and individuality	Self-maintaining organization can help individuate a causal bearer	Autonomy as sufficient for consciousness	Supplies a theory-independent candidate-generation discipline
Markov blanket / active-inference approaches	Conditional-independence structure can nominate candidate boundaries	A statistical blanket as proof of a subject	Useful proposal method followed by intervention and role tests
Extended-mind and group-mind approaches	The skull is not an a priori causal boundary	Any reliable coupling, tool, or communication channel as part of one subject	Requires constitutive contribution, autonomy, minimality, and separability
Electromagnetic or synchrony theories	Temporally coordinated distributed dynamics may help realize N_1 or N_3	A field, frequency band, or synchrony relation as the identity condition	Weak implementation adjacency only
Dualism, idealism, quantum, and anomalous theories	No core commitment is borrowed	Nonphysical substances, consciousness as fundamental reality, quantum collapse, or anomalous access	Outside the model's intended taxonomy

The most important teaching point is that CØ / N* is a **mid-level bridge theory**. It is not a metaphysical solution to why experience exists, not a theory of report, not a complete model of self-consciousness, and not initially a full theory of phenomenal character. It proposes a causal-dynamical identity for minimal phenomenal presence and a disciplined research program for testing and extending that identity.

3. Course-level learning outcomes

By the end of the course, students should be able to:

1. distinguish phenomenal presence, access, report, self-consciousness, sentience, wakefulness, and phenomenal character;
2. reconstruct the CØ / N* biconditional and explain both directions of empirical risk;
3. explain the distinct roles of boundary qualification, viability, integration, availability, and recurrence;
4. situate the theory among major scientific and philosophical approaches without reducing it to a collage of IIT, GNWT, and RPT;
5. apply the boundary-first, four-outcome, and surrogate-conflict rules to difficult cases;
6. distinguish correlation, criterion ablation, mechanism ablation, and measurement ablation;
7. use the diagnostic-set theorem to determine whether a study can identify component necessity;
8. explain the two-layer architecture for phenomenal presence and phenomenal character;
9. formulate the strongest objections to the theory and identify which are conceptual, empirical, or metaphysical;
10. design an adversarial, preregisterable study whose result could support, revise, or refute part of the program.

4. Suggested course format and assessment

Recommended meeting: one 110-minute seminar per week for 14 weeks.

Prerequisites: one prior course in philosophy of mind, cognitive neuroscience, philosophy of science, complex systems, or machine learning. No advanced mathematics is required, but students should be willing to interpret formal notation and causal diagrams.

Assessment scheme:

Item	Weight	Purpose
Weekly analytic memos	15%	Translate one formal claim, identify its hidden assumption, and name a falsifier
Theory-comparison brief	15%	Compare CØ / N* with one neighboring theory at the level of target, mechanism, evidence, and prediction
Collaborative case portfolio	15%	Revise four group artifacts with individual addenda, dissent records, and instructor feedback
Boundary and classification practicum	10%	Apply the boundary-first and four-outcome protocols to a new case
Adversarial ablation critique	20%	Determine whether a proposed study genuinely tests a conjunct
Capstone preregistration and oral defense	25%	Design and defend a falsifiable study or conceptual audit

Seminar norm: every positive claim must name its scope; every negative claim must state its negative-license conditions; every comparison must distinguish different explananda; and every proposed test must identify the case on which rival models disagree.

5. Collaborative learning architecture

Collaboration in this course is part of the epistemology, not a break from it. Consciousness research is unusually vulnerable to hidden shifts in target, bearer, measurement, and evidential standard. A well-designed group task makes those shifts visible because students must coordinate distinct forms of expertise and produce one auditable conclusion. Every collaborative exercise should therefore end in a durable artifact: a classification ledger, boundary map, intervention design, argument diagram, comparison grid, or preregistration revision.

Stable teams and rotating roles

Use teams of four whenever enrollment permits. Keep teams stable for two or three sessions so that students learn one another's reasoning styles, then reshuffle them to prevent fixed intellectual camps. Assign roles before students receive the case. Rotate the roles every week.

Role	Responsibility during the task	Question the role must answer
Model reconstructor	States the target, variables, logical form, and scope without advocacy	What exactly is the claim under evaluation?
Method and validity auditor	Tracks operationalization, confounds, boundary, uncertainty, and transport	Does the proposed evidence bear on the stated claim?
Adversarial rival	Develops the strongest live alternative or counterexample	What else could explain this result, and where do predictions diverge?
Synthesis reporter	Maintains the shared record, marks dissent, and presents the revised conclusion	What can the group jointly license, and what remains unresolved?

For teams of five, add a **process observer** who records target drift, unearned inference, suppressed disagreement, and uneven participation. For teams of three, combine model reconstruction with reporting. No student should occupy the adversarial role for the entire course; criticism and constructive synthesis are both learned practices.

The collaboration cycle

Most exercises use a six-stage cycle:

1. **Private commitment, 2-4 minutes.** Each student records an initial classification or argument before discussion. This prevents rapid consensus from erasing genuine disagreement.
2. **Model reconstruction, 3-5 minutes.** The team agrees on the exact explanandum, candidate bearer, variables, and rival claims.
3. **Distributed analysis, 8-15 minutes.** Role-holders complete separate portions of the common worksheet.
4. **Adversarial exchange, 5-10 minutes.** The rival challenges the inference; the auditor identifies licensing conditions; the reconstructor may clarify but may not redefine the target to escape criticism.
5. **Revision with dissent, 4-8 minutes.** The group produces one conclusion, one residual uncertainty, and, where necessary, a signed minority note.
6. **Public debrief, 8-15 minutes.** Groups compare not only answers but the assumptions that generated different answers.

Do not grade students for reaching the theory-friendly conclusion. Grade them for accuracy, validity discipline, rival quality, revision under criticism, and the explicit separation of scientific status from practical action.

Reusable discussion formats

Format	Best use	Procedure	Required output
Think-pair-specify	Clarifying a formal or conceptual distinction	Private answer, paired comparison, then one jointly specified claim and falsifier	Two-sentence claim plus failure condition
Structured academic controversy	Questions with serious arguments on both sides	Pairs defend assigned positions, switch sides, then write a synthesis neither side could have written initially	Agreement, strongest dissent, decisive evidence
Jigsaw	Multi-component models	Home-team members become temporary experts on different components, meet expert peers, then teach their home team	Completed component matrix with dependencies
Evidence tribunal	Positive, negative, indeterminate, and unlicensed decisions	Advocates argue competing outcomes before a licensing panel	Verdict, cause profile, and next study
Fishbowl	Conceptual issues that benefit from listening and revision	Inner circle discusses; outer circle tracks assumptions and inferential moves; circles switch	Assumption ledger and revised positions
Gallery walk	Comparing designs or schematics	Teams post a one-page artifact; peers attach one validity challenge and one constructive improvement	Revised artifact with response log
Case conference	Clinical, animal, organoid, or AI assessment	Disciplinary roles submit evidence to a chair who must separate science from action	Scientific tuple, action note, dissent record
Red-team / blue-team review	Preregistration and falsification	One team proposes; another identifies circularity, hidden flexibility, and an omitted outcome; proponents revise	Before-and-after design plus change log

Discussion quality and psychological safety

Require students to criticize a claim at its strongest plausible interpretation. A student may request a **target check** whenever the conversation shifts among phenomenal presence, access, report, character, wakefulness, or moral status. A

student may request a **license check** whenever a group moves from a weak or invalid indicator to a strong conclusion. These interventions pause the discussion for sixty seconds while the synthesis reporter restates the inference.

Because the course includes disorders of consciousness, animal welfare, and possible artificial consciousness, remind students that intellectual disagreement need not imply indifference to affected beings. Scientific classification and practical precaution are separated analytically so that both can be discussed more responsibly. Students may analyze a sensitive clinical case through a written role rather than personal disclosure.

Individual accountability within group work

At the end of each major exercise, collect three items:

1. the team's shared artifact;
2. a sixty-second individual addendum stating one point the student would preserve and one point they would revise; and
3. a role rotation record.

For graded collaborations, a useful allocation is 60% shared artifact, 25% individual addendum, and 15% documented contribution to the team's reasoning. Peer ratings should be diagnostic rather than popularity-based: preparation, epistemic contribution, responsiveness to criticism, and equitable participation.

Instructor debrief protocol

The instructor should resist announcing the “correct” answer immediately. First compare groups along four axes:

- Did they hold the target fixed?
- Did they evaluate the same candidate system and time window?
- Did they separate measurement failure from theory failure?
- Did they name evidence that would change their conclusion?

Then model a calibrated verdict. Use phrases such as **supported under the declared assumptions, negative prediction rather than observed absence, indeterminate because the valid interval straddles**, and **unlicensed until the boundary or transport bridge is repaired**. End every debrief by identifying the smallest next observation that would reduce the most important disagreement.

6. The 14-session sequence

Session	Central question	Main paper contribution
1. The explanandum and landscape	What exactly is the theory trying to explain?	Core scope and theory placement
2. The core biconditional	What does $V \wedge N^* \Leftrightarrow C_0$ commit us to?	Core model, logic, temporal grain, falsifiers
3. Where is the subject?	Which system is the bearer of the proposed state?	Boundary-first criterion
4. Viability and integration	What makes the organization real and unified?	V and N_1
5. Availability without report	What does system-wide availability actually require?	Full N_2 operationalization
6. Recurrent stability	Why must a present content persist through feedback?	N_3 and temporal presence
7. What does the evidence license?	When are positive, negative, indeterminate, and unlicensed warranted?	Four-outcome framework
8. Does every conjunct earn its place?	How can necessity be audited without a consciousness oracle?	Diagnostic theorem and firewall
9. From presence to character	How can the model explain what an experience is like?	Dynamical-geometry bridge
10. Reading consciousness from the schematic	What would make the identity structurally intelligible?	Schematic legibility
11. Biological and clinical cases	How does the framework classify variable human states?	Sleep, anesthesia, rivalry, brain injury, split systems
12. Animals, organoids, and artificial systems	How far can the model transport?	Substrate discipline and welfare-relevant indeterminacy
13. Differentiation and validation	Why this theory rather than its neighbors or simpler rivals?	Comparative arguments and empirical program
14. Capstone research design	What study would genuinely change our view?	Synthesis and adversarial preregistration

Part II. Lecture Notes and Teaching Materials

Session 1. The Explanandum and the Theory Landscape

Learning objectives

Students should be able to distinguish at least six consciousness targets, explain why apparent theory conflict can result from target mismatch, and place CØ / N* in the contemporary landscape without treating it as a simple average of other theories.

Instructor preparation

Assign the abstract and Sections 1-3 and 15 of *CØ as N*: A Minimal Network-Dynamics Model of Phenomenal Consciousness*. Ask students to bring one sentence stating what they previously thought a theory of consciousness must explain.

Timed plan

	Time	Teaching move
	0-8 min	Opening poll: What is the target of a consciousness theory?
	8-28 min	Lecture: the target ladder and category errors
	28-44 min	Theory-family map and matched-target comparison
	44-68 min	Collaborative target-sorting and inference-control exercise
	68-82 min	Worked examples and the mid-level bargain
	82-104 min	Structured academic controversy: Is the target too thin?
	104-110 min	Exit ticket and preview of the biconditional

Lecture notes

Begin with the claim that consciousness science does not have one uncontested explanandum. The word *consciousness* can refer to wakefulness, phenomenal presence, conscious access, reportability, self-awareness, a particular conscious content, sentience, or the detailed character of an experience. Theories may disagree because they answer different questions.

Introduce a target ladder:

1. **Arousal or wakefulness:** whether the organism occupies a globally responsive state.
2. **Phenomenal presence, C_0 :** whether there is anything it is like for the candidate at all.
3. **Content presence:** whether a particular content is phenomenally present.
4. **Access and report:** whether a content can guide reasoning, deliberate control, memory, or overt response.
5. **Reflective consciousness, C_1 :** whether the system represents or owns its own state.
6. **Phenomenal character:** what the admitted experience is like and how its qualities are related.

Stress that the ordering is conceptual rather than a proven biological hierarchy. The purpose is to prevent an explanation at one level from being credited or criticized as an explanation at another. A theory of report can be excellent and still fail as a theory of minimal phenomenal presence. A theory of presence can be valuable and still be incomplete as a theory of pain, color, ownership, or selfhood.

C_0 / N^* fixes its initial target at level 2: minimal phenomenal presence. This is what allows it to treat report, attention, metacognition, and intelligence as evidence or enrichment rather than as constituents. The move creates a burden: if minimal presence is defined too thinly, critics may say that the theory evades the most interesting features of consciousness. The proper response is not that richer features do not matter. It is that distinct explananda should be modeled in distinct layers and connected explicitly.

Now situate the main theory families by the pressure they place on an adequate account:

- IIT says unity and intrinsic causal structure matter.
- GNWT says a conscious content must have system-wide availability.
- RPT says feedback and recurrent stabilization matter.
- Higher-order theories say awareness of a state requires representation of that state.
- Predictive approaches say conscious states occur in hierarchical, precision-weighted inference.
- Quality-space and structural approaches say phenomenal character is organized relationally.
- Embodied and affective approaches say regulation, action, bodily state, and value cannot be ignored.
- Illusionism says the scientifically tractable target may be why systems judge themselves to have phenomenality.
- Panpsychist and fundamentalist approaches shift the question from which organization generates consciousness to how a fundamental consciousness property is structured or combined.

C_0 / N^* does not claim that all of these are equally right. It accepts three mechanistic pressures and turns them into one exposed conjunction. It then assigns other pressures to neighboring layers: metacognition to C_1 , quality structure to Q_ϕ , organismic support provisionally to V and boundary selection, and metaphysical explanation to an open question.

Explain the taxonomy placement. The theory is physicalist because it proposes a causally realized network-dynamical identity. It is functional because the conditions are defined by roles that can in principle have multiple realizations. It is information-dynamical because irreducible joint structure and time-dependent causal organization are central. It is not unrestricted functionalism: a lookup table, behavioral duplicate, or static graph does not pass. Effective coupling, correct temporal grain, operating regime, and substrate-appropriate viability are required.

End with the theory's division of labor. The boundary paper asks *which system*. The core model asks *whether presence*. The availability paper defines one gate. The indeterminacy paper governs *what the evidence licenses*. The ablation paper asks *whether every gate is needed*. The character paper asks *what the present experience is like*. The schematic paper asks *what would make the identity intellectually readable*.

Worked teaching sequence: the same observation under different targets

Students often understand the target ladder abstractly but collapse it as soon as a vivid case appears. Work through the following examples slowly. For each, write the observation first and prohibit use of the word *conscious* until the bridge premise is explicit.

Observation	Directly supports	Additional bridge needed for C_0	Rival interpretation to keep live
A behaviorally unresponsive patient follows commands by modulating imagery-related activity	Comprehension, task maintenance, volitional imagery, and covert command following in the tested runs	The capacities are present only if there is phenomenal presence, or the full $V \wedge N^*$ profile is independently supported	The task could be performed by nonphenomenal control; false positives or sensory misunderstanding must be assessed
A stimulus category is decoded from early visual activity without report	Content-specific neural information is present in the measured population	The decoded representation participates in the organization claimed to constitute presence	Unconscious representations can be decodable; the signal may reflect feedforward registration only

Observation	Directly supports	Additional bridge needed for C_0	Rival interpretation to keep live
A fluent AI says that it has a rich inner life and accurately discusses consciousness theory	Linguistic competence, self-description, context use, and perhaps a functional self-model	The running candidate has a qualified boundary, viability, integration, causal availability, recurrence, and a licensed transport bridge	Training-data imitation, policy optimization, or an external orchestration loop may explain the behavior

The exercise exposes three different mistakes. The first treats a demanding capacity as a direct observation of phenomenality. The second treats readable information as phenomenally present information. The third treats a system's semantic claim as privileged evidence about its causal realization. None of this makes the observations irrelevant. It specifies the inferential work still required.

The mid-level bargain

Explain the methodological bargain explicitly. A mid-level theory gives up the ambition to derive why anything exists phenomenally from nonphenomenal premises. In exchange, it offers a public constitutive hypothesis precise enough to compare candidates, design interventions, and state failure conditions. It also brackets the detailed character of admitted experiences until a second bridge is tested. This is productive only if the minimal target supports stable measurement and informative prediction. If every putative C_0 case can be redescribed only by importing character, ownership, or report into the evidence, the target may be too thin to function independently. That would be a problem for the research strategy, not merely a request for one more module.

Display plan

Write three columns:

Target	Typical evidence	Common mistake
Presence	report, no-report markers, causal dynamics	equating evidence with the target
Access / report	task use, memory, decision, output	treating access as presence by definition
Character	similarity, discrimination, deformation	assuming presence already explains quality

Finish with the one-line label: **physicalist ♦ information-dynamical ♦ functional ♦ mid-level bridge theory**.

Slide sequence

1. One word, many targets
2. The target ladder
3. $C_0 \neq R, C_0 \neq A, C_0 \neq C_1$
4. IIT: unity pressure
5. GNWT: availability pressure
6. RPT: temporal pressure
7. Other theories as implementation, enrichment, rival target, or metaphysics
8. The CØ / N* taxonomy position
9. Seven papers, seven jobs
10. The course's recurring four questions

Collaborative exercise: Target sorting and inference control

Purpose. Students learn that one observation can be relevant to several consciousness questions without directly answering all of them. The task separates the evidential target from the conclusion a researcher may wish to draw.

Setup. Give every team the following six claims on separate cards: “The patient can follow a command in motor imagery”; “The stimulus identity can be decoded from V1”; “The subject says that the pain is unpleasant”; “A recurrent loop persists for 250 ms”; “The model represents its own uncertainty”; and “The system is awake but behaviorally unresponsive.” Add two blank cards on which the team must invent difficult cases.

Roles and procedure. The model reconstructor lays out six columns: wakefulness, C_0 , content presence, access, C_1 , and phenomenal character. The validity auditor places each card first under the target to which it bears most directly and lists its auxiliary demands. The adversarial rival moves one card to the strongest alternative interpretation and explains the possible confound. The reporter then records the strongest licensed conclusion and one illicit stronger inference for every card. After twelve minutes, pair teams and ask them to challenge two placements without merely appealing to intuition.

Deliverable. Each team submits a target-evidence-inference grid containing: direct target, possible relevance to C_0 , auxiliary capacities, alternative explanation, strongest licensed conclusion, and forbidden inference. The two invented cases must be difficult because they mix targets, not because they omit all information.

Debrief. Compare the imagery-command and recurrent-loop cards. The first is behaviorally silent but depends on comprehension, task maintenance, and imagery control; the second is mechanistically suggestive but may occur in an isolated nonconscious circuit. Ask which case tempts students to infer too much and why. Conclude that evidence can be strong without being constitutive and mechanistic without being sufficient.

Collaborative discussion: Is the target too thin?

Use a structured academic controversy. One pair argues that beginning with minimal phenomenal presence improves tractability and prevents report or selfhood from being smuggled into the target. The other argues that a target stripped of character, ownership, and affect is too underspecified to identify scientifically. After six minutes, pairs switch positions and must improve the argument they initially opposed. The four-person team then writes:

1. one respect in which narrowing the target increases falsifiability;
2. one respect in which it risks explanatory deflation;
3. whether a theory of C_0 could be complete while leaving phenomenal character unexplained; and
4. one observation that would decide whether the mid-level target is scientifically productive.

The reporter must preserve one unresolved dissent. In the plenary debrief, distinguish **a legitimate partial theory** from **a relabeled proxy with no independent target**.

Common misconceptions

- **“The model combines three theories, so it has no distinctive claim.”** Correct by identifying the distinct conjunction, viability gate, boundary order, conflict policy, and two-directional falsifiers.
- **“Physicalist means neuron-specific.”** Clarify constrained multiple realization.
- **“Report-independent means reports are useless.”** Reports remain powerful evidence in appropriate calibration cases.
- **“Minimal means simple or low-level.”** Minimal names the proposed smallest adequate condition set, not a simple system.

Exit ticket

In three sentences, state the target of C_0 / N^* , name one neighboring theory that targets something richer or different, and give one observation that would be relevant evidence without being constitutive.

Readings

Required: Stilwell, C_0 as N^{**} , Sections 1-3 and 15; the project taxonomy note.

Recommended: Block (1995); Seth and Bayne (2022); Storm et al. (2024).

Session 2. The Core Biconditional and Network-Dynamics Model

Learning objectives

Students should be able to read the central formula in both directions, distinguish structural, functional, and effective connectivity, explain the role of temporal grain, and identify a valid sufficiency or necessity threat.

Instructor preparation

Assign Sections 4-14 and Appendices A-C of the core paper. Prepare the interactive Phenomenal Presence Lab if available, but do not allow its verdict to substitute for reasoning about the model.

Timed plan

	Time	Teaching move
	0-12 min	Logic warm-up: necessary, sufficient, conjunction, biconditional
	12-32 min	Causal graphs, indexing discipline, and temporal windows
	32-50 min	Four gates, truth table, and noncompensatory logic
	50-76 min	Component-matrix jigsaw and case adjudication
	76-92 min	Collaborative hearing: sharp law or probabilistic model?
	92-104 min	Falsifiers, auxiliary hypotheses, and surrogate conflict
	104-110 min	Exit ticket

Lecture notes

Write the model twice:

$$N^* := N_1 \wedge N_2 \wedge N_3$$

and

$$H_{C_0} : V \wedge N^* \Leftrightarrow C_0.$$

The first line defines the bundle. The second is a hypothesis about consciousness. This distinction is crucial. The theory may stipulate what the notation N^* abbreviates, but it cannot stipulate that N^* is consciousness and then cite the abbreviation as empirical evidence.

Review the two directions:

$$V \wedge N^* \Rightarrow C_0$$

is the sufficiency claim. A strong counterexample is a boundary-qualified, viable system that robustly passes N_1 , N_2 , and N_3 while independent evidence strongly supports absence of phenomenal presence.

$$C_0 \Rightarrow V \wedge N^*$$

is the necessity claim. A strong counterexample is a case of phenomenal presence with one component genuinely below threshold after boundary, viability, measurement, transport, and auxiliary-demand issues have been controlled.

Move to the graph framework $G = (V, E)$. Nodes can be neurons, populations, regions, subcortical nuclei, bodily control variables, or computational modules. Edges should represent effective causal influence, not merely anatomical connection

or statistical correlation. Structural connectivity asks what pathways exist. Functional connectivity asks which variables covary. Effective connectivity asks what makes a directed difference to what.

The theory is dynamical. Every conclusion is indexed to a time window Δ . Integration, availability, and recurrence must hold for the same candidate, content, grain, regime, and interval. Students should reject a study that measures integration over the whole brain at one second, availability over frontoparietal cortex at 300 ms, and recurrence within visual cortex at 100 ms, then combines the results as though one system satisfied N^* .

Explain each role in one phrase:

- N_1 supplies **unity**.
- N_2 supplies **system-level causal presence**.
- N_3 supplies **bounded temporal presence**.
- V supplies **actual realizability in the declared regime**.

None is sufficient alone. High integration can remain local or unstable. Broad broadcast can be diffuse, homogeneous, or feedforward. Recurrence can occur in an isolated loop. Viability without the three roles merely shows that the system can operate.

Introduce thresholding carefully. Measures can be continuous while the hypothesis uses a thresholded decision rule. The parameters $\tau_1, \theta, \alpha, \lambda, t_{min}, t_{max}, f$, and κ_i are paradigm-specific hypotheses, not universal constants or natural joints in phenomenology. Their legitimacy depends on preregistration, calibration, uncertainty bounds, held-out validation, and sensitivity analysis.

The surrogate-conflict rule is a scientific refusal to cherry-pick. If credible preregistered measures of one component disagree beyond tolerance, that component becomes indeterminate. This does not weaken the underlying biconditional. It distinguishes a sharp ontological hypothesis from a limited experiment.

Work through five cases:

1. **Ordinary waking perception:** likely $V +$ and $N^* +$ for the dominant content; predicted C_0 .
2. **Locked-in syndrome:** motor report fails, but V and N^* may remain; predicted C_0 if the causal bundle passes.
3. **Deep anesthesia:** viability in the biological sense may persist while the relevant dynamical regime and multiple N_i collapse; C_0 not predicted for that window.
4. **Message-bus AI:** widespread copied tokens do not establish N_1 , content-sensitive N_2 , N_3 , or substrate-appropriate V ; likely unlicensed or negative component results rather than an immediate consciousness verdict.
5. **Conflicting integration metrics:** no positive N^* classification even if one metric is high; return indeterminate.

End by emphasizing the model's exclusions. It does not infer consciousness from high intelligence, linguistic fluency, graph shape, raw complexity, widespread activation, or recurrence alone. Its claim is narrower and therefore more exposed.

Worked teaching sequence: from logic to experimental verdict

First contrast a noncompensatory conjunction with a weighted score. If a toy system has normalized values $N_1 = .98$, $N_2 = .95$, and $N_3 = .20$, an average score can exceed a permissive threshold. The conjunction does not allow excellence in unity and availability to compensate for failed temporal presence. This is not a mathematical necessity; it is a substantive commitment of the theory.

Use the following compact truth table, assuming the candidate boundary qualifies and the licensed viability result is $V = 1$:

N_1	N_2	N_3	N^*	Model prediction
1	1	1	1	C_0 predicted
0	1	1	0	C_0 not predicted; diagnostic for N_1 if the case is independently anchored
1	0	1	0	C_0 not predicted; diagnostic for N_2 if the failure is licensed
1	1	0	0	C_0 not predicted; diagnostic for N_3 if encoding and report confounds are controlled
0	0	0	0	C_0 not predicted, but the case does not identify which conjunct is necessary
?	1	1	?	Indeterminate at the component and bundle levels

Then separate four levels of claim:

1. **Ontological hypothesis:** $V \wedge N^* \Leftrightarrow C_0$.
2. **Component criterion:** for example, N_2 requires sufficiently diverse, content-specific causal availability.
3. **Surrogate:** a particular intervention-derived reach score estimates that role in one paradigm.
4. **Experimental verdict:** the score and uncertainty pass, fail, conflict, or cannot be interpreted in this case.

Failure at level 3 does not automatically refute level 1. Repeated inability to construct valid level-3 measures is nevertheless scientifically costly: a theory whose variables cannot constrain observation loses usefulness even if it remains logically possible.

Walk through this illustrative record. Thresholds and values are pedagogical only.

Gate	Estimate and interval	Validity information	Status
Boundary	stable for 92% of held-out perturbations	discovery and test data separated	Pass
V	.81 [.74, .86], threshold .65	regime and horizon declared	Pass
N_1	surrogate A: .72 [.66, .78]; surrogate B: .49 [.41, .57]	both preregistered and credible; conflict tolerance exceeded	Indeterminate
N_2	.69 [.63, .75], threshold .60	report-excluded family still passes	Pass

Gate	Estimate and interval	Validity information	Status
N_3	.58 [.52 , .64], threshold .50	content fidelity and late-report control pass	Pass

The bundle is indeterminate despite three positive gates and one high N_1 measure. Choosing surrogate A after seeing the result would be cherry-picking. The next study should adjudicate the construct or measurement genealogy, not simply add more cases to the same unresolved aggregate.

Finally, explain auxiliary hypotheses symmetrically. Boundary validity, content fidelity, transport, measurement sensitivity, and intervention selectivity can block a purported counterexample. They cannot become costless rescue clauses. A credible program preregisters them, reports their failure rates, and revises when the same auxiliary repeatedly shields the theory from diagnostic cases.

Display plan

Draw a four-gate pipeline terminating in C_0 , with a separate side arrow labeled **surrogate conflict -- indeterminate**. Under the pipeline write:

Same S, same C, same g, same R, same Δ.

Slide sequence

1. Bundle versus hypothesis
2. Necessary and sufficient directions
3. What would defeat each direction?
4. Causal graph and effective connectivity
5. The time-window requirement
6. N_1 : unity
7. N_2 : system-level availability
8. N_3 : bounded recurrence
9. V : independent viability
10. Thresholds and uncertainty
11. Surrogate conflict
12. Five worked profiles

Collaborative exercise: Component-matrix jigsaw

Purpose. Students translate the biconditional into a decision procedure while keeping boundary, measurement quality, and ontological interpretation separate.

Setup. Prepare five one-page case profiles: an ordinary positive case; a licensed N_2 failure with every other gate retained; a case with unassessed viability; a case in which two credible N_1 surrogates conflict; and a case whose selected boundary changes across reasonable specifications. Assign one profile to each home team and provide a partially completed matrix. Each profile should include uncertainty intervals rather than only binary values.

Jigsaw phase. In each home team, assign one student to B_S and V , one to N_1 , one to N_2 , and one to N_3 plus the final logic. Component experts have four minutes with peers from other teams to agree on the minimum information required for their column. They return to teach their home team, not merely announce a value.

Decision phase. The team completes columns for applicability, validity, estimate and interval, gate status, surrogate conflict, full-model prediction, evidential outcome, and ontological caution. For the component-failure case, the team must write both: “the model predicts no C_0 in this specified case” and “the experiment does not directly observe absence of C_0 .” For the boundary-unstable case, students must explain why calculating every component cannot repair the upstream license failure.

Deliverable. Submit the completed matrix plus one genuine threat to an assigned direction of the biconditional. The threat must specify candidate, content, regime, interval, and independent anchor pattern. Collect the two directions across the class rather than requiring both from every team.

Debrief. Ask teams that reached different outcomes to locate the first row at which they diverged. Do not let them debate consciousness until they agree on the input record. The main lesson is that **not predicted**, **indeterminate**, and **unlicensed** encode different scientific situations and imply different next studies.

Collaborative discussion: Sharp law or probabilistic model?

Place three positions around the room: retain the biconditional, replace it now with a probabilistic model, or treat the biconditional as an idealized limit of a graded relation. Students choose privately, then form mixed-position groups. Each group must specify the strongest empirical pattern for each position and answer three questions:

1. Is a sharp early-stage claim recklessly strong or productively exposed?
2. Could N_1 , N_2 , and N_3 be observable dimensions of one deeper process rather than distinct conditions?
3. What pattern of threshold instability, graded anchors, or systematic exceptions would force probabilistic revision?

The group may not conclude “more data are needed” without naming the discriminating data. Collect a one-paragraph joint recommendation and an individual vote after discussion; the change between initial and final votes is evidence of learning, not a grading target.

Common misconceptions

- **“If one term fails, the system is known to be unconscious.”** Only if the full test is licensed and the biconditional is true; the immediate result is the model’s negative prediction.
- **“The conjunction is confirmed when all terms correlate with wakefulness.”** Correlated movement does not identify component necessity.
- **“Viability just means alive.”** Viability is substrate- and horizon-relative capacity to realize the relevant organization.
- **“A threshold makes consciousness metaphysically binary.”** The threshold is a decision rule; the underlying measures and consciousness may be graded.

Exit ticket

Give one sufficiency threat and one necessity threat. For each, name one condition that must be met before the case genuinely bears on the biconditional.

Readings

Required: Stilwell, CØ as N^{**} ; Sections 4-14 and Appendices A-C.

Recommended: Casali et al. (2013); Doerig et al. (2021); Cogitate Consortium (2025).

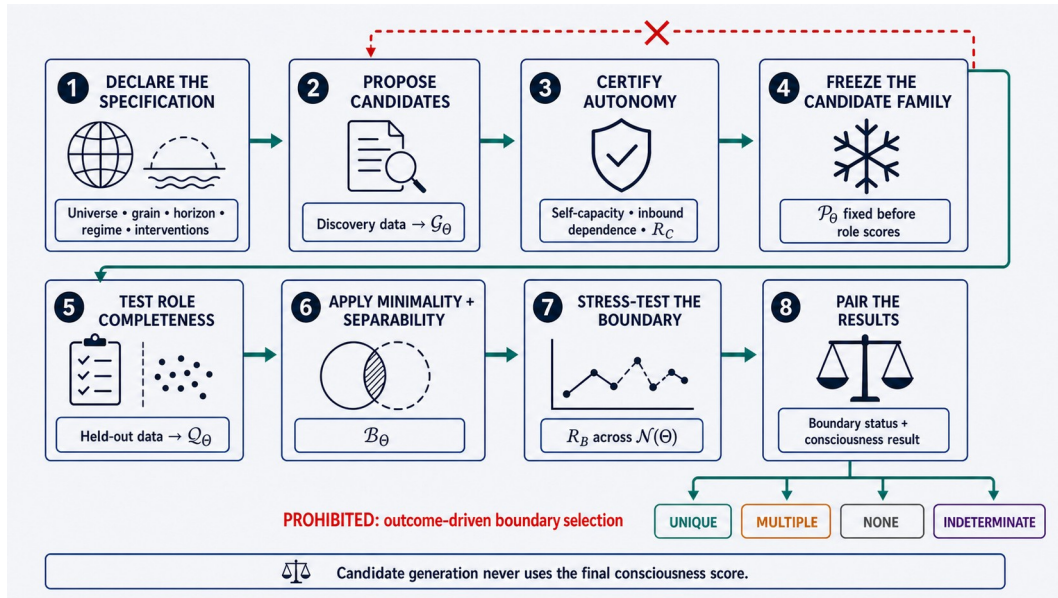
Session 3. Where Is the Conscious Subject?

Learning objectives

Students should be able to explain why every system-level measure is boundary-relative, reconstruct the Dynamical Subject Criterion, distinguish constituents from enabling supports, and assign one of the five boundary statuses.

Instructor preparation

Assign *Where Is the Conscious Subject?*, Sections 1-5 and 7, plus one worked application from Section 6. Display the existing boundary-first workflow figure.



Boundary-first workflow and prohibited feedback

Timed plan

Time	Teaching move
0-10 min	Boundary-reversal thought experiment
10-28 min	Why a node list is already a hypothesis
28-45 min	Autonomy, role completeness, minimality, and candidate lattice
45-73 min	Collaborative boundary adjudication hearing
73-89 min	Fishbowl: minimality, multiplicity, and persistence
89-103 min	Split-brain, brain-body, and distributed-AI applications
103-110 min	Boundary-status synthesis and exit ticket

Lecture notes

Start with a two-module system. Measured separately, each module may have high integration and recurrence but low reach. Measured together, reach may rise while normalized integration falls. Nothing physical changed; only the modeled boundary did. Therefore the proper notation is $N_i(S; g, \Delta, I)$, not a boundary-free N_i .

State the methodological rule:

Individuate S before testing $N^(S)$.*

Retrospective selection is circular when the consciousness-linked score selects the system that the same score then classifies:

$$S^{post} = \arg \max_{S \subseteq \Omega} M_C(S).$$

The alternative uses discovery evidence to propose candidates and held-out evidence to apply theory-specific roles. Candidate generation may use causal community detection, effective connectivity, perturbational clustering, or Markov-blanket proposals. None of those methods by itself proves consciousness.

Present the Dynamical Subject Criterion:

At a preregistered grain and horizon, the candidate conscious system is the inclusion-minimal causally admissible network that is sufficiently autonomous and jointly realizes the required roles of integration, availability, recurrence, and viability.

Break the criterion into three stages.

First, **causal admissibility**. A candidate must have substantial endogenous causal capacity J_{self} and limited enough inbound imposition J_{in} . The autonomy ratio is:

$$A_{\theta}(S) = \frac{J_{self}(S)}{J_{self}(S) + J_{in}(S) + \epsilon}.$$

A high ratio is not enough if both causal effects are trivial, so require $J_{self} \geq \sigma$. Autonomy nominates a bearer; it is not a consciousness score.

Second, **role completeness**. Frozen causally admissible candidates are evaluated for integration, availability, recurrence, and viability using uncertainty-aware thresholds. The same candidate must realize all roles. A role-indeterminate candidate is not silently treated as a failure or redrawn until it passes.

Third, **inclusion-minimality**. Retain every qualifying system with no qualifying proper subsystem at the same specification:

$$B_{\theta} = \text{Min}_{\subseteq}(Q_{\theta}).$$

This rule is set-theoretic, not a contest for the fewest nodes. It prevents a whole brain, organism, dyad, or data center from winning merely by containing a smaller role-complete candidate. At different grains or horizons, nested candidates can remain in an indexed hierarchy; they should not be forced into direct subset comparison without a mapping.

Explain the replacement test. Removal alone confuses constitution with dependence. Oxygen delivery is necessary for continuing brain function, but over a subsecond window its generic support may be externally replaced without changing the fast neural role profile. That supports an enabling interpretation. By contrast, replacing a patterned thalamic loop with constant drive may fail to restore availability and recurrence; that supports a constitutive interpretation.

Use three categories:

- **Constituent:** its specific dynamics are required and generic replacement fails.
- **Interface:** it transmits bounded input or output while the candidate preserves autonomous role completeness.
- **Enabling support:** generic replacement restores the role profile within tolerance.

Then explain the possible boundary outputs:

1. unique stable candidate;
2. multiple stable, pairwise separable candidates;

3. nested candidates across mapped scales;
4. no decisively qualifying candidate; or
5. boundary-indeterminate.

The distinction between “no qualifying candidate” and “boundary-indeterminate” matters. In the first, decisive tests find none within the declared universe and proposal family. In the second, confidence intervals, mappings, candidate reliability, or separability remain unresolved. Neither automatically means no consciousness. Without a stable bearer, a system-level consciousness inference is normally unlicensed.

Apply the method to split brains. Do not count subjects from the corpus callosum alone. Test left, right, and residual whole-system candidates. Ask whether each side retains endogenous capacity and role completeness, whether subcortical loops preserve a joint candidate, and whether perturbing one side leaves the other stable. Multiple subjects are supported only if multiple inclusion-minimal candidates are pairwise separable under intervention.

Apply it to distributed AI. Product names are not boundaries. A model call, persistent memory, scheduler, tools, evaluators, and human interventions may form different causal units at different runtime horizons. Service ablations, memory substitution, state resets, and controller perturbations determine which components are constitutive, enabling, or environmental. Shared storage and API connectivity do not establish one subject.

Worked teaching sequence: a candidate lattice

Use an illustrative split system with preregistered thresholds $A_\theta \geq .70$ and $J_{self} \geq .10$. The numbers are not proposed biological constants; they make the selection logic visible.

Candidate	J_{self}	J_{in}	A_θ	Role complete?	Initial result
Sensor cluster S_C	.05	.01	.83	No	Fails the absolute-capacity floor and role completeness
Left system S_L	.32	.10	.76	Yes	Qualifies
Right system S_R	.30	.08	.79	Yes	Qualifies
Whole system S_W	.64	.25	.72	Yes	Qualifies before minimality
System plus apparatus S_E	.66	.70	.49	Yes on raw role scores	Fails autonomy

This table demonstrates why a ratio alone is inadequate: the tiny sensor cluster has the highest autonomy ratio but negligible endogenous effect and no role completeness. It also shows what inclusion-minimality does. Because S_L and S_R are qualifying proper subsets of S_W , the whole system is not retained at the same specification. If interrupting specific left-right coupling leaves each side's role profile within equivalence bounds, the result supports multiple pairwise separable candidate bearers. If interruption collapses both, the apparent two-candidate interpretation weakens.

Now reveal that the role scores were computed before the candidates were frozen. Ask students to identify the circularity. The correct sequence is:

1. use discovery data to propose S_C, S_L, S_R, S_W , and S_E ;
2. freeze candidates, grain, horizon, autonomy rules, and mappings;
3. use held-out perturbations to test admissibility, role completeness, minimality, and separability; and
4. only then apply the consciousness-linked inference.

Compare six boundary notions. Anatomy supplies a convenient node set. Statistical communities supply covariance structure. Markov blankets nominate conditional-independence partitions. Organismic accounts emphasize self-maintenance. Causal boundaries identify endogenous difference-making units. Subject boundaries add the full admissibility, role, and minimality criteria. Earlier notions can propose candidates, but no proposal method is allowed to validate itself by the final consciousness score.

Temporal persistence requires an additional mapping. A candidate at t_1 and one at t_2 count as stages of one persisting bearer only if their causal organization, memory or state continuity, and boundary mapping satisfy a declared identity rule. Rapidly changing membership need not imply a new subject at every millisecond, but neither may personal persistence be assumed from a stable product label or organism name.

Display plan

Use the following sequence:

$$\Omega \rightarrow G_{\theta} \rightarrow P_{\theta} \rightarrow Q_{\theta} \rightarrow B_{\theta} \rightarrow \text{consciousness test}.$$

Label the forbidden arrow from final consciousness score back to candidate selection.

Slide sequence

1. Every measure is a function of a boundary
2. Boundary reversal
3. Physical, causal, and subject boundaries
4. The circular *arg max* procedure
5. Discovery / test separation
6. Endogenous capacity and external dependence
7. Autonomy is not consciousness
8. Role completeness
9. Inclusion-minimality
10. Constituent, interface, enabling support
11. Five boundary outputs
12. Split brain and distributed AI

Collaborative exercise: Boundary adjudication hearing

Purpose. Students experience boundary selection as a causal research problem rather than an anatomical or intuitive stipulation.

Case pool. Assign one case to each team: a patient on cardiopulmonary support; a brain-computer interface; an organoid coupled to a recurrent digital controller; a tightly coordinated two-agent conversation; or an AI service with a base model, persistent memory, scheduler, tools, and human oversight. Give a common observation window and require teams to state when a different horizon would change the candidates.

Preparation. Within each team, the candidate advocate proposes two or three plausible boundaries. The autonomy auditor evaluates endogenous and inbound causal influence. The constitution specialist designs removal and generic-replacement tests. The reporter applies role completeness, inclusion-minimality, stability, and separability. Every proposal must list omitted variables and interfaces.

Hearing. Pair teams. One presents its preferred boundary in four minutes; the other serves as an independent panel and asks only causal questions: What is imposed from outside? Which support can be generically replaced? Does a proper subsystem remain role-complete? What perturbation would distinguish one bearer from two? The presenting team then

issues one of five statuses: unique, multiple, nested, none, or indeterminate. It must state what the result does **not** yet establish about consciousness.

Deliverable. Submit a one-page boundary docket containing the observational universe, candidate family, grain and horizon, autonomy evidence, replacement test, qualifying subsets, separability result, boundary status, and most informative next perturbation.

Debrief. Put the same case at two horizons on the board. Show how a ventilator can be an enabling support over seconds yet indispensable to longer-horizon biological viability, or how persistent memory can be an interface for a single model call but a constituent of an agent-scale process. Boundary indexing is principled when declared prospectively; it is evasive when changed only after seeing the consciousness score.

Collaborative discussion: Minimality, multiplicity, and persistence

Use a sixteen-minute fishbowl. The first inner circle has five minutes to address whether inclusion-minimality unfairly privileges smaller candidates while the outer circle tracks set-theoretic minimality versus fewest nodes, simultaneous candidates versus one changing candidate, and causal individuality versus phenomenal unity. Switch for a five-minute second round addressing:

1. Can one biological individual contain multiple simultaneous candidate bearers?
2. What evidence would distinguish multiple subjects from a nested hierarchy at different grains?
3. How should temporally changing boundaries relate to persistence of a subject?

Reserve four minutes for each circle to contribute to a consensus-with-dissent statement and two minutes for report-out. The statement includes one case where minimality clarifies the bearer and one where it leaves the bearer legitimately unresolved.

Common misconceptions

- **“The most integrated set is automatically the subject.”** That is a theory-internal rule, not an independent boundary result.
- **“Everything causally necessary belongs inside.”** Remote necessity must be separated from specific constitutive dynamics.
- **“Multiple candidates means absurd subject proliferation.”** Multiplicity requires role completeness, minimality, stability, and separability.
- **“No qualifying candidate means unconscious.”** It may only show that individuation failed within the tested universe.

Exit ticket

State one boundary reversal and one intervention that could discriminate the rival boundaries without using a consciousness score to select between them.

Readings

Required: Stilwell, *Where Is the Conscious Subject?*, Sections 1-5 and one application from Section 6.

Recommended: Albantakis et al. (2023) on exclusion; Friston et al. (2021) on Markov blankets; Moreno and Mossio (2015) on autonomy.

Session 4. Viability and Integration: Realization and Unity

Learning objectives

Students should be able to explain why V and N_1 answer different questions, distinguish synergy from activity and correlation, formulate a non-circular viability gate, and design a diagnostic case for integration.

Instructor preparation

Assign Sections 6 and 10-12 of the core paper and Sections 8.1 and 8.4 of *Ablating N**. Prepare one example of high correlation caused by common input and one example of a synergistic computation, such as XOR.

Timed plan

Time	Teaching move
0-12 min	XOR demonstration: correlation, redundancy, and synergy
12-30 min	N_1 as a unity condition without overclaiming
30-44 min	Estimator families, causal evidence, and confounds
44-56 min	V as a prospectively frozen regime-licensing gate
56-83 min	Collaborative single-failure design sprint and audit
83-99 min	Structured controversy: What kind of unity does N_1 buy?
99-110 min	IIT comparison, falsifiers, and exit ticket

Lecture notes

N_1 addresses the unity problem. A conscious content is not merely a list of local signals. The theory requires that some subset U jointly carry or realize information in a way not reducible to the parts considered independently:

$$\exists U \subseteq S \text{ such that } Synergy_{\Delta}(U) \geq \tau_1.$$

The formula does not settle which synergy measure is best. It declares the target role and a decision policy. Candidate measures include multivariate synergy, partial-information-decomposition families, O-information-related measures, perturbational complexity, effective-connectivity models, and causal-cut sensitivity. Each captures part of the construct and each has failure modes.

Use three contrasts.

Activity versus integration. Two modules can be highly active while causally independent. Activity magnitude does not establish unity.

Correlation versus integration. Two modules driven by a common source can covary almost perfectly while making no joint causal contribution beyond that source. Sensor mixing and volume conduction can produce the same illusion.

Redundancy versus synergy. Redundant channels each carry the same information; synergistic organization contains information available only from the joint state. The XOR example is useful because neither input alone predicts the output, while the pair does.

Explain why N_1 is not simply IIT. IIT treats intrinsic integrated information as central to both phenomenal existence and structure. CØ / N* extracts a unity pressure without adopting IIT's full axiomatic identity or treating integration as

sufficient. In the present program, high N_1 can occur without consciousness when N_2 or N_3 fails. The empirical question is whether integration adds indispensable discrimination once the other conditions are preserved.

Now introduce V . Viability answers whether the candidate is in an operating regime capable of realizing and sustaining the measured organization. In biological systems, relevant indicators may include arousal-system integrity, metabolic support, perturbational responsiveness, noise bounds, effective coupling, and recovery capacity. In artificial systems, viability may concern persistence of the relevant control organization, energy and hardware support, state retention, scheduler continuity, and robustness across the declared runtime. Biological survival is not the universal definition.

V is dangerous because it could explain away any counterexample. The safeguards are:

1. define the functional role of viability independently of consciousness;
2. choose substrate-specific indicators and thresholds before the critical verdict;
3. state the interpretability consequence of each failed indicator;
4. analyze borderline cases with uncertainty and recovery tests;
5. version any revised viability rule and test it prospectively; and
6. penalize a theory that requires a bespoke new definition for every adverse case.

Clarify the relation between boundary and viability. Boundary qualification says what unit is eligible to be tested. Viability says whether that unit is actually operating in the relevant regime. Neither is itself phenomenal evidence. A stable boundary can exist in a nonviable preparation, and a viable organism can contain no boundary-qualified role-complete candidate at a particular grain.

The key integration audit asks whether a valid case can occupy:

$$V = 1, N_1 = 0, N_2 = 1, N_3 = 1.$$

Such a case is difficult because interventions that reduce synergy often also disrupt availability or recurrence. The proper outcome may be **causally coupled**, not a forced verdict about necessity. Promising designs preserve local information and aggregate activity while perturbing cross-module timing, effective coupling, or causal coordination. Required controls include common-drive models, SNR matching, source reconstruction, boundary stability, and direct verification that N_2 , N_3 , and V remain within equivalence bands.

Use the phrase **unity without overclaiming**. N_1 does not prove a subject, because subject individuation comes first. It does not prove presence, because the other gates matter. It does not determine character, because the character bridge is separate. It claims only that irreducibly joint realization is a necessary part of a present content.

Worked teaching sequence: viability is not integration

Begin with a contrast table.

Question	V	N_1
Core concern	Can this candidate actually sustain the relevant regime over the declared horizon?	Is the content realized as an irreducibly joint causal event?
Typical evidence	physiological stability, causal responsiveness, state retention, recovery, scheduler continuity	perturbational dependence, causal-cut sensitivity, synergy, nonfactorizable joint influence
Common false positive	an inert formal description or failing preparation is treated as a realized system	common input, duplicated information, coactivation, or raw complexity is treated as unity
Inferential role	licenses application of the dynamical test	supplies one necessary phenomenal-presence gate

Question	V	N_1
Failure does not show	that the substrate type can never realize consciousness	that the candidate lacks all complex or useful processing

Use XOR to illustrate synergy without phenomenological inflation. Let independent fair inputs X and Y determine $Z = X \oplus Y$.

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

Knowing X alone or Y alone does not reduce uncertainty about Z , while knowing the pair determines it. The joint state carries predictive structure unavailable in either part. This is a clean example of formal synergy. It is not evidence that a three-bit XOR gate has a unified experience. The candidate boundary, causal realization, viability, availability, recurrence, and phenomenal bridge remain unestablished.

Then contrast XOR with two misleading cases. In **redundant copying**, two modules carry the same signal; joint information is high but adds little beyond either module. Under **common drive**, two modules correlate because a third source controls both; observed dependence need not reflect joint causal integration. Interventions that cut one pathway, perturb one source, or compare the intact system with a partitioned causal model help distinguish these structures.

No single estimator should be equated with N_1 .

Estimator family	What it may reveal	Main limitation to teach
PID-style synergy	information available only from joint sources	decomposition choices vary and observational synergy need not be intrinsic causation
O-information or high-order dependence	balance of redundancy and synergy at scale	sign and magnitude can be boundary- and sample-sensitive
Perturbational complexity	differentiated, integrated causal response	complexity may combine several components and depends on stimulation and recording routes
Causal-cut sensitivity	loss caused by partitioning the system	partitions, interventions, and normalization encode assumptions
Effective-connectivity models	directed influence and possible common-drive control	model misspecification and latent causes can create false integration

The operational strategy is triangulation with prospective conflict rules. Agreement among estimators with different failure modes strengthens construct validity. Credible disagreement returns indeterminate and motivates a construct audit.

For V , have students write a multidimensional operating envelope rather than a life/death flag. A biological study might include perfusion, temperature, arousal regime, evoked responsiveness, and recovery within a ten-minute horizon. An artificial study might include power, scheduler continuity, writable state, message integrity, and recovery from bounded perturbation over one hundred processing cycles. The envelope must be frozen before the diagnostic case; otherwise every failure can be reclassified as “nonviable” after the fact.

Display plan

Draw four panels:

1. high activity, no coupling;
2. high correlation from common input;
3. redundant copies;
4. synergistic joint influence.

Beside them write: V licenses the regime; N_1 tests unity.

Slide sequence

1. What unity must exclude
2. N_1 in formal and plain language
3. Activity is not integration
4. Correlation is not integration
5. Redundancy is not synergy
6. Candidate estimator families
7. Why N_1 is not IIT
8. The role of V
9. The viability rescue-clause danger
10. A valid N_1 diagnostic case
11. Causal coupling as a result

Collaborative exercise: Single-failure design sprint

Purpose. Students learn why component necessity is identified in disagreement regions, not by showing that all measures decline together in anesthesia or injury.

Design challenge. Teams must engineer the profile $V = 1, N_1 = 0, N_2 = 1, N_3 = 1$ without using global suppression. The model reconstructor defines which aspect of joint structure is removed. The intervention engineer chooses a manipulation of cross-module timing, effective coupling, or causal coordination. The validity auditor specifies common-input, source-fidelity, SNR, task-demand, and measurement-route controls. The rival advocate states the predictions of the full theory, the N_1 -ablated model, and one nonconjunctive rival.

Constraint cards. Midway through the exercise, give each team one complication: N_2 falls slightly; the boundary splits; a second N_1 surrogate remains high; recurrence appears slower; or the anchor changes only during report trials. Teams must apply any preregistered contingency; otherwise they classify the current result as coupled, indeterminate, or unlicensed and may version a wider equivalence margin only for a new confirmatory study. They may not repair the current verdict by changing a margin after seeing the complication.

Deliverable. Produce a single-failure study canvas with manipulation, intended mechanism, retained-gate checks, equivalence margins, independent anchors, rescue condition, rival predictions, outcome rules, and one reason the study might remain inconclusive. Another team performs a five-minute selectivity audit.

Debrief. Compare designs that reduce statistical dependence with designs that change causal synergy. Ask whether local information and aggregate activity were preserved. Stress that an intervention named “integration disruption” earns no special evidential status until its selectivity is demonstrated.

Collaborative discussion: What kind of unity does N_1 buy?

Run a structured controversy. One side argues that irreducible joint information is a plausible necessary condition for a unified experience; the other argues that it is a measure of complex coordinated processing that may be neither necessary nor phenomenally specific. Switch sides, then require a synthesis answering:

1. Which unity claim is phenomenological, which is causal, and which is merely statistical?
2. Should V remain an external realization gate or become a fourth member of N^* ?
3. If licensed interventions repeatedly show N_1 and N_3 to be inseparable, should the theory combine them, reinterpret them, or retain the factorization for explanatory reasons?

The synthesis must name a result that would support each revision. A bare appeal to intuition about unity is insufficient.

Common misconceptions

- **“More integration means more consciousness.”** The theory states a thresholded necessary role, not a general consciousness quantity.
- **“PCI is N_1 .”** PCI is one candidate surrogate and may reflect multiple constructs.
- **“A viable system is a conscious candidate.”** It is only in an interpretable operating regime.
- **“Failure to selectively ablate N_1 confirms it.”** It may show causal coupling or inadequate intervention.

Exit ticket

Explain why a highly correlated network can fail N_1 , and state one rule that prevents V from becoming a post hoc escape hatch.

Readings

Required: Stilwell, *C $\emptyset$ as N^{**}* , Sections 6 and 10-12; Stilwell, *Ablating N^{**}* , Sections 8.1 and 8.4.

Recommended: Albantakis et al. (2023); Casali et al. (2013); selected work on synergy and partial information decomposition.

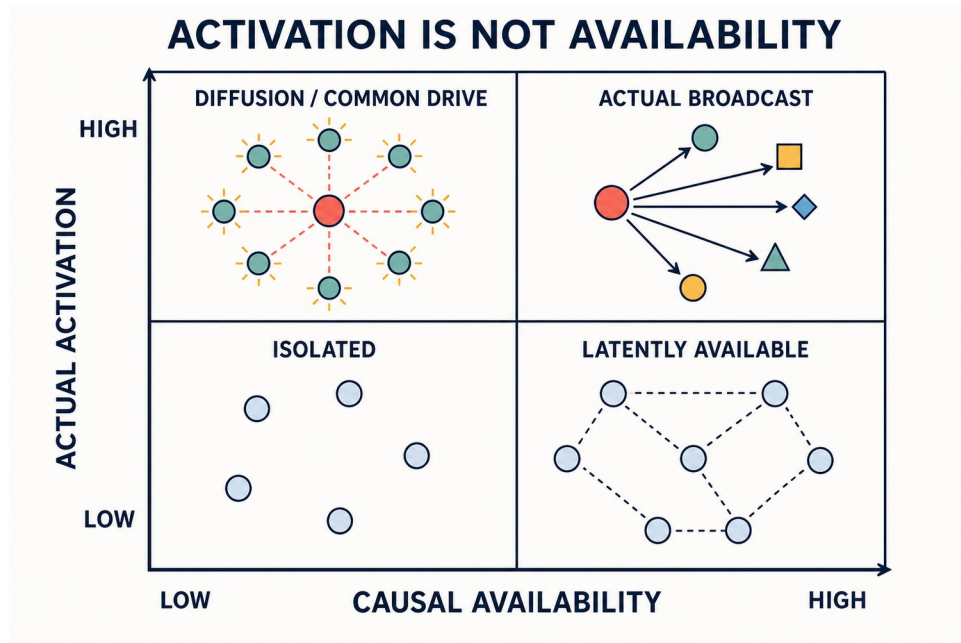
Session 5. Consciousness Without Report: System-Wide Availability

Learning objectives

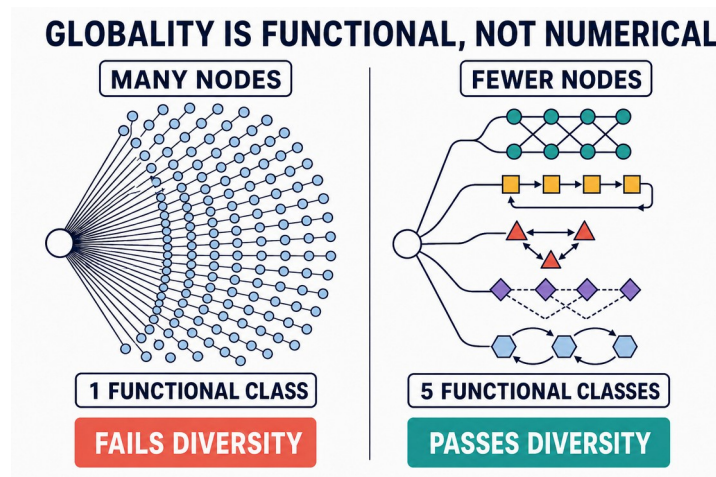
Students should be able to preserve the six central distinctions in the N_2 paper, reconstruct the recipient-qualification logic, distinguish functional diversity from node count, and explain how posterior or subcortical availability can qualify without automatic success.

Instructor preparation

Assign *Consciousness Without Report*, Sections 1-5, 7-9, and Appendices D and H. Display both existing contrast figures.



Activation and causal availability are logically independent



Functional globality is not a node-count statistic

Timed plan

Time	Teaching move
0-10 min	Four-quadrant activation / availability warm-up

	Time	Teaching move
	10-29 min	Six distinctions and the report-independent criterion
	29-45 min	Worked recipient ledger, diversity, and evidence ladder
	45-72 min	Collaborative causal-routing laboratory
	72-89 min	Structured discussion: Is availability already access?
	89-104 min	Rivalry, locked-in state, subcortical and AI cases; falsifiers
	104-110 min	Exit ticket

Lecture notes

Begin with the central correction: **availability is a disposition, not an output and not a census of current activation**. A recipient may be idle yet causally poised to respond to a content under a nearby admissible context. Conversely, a recipient may activate because of common input, arousal, diffusion, or volume conduction while lacking content-sensitive access.

Teach the six non-collapse rules:

1. actual activation is not causal availability;
2. availability is not report;
3. broadcast is not diffusion;
4. globality is not anatomical extent;
5. functional diversity is not node count; and
6. posterior or subcortical reach is not the same as executive access.

The proposed criterion is:

A content is system-wide available when admissible, content-specific interventions show that it could causally modulate multiple independently individuated and functionally distinct recipient classes within a preregistered latency bound, across a nontrivial mass of nearby contexts, without requiring overt report, executive uptake, or current task use.

Break this into the design sequence.

First, freeze the boundary-qualified system, content contrast, grain, time horizon, and operating regime. “Global” is always global relative to that candidate.

Second, build recipient classes independently of the target content. Provisional modules are grouped by their causal response repertoires under calibration interventions. Two parcels count as different recipient classes only if their response repertoires differ reliably. Different anatomical names are not enough.

Third, define admissible nearby contexts. The preparation-burden functional I' limits training time, energy, controller involvement, structural change, and other modifications. A context is inadmissible if it changes the operating regime or installs a new route. This prevents the experiment from creating the availability it claims to discover.

Fourth, perform matched content interventions. For recipient class k in context u , estimate a content-specific effect $E_{k,u}$ by changing the target content while matching intensity, duration, source support, and nonspecific global-state change. Source fidelity must show that the intended content contrast was preserved. Injection-site leakage cannot be counted as recipient reach.

Fifth, require five properties to concur for a qualifying recipient:

- effect above the class-specific threshold;
- latency below the maximum;

- content selectivity beyond matched null interventions;
- route dependence under a nominated pathway or route-family interruption; and
- robustness across at least the preregistered mass ρ_U of nearby contexts.

This concurrence matters. A large effect in one context, fast latency in another, and route support in a third do not jointly establish availability.

Sixth, aggregate over functional classes. Let a_k mark whether class k passes all gates. Weighted coverage is

$C_A = \sum_k w_k a_k$; class count is $K_A = \sum_k a_k$; diversity evenness H_A prevents one recipient from carrying nearly all the effect while token effects barely cross thresholds elsewhere. Weights cannot be proportional to raw node count alone.

Finally, repeat the analysis after removing report classes and after removing executive classes. A positive result must survive both reduced-family tests:

$$N_2^{-R} = 1, N_2^{-E} = 1.$$

This does not require destructive lesions. The study can omit report by design, use natural motor dissociations, transiently interrupt channels, or model their contribution while preserving arousal and generic support.

Explain posterior and subcortical possibilities. The theory is anatomically flexible but causally strict. A posterior-dominant visual network can pass if it reaches genuinely distinct sensory, spatial, orienting, affective, or visuomotor classes with timely, content-specific, route-supported effects. A thalamic pulse that merely changes arousal fails content selectivity. Prefrontal activity can count when it plays a qualifying role, but it is not mandatory by fiat.

Use the four component outputs. **Positive** means every validity and causal gate passes. **Negative** means the test is licensed and an upper uncertainty bound falls below a required gate. **Indeterminate** means valid evidence straddles thresholds, conflicts, or lacks discriminating coverage. **Unlicensed** means the boundary, content, recipient taxonomy, context, intervention, regime, source fidelity, or transport interpretation failed.

Worked teaching sequence: six ways availability can collapse

Use paired examples to keep the criterion from collapsing into a familiar proxy.

Tempting proxy	Counterexample	What N_2 additionally requires
Current activation	Arousal drives many regions without content-specific effects	Matched content interventions and selective downstream change
Current activation	An idle controller is not active now but is demonstrably reachable in a nearby low-burden context	Counterfactual receptivity can count if it is not manufactured by retraining
Spatial spread	A signal passively diffuses through tissue or a shared memory bus	Route-dependent causal modulation of recipient function
Node count	One compact circuit affects four distinct functional classes	Functional diversity rather than anatomical extent
Many recipients	One hundred identical loggers store the same copy	Nonredundant recipient families and an evenness or diversity constraint
Executive uptake	A report module receives the content and dominates the aggregate	Positive reduced-family results after report and executive removal

Work through an illustrative recipient ledger with frozen weights. A class qualifies only when effect, latency, selectivity, route, and context tests all pass.

Recipient class	Weight	Effect and interval	Latency	Selectivity / route / context	Class result
Sensory comparison	.25	+ .18 [.14 , .22]	85 ms	Yes / yes / yes	Qualifies
Value updating	.20	+ .12 [.08 , .16]	110 ms	Yes / yes / yes	Qualifies
Memory indexing	.20	+ .04 [- .01 , .09]	125 ms	Unresolved / yes / yes	Indeterminate
Orienting / visuomotor	.20	+ .15 [.10 , .20]	95 ms	Yes / yes / yes	Qualifies
Report preparation	.15	+ .20 [.16 , .24]	160 ms	Yes / no / yes	Fails

The qualifying weight is .65 across three distinct classes. Renormalized qualifying shares are .385, .308, .308, giving normalized Shannon evenness $H_A \approx .99$. If the preregistered coverage threshold is .60, minimum class count is three, latency ceilings are 180 ms, and minimum evenness is .80, the full-family aggregate is positive despite one indeterminate and one failed class. In this toy taxonomy, report removal excludes the .15 report class; renormalized qualifying coverage is $.65 / .85 = .765$, so N_2^{-R} remains positive. Executive removal excludes memory indexing and report preparation; the three qualifying nonexecutive classes exhaust the remaining .65 weight, so N_2^{-E} is also positive after the preregistered renormalization. The unresolved and failed classes remain visible. If qualifying weight had been concentrated in executive subclasses, the reduced-family result could fail even when the full aggregate passed.

Use a causal-evidence ladder:

1. decoding or covariance shows that information is present;
2. temporal precedence constrains direction but not common cause;
3. matched content intervention shows selective downstream difference;
4. route interruption tests mediation rather than parallel drive; and
5. rescue strengthens the causal-role interpretation.

Not every experiment must reach the top rung, but the conclusion must match the rung achieved. A large decoding effect cannot be described as proven broadcast.

Finally, differentiate CØ / N* from a standard GNWT claim prospectively. Look for a case in which content-specific causal reach among diverse posterior, subcortical, affective, orienting, or visuomotor recipients passes while report and executive families do not. CØ / N* can count that as positive N_2 . A version of GNWT requiring a particular frontoparietal ignition or executive access should not. If the theories make identical predictions after their terms are operationalized, the disagreement is terminological rather than diagnostic.

Display plan

On the board, use the pipeline:

content fidelity $\rightarrow$ effect + latency + selectivity + route + context $\rightarrow$ coverage + count + diversity $\rightarrow (-R, -E) \rightarrow N_2$

Slide sequence

1. Availability is not an output
2. Activation / availability matrix
3. Broadcast / diffusion controls
4. Globality / anatomical extent
5. Recipient classes from causal repertoires

6. Admissible contexts and preparation burden
7. Five gates for one recipient
8. Coverage, class count, and diversity
9. Report and executive exclusion
10. Posterior and subcortical implementations
11. Four N_2 outcomes
12. Direct falsifiers

Collaborative exercise: Causal-routing laboratory

Purpose. Students operationalize report-independent availability and learn why node count, coactivation, copying, and potential connectivity are inadequate substitutes.

Architectures. Architecture A copies every token to one hundred identical logging processes. Architecture B routes one content to five modules with independently demonstrated sensory comparison, value updating, memory indexing, conflict monitoring, and action-selection repertoires. Architecture C is a compact biological circuit with only twelve measured nodes but four causally distinct recipient families. Assign one architecture, not all three, to each team and provide a partially completed ledger; the class compares architectures during debrief.

Roles and procedure. The recipient-taxonomy lead decides which recipients are genuinely functionally distinct. The intervention lead specifies target-content and matched-control perturbations. The route auditor designs mediation, interruption, latency, and alternative-path tests. The report-exclusion lead recalculates the aggregate after removing report and executive classes and checks preparation burden. Teams then draw a routing graph that marks actual effects, idle reachability, common-drive paths, and invalid recipients.

For every qualifying class, require a ledger entry for effect magnitude, content selectivity, latency, route dependence, context robustness, and uncertainty. A current absence of activation may count only if a nearby low-burden intervention demonstrates counterfactual receptivity without constructing a new architecture. A recipient reached only after extensive retraining fails the preparation-burden rule for the original state.

Deliverable. Each team submits an N_2 recipient ledger for its assigned architecture, diversity score rationale, reduced-family results N_2^{-R} and N_2^{-E} , final component status, and the smallest missing test. During a five-minute gallery comparison, teams add one contrast with each other architecture. They must say why a nonqualifying architecture fails N_2 without concluding that it is unconscious.

Debrief. Put “global” on trial. Contrast spatial extent, number of recipients, functional diversity, and causal influence. Architecture C may be more system-wide in the model’s sense than Architecture A. The criterion concerns the range of roles the content can causally modulate inside the candidate system.

Collaborative discussion: Is availability already access?

Use four corner positions: N_2 is covert access consciousness; N_2 is a weaker causal prerequisite of access; N_2 is part of phenomenal presence but not access; or the distinction is unstable. Mixed groups must formulate the best version of every position and then answer:

1. Is a spatially compact but functionally diverse circuit genuinely system-wide?
2. Does counterfactual availability import the very access capacity the theory claims not to require?
3. How much perturbation or preparation is allowed before the experiment creates a new state rather than revealing the old one?

The joint conclusion must specify at least one empirical case on which report-independent N_2 and a standard access criterion diverge.

Common misconceptions

- **“Decoded everywhere means globally available.”** Decodability can reflect common input, leakage, or redundant representation.
- **“No report means no evidence.”** It means report is not a required recipient; other causal and content-indexing evidence remains possible.
- **“Prefrontal inactivity disproves broadcast.”** The criterion is functional and candidate-relative, not fixed to one region.
- **“A message bus establishes globality.”** Subscribers must undergo content-sensitive, class-native state transitions.
- **“One favorable readiness condition is enough.”** Context mass and preparation burden are preregistered gates.

Exit ticket

State one case of activation without availability and one case of availability without current activation. For each, name the intervention that distinguishes it.

Readings

Required: Stilwell, *Consciousness Without Report*, Sections 1-5, 7-9, and Appendices D and H.

Recommended: Tsuchiya et al. (2015); Mashour et al. (2020); Kronemer et al. (2022); Cogitate Consortium (2025).

Session 6. Recurrent Stability and Temporal Presence

Learning objectives

Students should be able to distinguish recurrence from persistence and report-related late activity, explain the bounded-dwell condition, design a selective N_3 perturbation, and state how temporal presence differs from temporal character.

Instructor preparation

Assign Section 8 of the core paper, Sections 8.3 and 9.3 of *Ablating N**, and Sections 4.9 and 6.5 of the phenomenal-character paper. Prepare timelines for feedforward sweep, recurrent stabilization, working-memory maintenance, and motor report.

Timed plan

Time	Teaching move
0-10 min	Timeline sorting: encoding, recurrence, report, memory
10-27 min	N_3 as bounded temporal presence
27-43 min	Time-unrolled architectures: recurrence versus persistence
43-70 min	Collaborative temporal-intervention studio
70-87 min	Fishbowl: topology, duration, and substrate generality
87-104 min	Temporal character, RPT comparison, edge cases, and falsifiers
104-110 min	Exit ticket

Lecture notes

N_3 is the claim that a phenomenally present content is not an instantaneous feedforward spark. It is maintained through recurrent or metastable organization for a bounded interval:

$$\exists L \subseteq S \text{ such that } t_{\min} \leq \text{dwell}(c; L) \leq t_{\max}.$$

The lower bound blocks momentary transients. The upper bound blocks the assumption that more persistence is always better. Seizure-like fixation, uncontrolled reverberation, and an inert stored state are not thereby more conscious. The target is metastability: sufficient maintenance for system-level temporal presence with capacity for updating and decay.

Distinguish four phenomena:

1. **Feedforward duration:** a slow cascade can persist without returning causal influence.
2. **Recurrent processing:** later states causally influence earlier or mutually connected processing stages.
3. **Working-memory maintenance:** a content may remain usable after the proposed phenomenal interval and include executive or report demands.
4. **Stimulus trace:** passive persistence in receptors, sensors, or slow variables can outlast the content without recurrent organization.

For the primary operational contrast in this manual, **causal recurrence** means that an earlier content-supporting state makes a difference to a later state through a directed return relation in the time-unrolled causal graph. The return may be an inter-node feedback loop or genuine state-mediated self-feedback, provided intervention on the earlier state changes the later content state beyond continuing input and passive decay. A slow unit whose state merely fades forward is the

duration-matched rival. Metastable return without an identifiable loop can be evaluated as an alternative N_3 surrogate, but it should not be silently equated with the topology claim; credible disagreement invokes the surrogate-conflict rule.

N_3 is content-indexed. A globally recurrent or oscillatory state with no validated content c does not pass $N_3(S, c)$ merely because something recurs. It may be relevant to background state or viability and still leave the content gate unassessed.

An estimator for dwell might use survival time before trajectory autocorrelation drops below a frozen fraction, a metastable hidden-state plateau, recurrent effective connectivity, or causal-loop response under perturbation. None is automatically N_3 . The measure must track content-relevant feedback rather than generic slow activity.

Explain the relationship to Recurrent Processing Theory. RPT provides the strongest neighboring pressure: feedback within sensory systems may be central to conscious perception. CØ / N* accepts recurrence as necessary for temporal presence but denies that a stable local loop is sufficient. A recurrent process that is causally isolated or unintegrated fails N_2 or N_1 .

The selective test for N_3 is unusually time-sensitive. An early perturbation may prevent encoding, thereby reducing content fidelity rather than recurrence. A late perturbation may disrupt report or memory after the phenomenal interval. The intervention must target recurrent return paths or effective gain during the preregistered stabilization window while preserving initial content information, integration, causal reach, viability, boundary, and operating regime.

Feedforward architectures with long time constants are powerful adversarial cases. If a strictly feedforward system can match every independent anchor assigned to recurrent stabilization, the theory must explain why loop structure rather than temporal persistence is necessary. This is a direct differentiating test between a causal-topology claim and a duration claim.

Use rescue designs. Disrupt a recurrent pathway, verify that N_3 crosses its gate while the retained components remain within equivalence bands, then restore the pathway or its causal role without globally improving every measure. Recovery of the predicted anchor pattern is stronger evidence than a nonspecific before-and-after deficit.

Now separate temporal presence from temporal character. N_3 asks whether a content lasts through an appropriate recurrent interval. The character extension's path profile Π_c asks how the experience unfolds: direction, rate, curvature, dwell, ordering, and hysteresis in quality space. Two experiences can satisfy the same N_3 gate while differing as steady versus throbbing pain, ascending versus descending melody, or smooth versus stuttering apparent motion.

Discuss offset and onset evidence. Late signals may reflect report, working memory, task termination, or post-perceptual processing. A valid N_3 measure must be tied to the content and temporal role, not merely to lateness. Conversely, absence of a familiar late component does not establish absence of all recurrent stabilization if another causal route supports it.

Direct challenges include well-supported conscious contents under genuine feedback absence, valid N_3 ablations that leave all phenomenal anchors unchanged while other conditions remain intact, and recurrence measures that fail to outperform a simpler persistence model on held-out cases.

Worked teaching sequence: recurrence, persistence, and report

Draw four time-unrolled architectures.

Architecture	How the state lasts	Diagnostic perturbation	What duration alone misses
Recurrent network	Later states causally feed back into the content-supporting organization	Interrupt the return path during the stabilization interval	Whether persistence depends on a causal loop
Feedforward chain with slow units	Each unit drives the next and decays slowly	Cut a purported feedback edge that is not causally used	The same dwell time can occur without recurrence

Architecture	How the state lasts	Diagnostic perturbation	What duration alone misses
Passive trace	A state remains stored but does not modulate ongoing processing	Probe causal influence while the trace is present	Storage is not recurrent maintenance
Executive working memory	Control circuitry refreshes a representation for later task use	Disrupt control after the proposed phenomenal interval	Late maintenance may support report rather than presence

Give students two systems with identical observable dwell of 240 ms. System R returns toward a content-specific attractor after a small perturbation, and cutting its feedback edge shortens dwell to 70 ms. System F has slow feedforward units; the same nominal edge cut changes nothing, while shortening a unit time constant reduces dwell. Autocorrelation alone treats the systems alike. Perturbational return and pathway dependence distinguish causal recurrence from duration.

Now show how timing changes the inference. Suppose feedforward identity information emerges at 60 ms, proposed recurrent stabilization occupies 110-220 ms, memory maintenance begins near 260 ms, and report preparation begins near 340 ms.

- A perturbation at 40-90 ms risks destroying encoding and is not selective for N_3 .
- A perturbation at 140-180 ms can test the return pathway if N_1 , N_2 , content fidelity, V , and boundary remain stable.
- A perturbation at 360 ms primarily tests report or late memory, not the proposed phenomenal interval.
- A rescue that restores the specific return path and the predicted anchor pattern is more informative than nonspecific recovery.

Explain the dwell bounds as operational hypotheses. The lower bound excludes evanescent transients that never stabilize. The upper bound is intended to capture metastable presence that remains updateable, not to declare every unusually persistent state unconscious. If pathological fixation or timeless absorption remains phenomenally present despite exceeding t_{max} , either the upper bound belongs to healthy cognition rather than C_0 , or the estimator must be revised. That empirical burden should be stated, not hidden.

The substrate-general challenge is direct. A nonrecurrent architecture with internal state, delays, or slow dynamics may reproduce every anchor currently assigned to N_3 . The theory must then show that loop topology predicts intervention and rescue results that duration-only models cannot, or weaken recurrence from a universal necessity to a common biological mechanism.

Display plan

Draw a timeline with four colored bands: feedforward encoding, recurrent stabilization, downstream access, and report. Mark two prohibited intervention errors: **too early -> content never formed** and **too late -> report or memory only**.

Slide sequence

1. Why presence is temporally extended
2. The bounded-dwell formula
3. Lower and upper bounds
4. Recurrence versus slow feedforward duration
5. Recurrence versus working memory
6. Candidate measures and confounds
7. Relation to RPT
8. The selective intervention window
9. Rescue designs
10. Temporal presence versus temporal character

11. Falsifiers

Collaborative exercise: Temporal intervention studio

Purpose. Students distinguish feedforward encoding, recurrent stabilization, persistence, working memory, and report by designing an intervention at the correct temporal grain.

Timeline packet. Supply a hypothetical trial with stimulus onset, early feedforward sweep, content-specific feedback, behavioral divergence, memory maintenance, report preparation, and response. Include uncertainty in the timing of each event. Give different teams slightly different causal evidence so the class must negotiate rather than assume one canonical window.

Roles and procedure. The temporal modeler proposes t_{min} , t_{max} , and the stabilization interval. The perturbation specialist selects the pathway, duration, and intensity of an intervention. The retained-component auditor specifies how initial content information, N_1 , N_2 , V , boundary, and regime remain within equivalence bands. The rival advocate designs both a purely feedforward long-time-constant model and a late report-disruption account.

Teams place an early, target, late, sham, and rescue intervention on the timeline. They predict the full anchor vector for each condition. The target manipulation is unlicensed if it prevents encoding, changes content identity, globally suppresses the system, or overlaps only with post-perceptual report. The late control must disrupt report while sparing the proposed phenomenal interval.

Deliverable. Submit an annotated causal timeline, predicted component table, temporal selectivity argument, rescue prediction, and a statement of what result favors recurrence over mere duration. Cross-team reviewers identify one timing ambiguity that could reverse the inference.

Debrief. Ask which conclusions depend on a particular phenomenal-onset assumption. Emphasize that “late” is not a functional role. The relevant question is whether feedback causally maintains the content in the interval at issue.

Collaborative discussion: Topology, duration, and temporal character

In a fishbowl, the inner circle compares recurrence with a strictly feedforward system having long intrinsic time constants. The outer circle records every point at which speakers infer topology from duration or phenomenal timing from report timing. Switch circles, then address:

1. Could recurrence be a common biological implementation without being substrate-general necessity?
2. Does an upper dwell bound belong to consciousness, healthy cognitive flexibility, or only the operational test?
3. How should a recurrently stabilized but apparently contentless state be classified?
4. What distinguishes the N_3 presence gate from the temporal path Pi_c of phenomenal character?

End with one discriminating intervention rather than a verbal definition.

Common misconceptions

- “Any late signal is recurrence.” Lateness can reflect many downstream processes.
- “Persistence proves a loop.” Slow feedforward or passive traces can persist.
- “More dwell means more presence.” The model requires bounded metastability, not indefinite fixation.
- “Disrupting recurrence and report together confirms N_3 .” The intervention is nonselective and may target the wrong interval.

Exit ticket

Give one experimental difference between recurrent stability and slow feedforward persistence, and one difference between N_3 and the temporal path profile Π_c .

Readings

Required: Stilwell, *$C\emptyset$ as N^{**}* , Section 8; Stilwell, *Ablating N^{**}* , Sections 8.3 and 9.3; Stilwell, *From Phenomenal Presence to Phenomenal Character*, Sections 4.9 and 6.5.

Recommended: Lamme (2006); Pitts et al. (2014); selected work on metastability and recurrent effective connectivity.

Session 7. What Does the Evidence License?

Learning objectives

Students should be able to distinguish positive, negative, indeterminate, and unlicensed outcomes; apply the precedence rule; formulate positive- and negative-license conditions; construct an indeterminacy profile; and separate scientific classification from precautionary action.

Instructor preparation

Assign *Indeterminacy as a Scientific Result*, Sections 1-4 and 6-9. Prepare four short cases that differ only in applicability, data quality, surrogate agreement, or score region.

Timed plan

	Time	Teaching move
	0-12 min	Four cases, four outcomes
	12-30 min	Assessment tuple, precedence, and license principles
	30-45 min	Identical-score examples and test-performance reporting
	45-72 min	Collaborative evidence tribunal
	72-89 min	Structured hearing: Does indeterminacy protect the theory?
	89-104 min	Cause profiles, science-action separation, and failure budgets
	104-110 min	Exit ticket

Lecture notes

Begin with the epistemic distinction that governs the entire program: **the state of the system is not the state of the evidence**. Consciousness may be present or absent even when the available assessment licenses neither conclusion. Indeterminacy is therefore not the assertion that the system is half conscious. It is a status of warranted attribution.

Index every assessment to the tuple:

$$A = \{ S, C, T, E, M, V \},$$

where S is the candidate system, C the target form of consciousness, T the test, E the evidence, M the inferential model, and V the validity envelope. The validity envelope records populations, substrates, tasks, boundaries, recording conditions, and auxiliary capacities for which the evidence-target interpretation is supported.

This tuple blocks inferential drift. Evidence of command following does not automatically establish continuous phenomenal richness. A cortical score does not automatically classify a whole brain-body system. A measure calibrated in adult mammalian brains does not automatically retain its meaning in an artificial network.

Present the precedence rule:

$$D(A) = \begin{cases} \text{Unlicensed,} & a=0 \text{ or } q=0, \\ \text{Indeterminate,} & a=1, q=1, c=1, \\ \text{Positive,} & a=1, q=1, c=0, L \geq \tau_+, \\ \text{Negative,} & a=1, q=1, c=0, L \leq \tau_-, \\ \text{Indeterminate,} & a=1, q=1, c=0, \tau_- < L < \tau_+. \end{cases}$$

a is applicability, q execution and data quality, c unresolved credible-surrogate conflict, and L a validated evidence grade or score. Validity and quality are checked before evidential strength. Conflict is checked before aggregation. Only a licensed, adequately executed, nonconflicted assessment reaches the rule-in and rule-out thresholds.

Define the four outcomes precisely.

- **Positive:** the evidence licenses attribution of the specified target under the declared model and validity envelope.
- **Negative:** an applicable, sufficiently sensitive, adequately executed test yields evidence in a validated rule-out region.
- **Indeterminate:** the assessment is valid and informative but does not discriminate enough, either because evidence lies in the gray zone or credible surrogates or models remain unresolved.
- **Unlicensed:** a condition required to interpret the assessment failed. This is a meta-output about the procedure, not negative evidence about the system.

Teach the Positive License Principle. An impressive marker supports consciousness only when its target relevance and specificity are validated, the test is applicable, quality is adequate, and credible nonconscious or artifactual explanations are controlled. Complexity, self-report, recurrent activity, or flexible behavior is not a default positive.

Teach the Negative License Principle. Non-detection supports absence only when the indicator would probably have been detectable if the target were present, auxiliary task demands were satisfied or controlled, data quality was adequate, the test applied to this system, and alternative causes of non-detection were excluded. “Not detected” is not “absent” by default.

Now build the cause profile:

$$P_R(A) = \{e, s, m, b, t\},$$

where e is evidential scarcity, s surrogate discordance, m model dependence, b boundary or grain sensitivity, and t transport uncertainty. The entries diagnose the inference, not the quantity or dimensions of experience.

Match each cause to a research response:

Cause	What it means	Proper next step
Evidential scarcity	Valid data do not discriminate	Increase power, trials, temporal coverage, or measurement precision
Surrogate discordance	Credible measures conflict	Decompose constructs, trace auxiliary demands, repeat or intervene to adjudicate
Model indeterminacy	Live inferential models yield different outputs	Design a discriminating prediction or justify model weighting
Boundary / grain	Reasonable system definitions reverse the result	Perform independent causal individuation and boundary perturbation
Transport	The indicator-target bridge may not carry to the new domain	Establish construct, causal-role, and calibration preservation

Explain resolution asymmetry. Indeterminate evidence can be resolved by more informative evidence. Unlicensed assessment first requires repair of its applicability, quality, target, boundary, or transport. More data from the same invalid procedure can produce a highly reproducible nonanswer.

Finally, separate science from action. An indeterminate welfare-relevant case can justify precaution without being relabeled positive. Let D be the scientific status and Λ a loss structure; a decision minimizes expected loss given both. High stakes, irreversibility, and asymmetry between false denial and false attribution can justify protective action. The action does not change the scientific output.

Introduce test-performance reporting. Coverage is the proportion receiving positive or negative classifications. r_I and r_U are the indeterminate and unlicensed rates. Selective risk is error among classified cases when an independent reference is

defensible. A test that looks accurate only because it refuses or invalidates the hard cases should not be presented as universally decisive.

Worked teaching sequence: identical score, different license

Place two assessments side by side. Both produce aggregate score $L = .72$ with a positive threshold of $.70$.

Feature	Assessment A	Assessment B
Applicability	Validated for the patient's injury class	Validated only in healthy adults
Sensory route	Hearing confirmed	Severe hearing loss unresolved
Data quality	Artifact checks pass	Muscle contamination overlaps the effect
Surrogates	EEG and imaging agree	Credible EEG and imaging measures conflict
Boundary	Stable candidate	Result reverses across reasonable candidates
Outcome	Positive if all other rule-in conditions pass	Unlicensed because applicability and quality fail; conflict and boundary also prevent a positive

The equal score does not imply equal evidence. Assessment A may enter the rule-in region. Assessment B cannot be repaired by collecting more contaminated trials through an unavailable channel. It needs a validated route, boundary audit, and conflict resolution.

Clarify rule-in and rule-out asymmetry. A highly specific marker can support a positive when credible nonconscious causes of the effect have been controlled. A negative needs strong sensitivity under the actual conditions, adequate detection opportunity, satisfied auxiliary task demands, and a validated rule-out region. A test can be useful for positive detection while being unsafe for negative exclusion.

Use a small performance report. Out of 100 applicable target cases, a tool classifies 30 positive and 20 negative, returns 25 indeterminate, and finds 25 assessments unlicensed. Its coverage is $(30 + 20) / 100 = .50$, $r_I = .25$, and $r_U = .25$. If 5 of the 50 classified cases disagree with a defensible independent reference, selective risk is $.10$. A second tool that classifies only 20 easy cases with no observed errors has lower measured selective risk but only $.20$ coverage. It is not automatically the better general test.

Now distinguish case-level and field-level indeterminacy. One theory may confidently classify a case positive and another negative because they assign different constitutive roles to the same data. Each internal verdict can be determinate while comparative adjudication remains model-indeterminate until a diagnostic prediction or justified model weighting is available.

Finish with a two-action loss sketch. Suppose an indeterminate patient assessment is paired with two choices: provide a low-burden communication trial or withhold it. If the cost of an unnecessary trial is small and the cost of missing covert command following is large, the trial can be justified under indeterminacy. The action follows from evidence plus asymmetric loss; it does not convert $D = I$ into $D = +$.

Display plan

Draw a decision tree in this order:

Applicable? -> Quality adequate? -> Credible conflict? -> Rule-in / gray zone / rule-out.

Beside it write: **Unlicensed is not weak evidence; it is failed interpretability.**

Slide sequence

1. World state versus evidence state

2. The assessment tuple
3. The validity envelope
4. Four outcomes
5. Precedence before thresholds
6. Positive License Principle
7. Negative License Principle
8. Five-part cause profile
9. Resolution paths
10. Scientific status versus action
11. Coverage and unresolved rates
12. Misuse guardrails

Collaborative exercise: Evidence tribunal

Purpose. Students practice the four-outcome framework as a rule-governed classification system and learn that uncertainty, invalidity, and negative evidence demand different responses.

Case packet. Prepare five variants of one locked-in patient assessment: a valid strong positive; a valid score in the gray zone; credible EEG and imaging surrogates that conflict; poor hearing during a command-following task; and a new neural measure with no validation in severe injury. Assign one variant to each team and provide a partially completed decision form. Add evidence on stakes and reversibility, but instruct teams not to use it in the scientific verdict.

Tribunal roles. A positive advocate states the strongest positive license. A negative advocate states the strongest negative license and rule-out region. A validity officer audits applicability, quality, boundary, transport, and auxiliary demands. A chair assigns the outcome and cause profile, records dissent, and recommends the next study. Rotate roles in the next tribunal exercise rather than forcing multiple cases into this block.

For each assigned case, advocates have ninety seconds; the validity officer then rules which evidence is admissible. The chair must choose positive, negative, indeterminate, or unlicensed and complete $\{e, s, m, b, t\}$. “Indeterminate” is not allowed when the assessment is invalid; “unlicensed” is not allowed merely because the result is inconvenient. After the scientific verdict, the group writes a separate practical recommendation using the supplied loss structure.

Deliverable. Each team submits one outcome record with decisive rule, cause profile, confidence statement, practical-action note, and smallest informative next study. The instructor combines the five records into a class table during debrief. Every record must say whether simply collecting more of the same data would help.

Debrief. Ask groups to defend the difference between a score in the gray zone and a test administered through an unavailable sensory channel. More precision may resolve the first; the second requires a different assessment. Then compare scientific and clinical recommendations to show why precaution does not turn an indeterminate result into a positive one.

Collaborative discussion: Does indeterminacy protect or expose a theory?

Use a structured hearing. Half the class argues that four outcomes create escape routes from falsification; half argues that collapsing every failure into negative or uncertain produces invalid certainty. Each side must state prospective conditions under which indeterminate and unlicensed rates would themselves count against the research program. Then mixed groups answer:

1. When should disagreement among valid measures be modeled as latent heterogeneity rather than returned as indeterminate?
2. Can a field-level comparison be indeterminate while one theory gives confident case-level verdicts?
3. What maximum unresolved-output rate would make a proposed C-test scientifically unhelpful?

The final synthesis must include a **failure budget**: a preregistered rate or pattern of unresolved cases that triggers revision of measures, boundary rules, or the theory.

Common misconceptions

- **“Indeterminate and unlicensed are synonyms.”** One is valid nondiscrimination; the other is failed inferential license.
- **“Negative means no positive marker.”** Negative requires a validated rule-out and adequate detection opportunity.
- **“Precaution implies a positive scientific finding.”** Action depends on stakes and error costs in addition to evidence.
- **“Probabilities make the four outcomes unnecessary.”** A precise probability can still rest on an invalid transport or target assumption.

Exit ticket

Write one case that is indeterminate and one that is unlicensed. State why collecting more of the same data helps the first but not necessarily the second.

Readings

Required: Stilwell, *Indeterminacy as a Scientific Result*, Sections 1-4 and 6-9.

Recommended: Bayne et al. (2024); Chow (1970); Shinkins et al. (2013); Birch (2024).

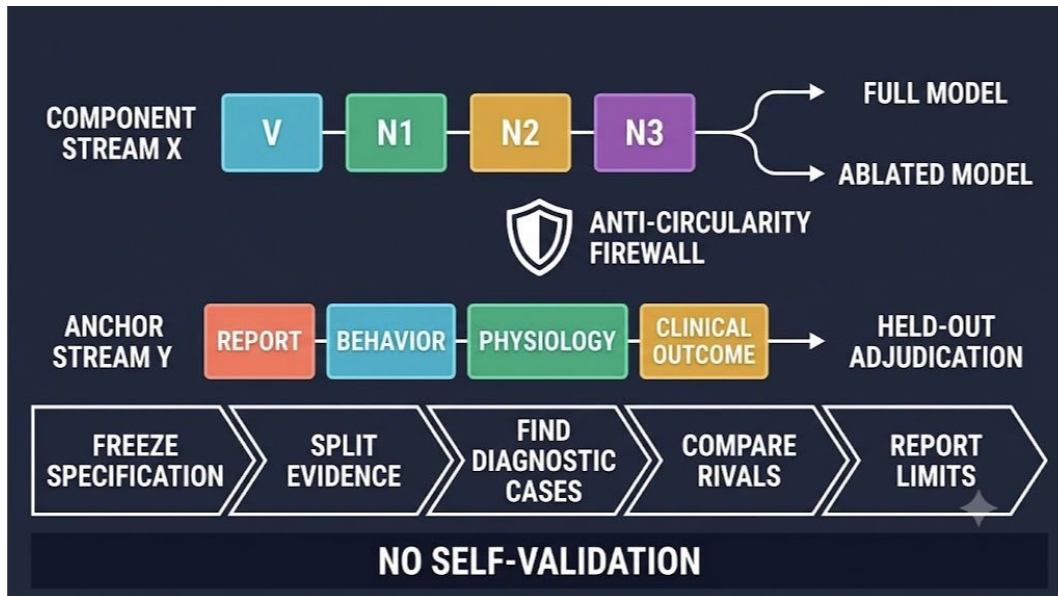
Session 8. Does Every Conjunct Earn Its Place?

Learning objectives

Students should be able to prove the diagnostic-set result informally, determine whether a dataset identifies component necessity, distinguish three kinds of ablation, apply the anti-circularity firewall, and assign one of five component-audit statuses.

Instructor preparation

Assign *Ablating N**, Sections 1-8 and 10, plus Appendices B and G. Display the existing anti-circularity figure.



Anti-circularity architecture for component audits

Timed plan

	Time	Teaching move
	0-15 min	Boolean-cube demonstration: why diagonal data are insufficient
	15-31 min	Three kinds of necessity and the diagnostic-set theorem
	31-45 min	Anti-circularity firewall and evidence partitions
	45-72 min	Collaborative diagnostic-cell design laboratory
	72-87 min	Structured discussion: No oracle, no necessity?
	87-104 min	Grouped ablations, selection bias, and strong minimality
	104-110 min	Exit ticket

Lecture notes

The conjunction claims more than bundle-level prediction. Every term is advertised as necessary. Yet if all positive cases have $(V, N_1, N_2, N_3) = (1, 1, 1)$ and all negative cases have $(0, 0, 0, 0)$, the full conjunction, every single component, and many pairs classify perfectly. More diagonal data cannot identify which component matters.

For case x , let $X = (X_0, X_1, X_2, X_3) = (V, N_1, N_2, N_3)$. The full model is:

$$M_*(x) = \bigwedge_{j=0}^3 X_j(x).$$

The model with component k removed is:

$$M_{-k}(x) = \bigwedge_{j \neq k} X_j(x).$$

The two models disagree only on:

$$D_k = \left\{ x : X_k(x) = 0 \wedge \bigwedge_{j \neq k} X_j(x) = 1 \right\}.$$

The proof is simple. If the full conjunction is true, the ablated conjunction must be true. Disagreement therefore occurs only when the full model is false and the ablated model true. That requires every retained term to pass and the omitted term to fail. No observation outside this cell can discriminate the two rules on that case.

Empirical measures are uncertain, so use a conservative diagnostic region. The upper bound of X_k must fall below its threshold, and the lower bounds of every retained component must clear their thresholds. Threshold-straddling cases do not become diagnostic by point estimate. The diagnostic margin records distance from the nearest relevant threshold.

Diagnostic coverage Γ_k reports how much of the target distribution occupies the discriminating region. A strong effect in a rare engineered cell does not establish broad necessity. An empty or tiny diagnostic set means necessity is underidentified, not confirmed.

Now address the missing consciousness oracle. The audit must never define labels from N^* and then use those labels to show that removing N_k worsens performance. Install the firewall:

1. Treat $C_0 \Leftrightarrow V \wedge N^*$ as a hypothesis, never a database definition.
2. Separate the component stream X from an independently constructed anchor vector Y .
3. Split development, calibration, and final adjudication evidence.
4. Prohibit one measure from constructing a component and validating that component's necessity in the same fold.
5. Freeze boundary, grain, content, regime, interval, thresholds, interventions, anchor battery, translation rule, scoring rule, and sensitivity neighborhood.
6. Compare full and ablated models only on held-out diagnostic cases.

An anchor vector may contain calibrated reports, discrimination, metacognitive structure, physiological relations, perturbational signatures validated outside the critical contrast, and theory-light clinical outcomes. It is not an oracle and must preserve disagreement. The audit supports a conjunct only relative to the declared anchors, target population, and rival class.

Held-out risk $R_k(m)$ evaluates how each model predicts the anchor pattern in diagnostic cases. Incremental value is:

$$\Delta_k = R_k(-k) - R_k(*).$$

Positive Δ_k favors retention; negative values favor removal; intervals spanning the minimum meaningful effect remain indeterminate.

Distinguish three ablations:

Type	What is removed	Inference
Criterion ablation	A term from the decision rule	Incremental predictive discrimination

Type	What is removed	Inference
Mechanism ablation	A proposed physical realization	Causal contribution, only if selectivity and identity are preserved
Measurement ablation	A sensor, estimator, feature family, or pipeline	Robustness to measurement choice

A lesion called “recurrence ablation” may reduce integration, availability, arousal, and the candidate boundary. Its name does not establish selectivity. A licensed mechanism ablation must move the target component by a minimum amount while retained components stay within equivalence bands and the boundary and viability remain licensed.

Teach the five audit outcomes:

- **Indispensable:** adequate coverage and a stable, meaningful held-out advantage, with a selective license when causal necessity is claimed.
- **Redundant:** valid diagnostic cases exist and the ablated model is equivalent or better without compensatory retuning.
- **Causally coupled:** the target component cannot be varied without moving retained components, despite valid intervention and measurement.
- **Indeterminate:** coverage is weak, intervals straddle, anchors conflict, or rivals tie.
- **Unlicensed:** boundary, viability, intervention, measurement, independence, or transport failed.

Finally, explain grouped ablation. One-at-a-time deletion tests weak minimality. Components can be jointly redundant or interact. For a removed set A , compare the full model with M_{-A} on cases where every retained term passes and at least one removed term fails. Strong empirical minimality requires the full model to outperform every preregistered proper submodel, with multiplicity control and matched flexibility. It still does not establish sufficiency against unrelated nonconjunctive theories.

Worked teaching sequence: three kinds of necessity

Begin by making the word *necessary* less ambiguous.

Claim	Meaning	Evidence required	Scope of conclusion
Logical necessity	The term appears in the written conjunction, so deleting it changes the formula	Inspection of the theory	Only what the theory says, not whether nature requires it
Incremental empirical necessity	The full model predicts held-out independent anchors better than a declared ablated rival in diagnostic cases	Frozen models, adequate diagnostic coverage, independent anchors, matched flexibility	Relative to the anchor family, population, specification, and rival
Causal necessity	Selectively changing the term's physical realization changes anchors while retained terms remain equivalent	Mechanism intervention, selectivity checks, rescue, boundary and viability preservation	Relative to the realized mechanism and tested domain

Use a numerical diagnostic example. Let every component threshold be .60. To test N_2 , suppose one case has upper bound $U_2 = .42$ while retained-component lower bounds are $L_V = .73$, $L_1 = .80$, and $L_3 = .69$. The conservative distance from every relevant decision boundary is:

$$m_2 = \min(.60 - .42, .73 - .60, .80 - .60, .69 - .60) = .09.$$

This is a genuine N_2 diagnostic case: N_2 is below threshold and the retained conditions clear theirs. Now suppose weighted diagnostic coverage is only .05, while the full model's held-out advantage over the ablated rival is $\Delta_2 = .13$ with interval $[.04, .21]$. The case offers meaningful local evidence that N_2 adds predictive value, but the population

claim must remain narrow because only five percent of the target weight occupies the disagreement region. A large effect does not manufacture coverage.

Contrast weak and strong minimality. With four terms V, N_1, N_2, N_3 , there are $2^4 - 1 = 15$ nonempty removal sets. Four one-at-a-time audits can all favor the full model while a two-term removal reveals joint redundancy. For example, if N_1 and N_3 are noisy estimators of one deeper recurrent-integration process, each may appear useful when the other remains, yet a matched composite can outperform both. Hierarchical testing should begin with scientifically motivated single removals and coupled sets, control multiplicity, and reserve exhaustive proper-submodel comparison for mature data.

Diagnostic conditioning can also create selection bias. If measurement error or a common cause affects both component scores and anchors, selecting cases because a score falls below threshold can enrich misleading anchor patterns. Safeguards include defining diagnostic membership before revealing held-out anchors, cross-fitting membership rules, probabilistic membership near thresholds, negative controls, and an unconditional sensitivity analysis.

Finally, distinguish a diagnostic case from a representative case. An engineered, off-manifold system can be excellent for construct validation while saying little about prevalence in biological populations. A successful audit should produce more than a favorable Δ_k : expect enrichment of informative near-misses, selective rescue, cross-component asymmetry, and invariance across defensible measurement routes.

Display plan

Draw the Boolean cube's all-pass and all-fail corners, then circle the four single-failure cells. Write:

No diagnostic cases -> no component identification.

Slide sequence

1. Bundle prediction versus component necessity
2. Why diagonal data hide redundancy
3. Full and leave-one-out models
4. Diagnostic-set theorem
5. Approximate regions and coverage
6. No consciousness oracle
7. The anti-circularity firewall
8. Independent anchor vectors
9. Criterion, mechanism, and measurement ablations
10. Five component statuses
11. Grouped ablations
12. What genuinely falsifies minimality

Collaborative exercise: Diagnostic-cell design laboratory

Purpose. Students distinguish logical inclusion, incremental empirical value, and selective causal necessity, then determine whether an apparently successful ablation study has examined cases on which its models disagree.

Materials. Give every team a twelve-case toy dataset containing component intervals, target-population weights, held-out anchor vectors, two intervention descriptions, and one measure that has been used both to define a component and to score the anchor. Include an anesthesia contrast in which every component, arousal, behavior, and data quality move together.

Roles and procedure. The component theorist writes the full and leave-one-out models. The measurement and statistics lead identifies conservative diagnostic cases and estimates margin and coverage. The intervention lead audits minimum

target change, equivalence of retained components, boundary and viability preservation, and rescue. The anti-circularity auditor traces every variable through development, calibration, selection, and final test.

At the midpoint, each team receives a hidden complication: the largest diagnostic case is off-manifold; membership was computed after examining anchors; the intervention changes the boundary; or a measure appears in both evidence streams. Teams must revise the status rather than explain the complication away.

Deliverable. Complete Worksheet 4 and attach a one-sentence claim whose scope names the component, anchor family, target population, specification neighborhood, rival, and type of necessity. Another team reviews the claim and marks any word that exceeds the license.

Debrief. Compare teams with the same held-out risk difference but different diagnostic coverage or selectivity. Classify the original anesthesia design: it may establish ordinary-state association or measurement utility, but diagonal movement does not identify the necessity of each conjunct.

Collaborative discussion: No oracle, no necessity?

In quartets, assign a defender of the objection, a defender of the audit, a methods referee, and a recorder. Each side gives a two-minute case, then switches positions and improves the opposing argument. The referee requires a shared premise and distinguishes direct phenomenal acquaintance from comparative prediction against independent anchors.

The group submits two columns:

- **The audit can establish:** anchor-relative incremental value, diagnostic prediction, and, with a selective intervention, causal contribution in a specified domain.
- **The audit cannot establish:** metaphysical necessity against every imaginable account, direct observation of another's experience, or sufficiency of the whole conjunction.

The group must also decide when causal coupling warrants more intervention work and when it should motivate a combined component. Finish by asking whether strong minimality against all preregistered proper submodels is an attainable confirmatory goal or a longer-term program standard.

Common misconceptions

- **“The conjunction logically entails necessity, so the empirical audit is redundant.”** Syntax is not evidence that the world requires the syntax.
- **“Performance loss after lesion proves phenomenal loss.”** It may reflect implementation damage, report failure, or regime collapse.
- **“An empty diagnostic set supports necessity because no counterexample exists.”** It leaves necessity unidentified.
- **“Synthetic ablation gives consciousness ground truth.”** It validates construct separability and engineering roles only.

Exit ticket

For N_2 , write the exact component profile on which the full model and the N_2 -ablated model disagree. Then name one forbidden dual-use measure.

Readings

Required: Stilwell, *Ablating N^{**}* ; Sections 1-8 and 10, Appendices B and G.

Recommended: Doerig et al. (2021); Kleiner and Hoel (2021); Yaron et al. (2022); Cogitate Consortium (2025).

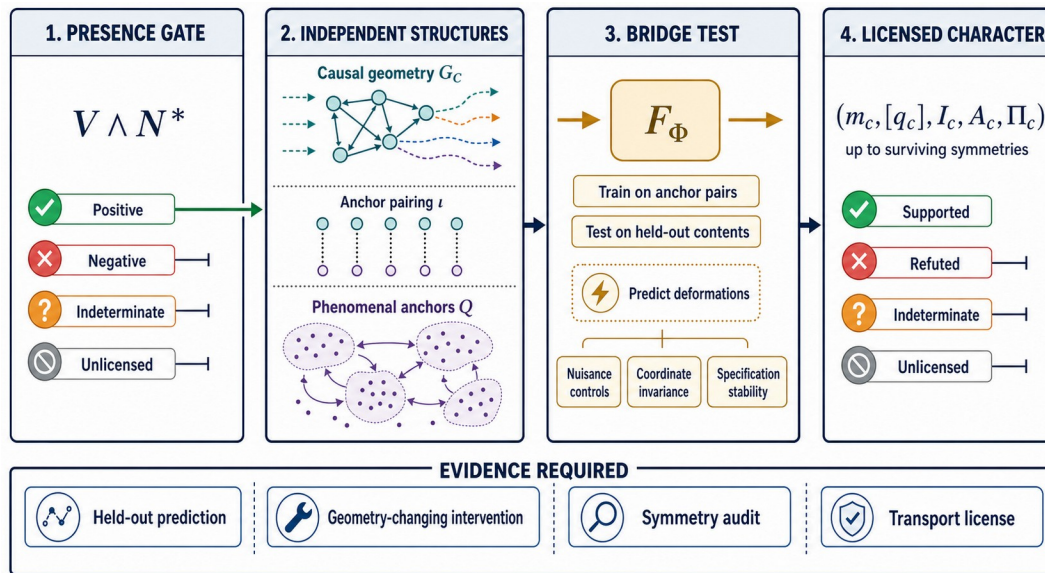
Session 9. From Phenomenal Presence to Phenomenal Character

Learning objectives

Students should be able to explain the two-layer architecture, distinguish causal-trajectory geometry from activation embeddings, describe the bridge-validation requirements, interpret character as a symmetry-qualified equivalence class, and separate bridge refutation from presence failure.

Instructor preparation

Assign *From Phenomenal Presence to Phenomenal Character*, Sections 1-5 and 7-9. Display the existing two-layer inference figure.



Two-layer inference architecture for presence and character

Timed plan

Time	Teaching move
0-12 min	Presence / character error-matrix warm-up
12-28 min	Two explananda and the frozen presence gate
28-44 min	Causal trajectories, independent quality anchors, and bridge logic
44-70 min	Collaborative character-bridge construction and break test
70-86 min	Fishbowl: Does relational structure explain character?
86-104 min	Symmetry, inversion, temporal character, pain, and transport
104-110 min	Exit ticket

Lecture notes

State the limitation that motivates the extension: $V \wedge N^*$ says whether experience is present, not why a present state feels red rather than green, painful rather than neutral, or continuous rather than stuttering.

The two-layer architecture is:

$$\text{presence gate} = V \wedge N^*,$$

$$\text{phenomenal character} = Q_\Phi(G_c, \Delta).$$

Presence is a gate, not another coordinate beside hue, pitch, intensity, or valence. Only a positive presence result licenses an ordinary phenomenal-character attribution. A rich geometry outside the gate remains scientifically meaningful representational structure, but the model does not call it phenomenal merely because it is structured.

Preserve both non-implications:

$$P(c_i) = P(c_j) = 1 \Rightarrow \not\Rightarrow \chi(c_i) = \chi(c_j),$$

and

$$G_{c_i} \cong G_{c_j} \Rightarrow \not\Rightarrow P(c_i) = P(c_j) = 1.$$

Define G_c carefully. It is not a single activation vector, a decoder embedding, or a convenient neural manifold. It is the content's neighborhood within a perturbation-derived geometry that contains:

- distances among content-specific response distributions;
- admissible causal transitions;
- trajectory timing, direction, and recurrence;
- occupancy under the declared regime; and
- intervention semantics and route sensitivity.

The geometry can be a metric space, pseudometric, directed graph, stratified manifold, or another relational structure. Do not force Euclidean structure when triangle, symmetry, or identity conditions fail.

Construct the phenomenal target independently. In report-capable anchor participants, use converging similarity judgments, discrimination thresholds, confusion matrices, generalization, adaptation, cross-context matching, and transition judgments. No single report defines the quality space. If anchor measures conflict beyond tolerance, the target geometry is indeterminate.

The bridge F_Φ maps frozen causal-geometry features to the independently estimated quality structure. Model family, dimensionality, nuisance controls, invariances, and thresholds belong to specification Φ and are selected within training data. A high in-sample correlation is insufficient.

A supported bridge should:

1. predict held-out phenomenal distances and neighborhoods better than declared nulls;
2. preserve transition structure and relevant asymmetries;
3. survive coordinate rotations, sensor relabeling, harmless rescaling, and reasonable preprocessing;
4. outperform nuisance models based on stimulus distance, semantics, arousal, or response preparation;
5. predict how geometry-changing interventions deform independently measured phenomenal relations; and
6. transport to new systems only under a separate bridge license.

The deformation test is the causal center. An intervention changes G_c in a preregistered direction. The frozen bridge predicts which phenomenal distances should contract, expand, or change transition order. The observed phenomenal deformation is collected independently. Baseline fit and intervention fit are reported separately so a good static correlation cannot hide a failed causal prediction.

Now introduce symmetry. If a transformation preserves every measured quality-space relation, the data cannot distinguish an absolute point q from its transformed counterpart. The warranted attribution is an equivalence class:

$$Q_{\phi}(G_c, \Delta) = [F_{\phi}(\psi_{\phi}(G_c, \Delta))]_{Aut(Q)}.$$

This is the disciplined answer to inverted-spectrum concerns. A global red-green permutation remains underdetermined if it preserves every distance, transition, valence, memory relation, and causal role. Additional asymmetrical anchors may shrink the equivalence class, but the theory cannot assign an absolute label by assertion.

Summarize the structured character profile:

$$\chi_c = (m_c, [q_c], I_c, A_c, \Pi_c),$$

where m_c is modality stratum, $[q_c]$ relational location, I_c intensity, A_c valence field or vector, and Π_c temporal path profile. The tuple is not assumed exhaustive and does not require all modalities to occupy one commensurable Euclidean space.

The bridge outcomes are **supported**, **refuted**, **indeterminate**, and **unlicensed**. A refuted bridge in a positive presence case means the experience is present but this proposed geometry does not explain its differentiation. It does not mean there is no character. Conversely, a beautiful bridge fit cannot rescue an unlicensed presence gate.

Worked teaching sequence: constructing and testing the bridge

Display the two-layer error matrix before presenting any geometry.

Presence result	Bridge status	Combined interpretation
Positive	Supported	Presence supported; tested character relation supported
Positive	Refuted	Presence supported; this character mapping fails
Positive	Indeterminate	Presence supported; tested character relation unresolved
Positive	Unlicensed	Presence supported; bridge procedure invalid
Negative prediction	Any	Character attribution is not licensed by this program; do not infer a specific absence of character
Indeterminate	Any	Do not interpret bridge fit or mismatch until presence is resolved
Unlicensed	Any	Repair the presence assessment before combined phenomenal inference

The disciplined construction pipeline is:

$$\text{content intervention} \rightarrow \text{trajectory distributions} \rightarrow \text{dissimilarities and directed transitions} \xrightarrow{F_{\phi}} G_c \rightarrow Q_{\phi}.$$

G_c is not a single activation vector and not any embedding that separates stimulus labels. It summarizes how content-specific causal trajectories relate under a frozen family of admissible perturbations. Freezing the perturbation mixture prevents the analyst from selecting whichever intervention produces the prettiest match to phenomenal reports.

Estimate the empirical phenomenal quality structure Q independently. Similarity judgments, discrimination thresholds, confusion matrices, adaptation effects, and transition judgments may converge, but they need not. Q_{ϕ} is the bridge-qualified character readout predicted from G_c and evaluated against Q . If only rank order is stable, use an ordinal structure. If some contents have undefined or asymmetric relations, use a graph or directed structure. Do not force a Euclidean space because it is easy to plot.

Use a four-content example. Fit F_ϕ on red, orange, and green; hold blue out. Baseline success requires predicting blue's neighborhood without training on its phenomenal relation. Stronger success requires a preregistered adaptation that contracts one causal distance and a matching contraction in the independent phenomenal relation, while stimulus, response, attention, and task-demand nuisance models fail. Report baseline fit, held-out neighborhood accuracy, deformation-direction agreement, and null-model comparison separately.

Make automorphisms concrete with an unlabeled equilateral triangle. Any permutation of its three vertices preserves every distance, so relational evidence cannot identify an absolute vertex label. Add a validated asymmetry, such as one vertex carrying negative valence or a distinctive transition, and fewer permutations remain. The theory identifies character only up to the symmetries left by all warranted relations. This is constraint, not a verbal solution to the inverted spectrum.

Include pain as a nonvisual case. Intensity, unpleasantness, location, urgency, controllability, and temporal throbbing can dissociate. A one-dimensional pain scale therefore hides structure. N_3 asks whether the pain content is recurrently present; Π_c distinguishes steady, pulsing, spreading, or anticipatory temporal paths within an admitted episode. A useful character theory may need stratified geometries, directed transitions, and value fields rather than one universal metric space.

Display plan

Use two separated streams:

$$\text{causal trajectories} \rightarrow G_c \parallel \text{phenomenal anchors} \rightarrow Q$$

followed by a bridge trained on anchors and tested on held-out contents and interventions. Draw no arrow from quality fit back to presence classification.

Slide sequence

1. Presence is not character
2. The two non-implications
3. Presence-gated domain
4. What G_c contains
5. What G_c is not
6. Independent quality-space anchors
7. The bridge F_ϕ
8. Held-out prediction
9. Geometry-changing intervention test
10. Symmetry-qualified character
11. Modality, location, intensity, valence, path
12. Four bridge outcomes

Collaborative exercise: Build and try to break a character bridge

Purpose. Students keep the presence gate independent from the character bridge and learn to distinguish baseline fit from held-out, intervention-sensitive relational prediction.

Materials. Provide a frozen positive presence-result card, four contents, a small perturbation-derived causal-distance matrix, an independently generated phenomenal-relation matrix, one held-out content, and a geometry-changing intervention. Give half the teams data in which causal and phenomenal structures deform coherently and half a valid mismatch. Include a decoder embedding that fits the stimulus labels beautifully but was trained on report data.

Roles and procedure. The causal-model lead constructs G_c from intervention-conditioned trajectories. The phenomenal-anchor lead chooses an ordinal, graph, pseudometric, or stratified quality representation and records its validation genealogy. The bridge analyst freezes training, validation, test, null, and deformation predictions. The leakage and nuisance auditor searches for stimulus geometry, report reuse, attention, task difficulty, and analysis selection.

Teams record the frozen positive presence result and may not reopen it to improve the bridge. They then fit only the permitted training relation, predict the held-out neighborhood and intervention deformation, and classify the bridge as supported, refuted, indeterminate, or unlicensed. A strong baseline correlation without held-out or causal success is not support.

Deliverable. Submit Worksheet 5 plus a two-stream diagram showing that causal and phenomenal evidence meet only at F_ϕ . State what remains intact if the bridge fails and list the surviving automorphisms.

Debrief. Ask each nuisance auditor to explain how an impressive colored embedding can be circular or stimulus-driven. Then compare the coherent and mismatch cases without changing the positive presence judgment in either.

Collaborative discussion: Does relational structure explain character?

Use a fishbowl. The inner roles are structural-identity advocate, predictive-only advocate, inverted-spectrum critic, and phenomenological pluralist. Outer observers each track one distinction: prediction versus explanation, absolute labels versus equivalence classes, geometric versus nongeometric structure, and presence versus character. Switch after seven minutes.

The class produces an argument map with four rows: common ground, deepest disagreement, possible empirical discriminant, and residual philosophical issue. Address whether lawlike mapping is explanation, whether automorphism-relative identity is adequate, and whether valence, meaning, or bodily urgency require structures richer than a single geometry. The goal is not forced consensus but a precise statement of what empirical success would and would not settle.

Common misconceptions

- **“Any neural manifold is a quality space.”** The predictor and phenomenal target must be independently constructed and causally bridged.
- **“A failed bridge means the state was unconscious.”** Character mapping and presence classification are modular.
- **“Similarity reports make the theory report-dependent.”** Reports are one anchor family in calibration cases, not the presence criterion.
- **“A relational theory rules out inversion.”** The framework explicitly preserves unbroken symmetries.

Exit ticket

State one reason decoding accuracy is weaker than bridge fit and one result that would refute Q_ϕ while leaving N^* intact.

Readings

Required: Stilwell, *From Phenomenal Presence to Phenomenal Character*, Sections 1-5 and 7-9.

Recommended: Fleming and Shea (2024); Kriegeskorte and Kievit (2013); Brouwer and Heeger (2009); Clark (1993).

Session 10. Reading Consciousness from the Schematic

Learning objectives

Students should be able to distinguish correlational, black-box, and constitutive readings of the theory; reconstruct the role vector; apply the four schematic-legibility conditions; and evaluate the analogy's force and limits.

Instructor preparation

Assign *Reading Consciousness from the Schematic* in full. Bring a simple electrical schematic or another expert-readable mechanistic diagram whose operating state changes under a localized fault.

Timed plan

	Time	Teaching move
	0-12 min	What does an engineer read from a schematic?
	12-28 min	Marker, black-box law, and constitutive schematic
	28-43 min	Role vector and the D, C, F, E legibility conditions
	43-69 min	Collaborative blind schematic-readout trial
	69-85 min	Constitutive-identity tribunal
	85-104 min	Circularity, observation, zombies, transport, and teachability
	104-110 min	Exit ticket

Lecture notes

The schematic paper begins after a conditional concession: suppose the biconditional survives empirical testing. Would it remain an opaque law that simply pairs a dynamical bundle with consciousness, or could a trained observer understand the organization as the phenomenal state itself?

Contrast:

$$\text{black-box necessity : } X \Rightarrow C_0$$

with

$$\text{structurally legible necessity : } X \Rightarrow \rho \Rightarrow C_0.$$

ρ is an interpretable role profile. For CØ / N* it contains:

$$\rho_{C_0} = \langle \text{bearer, realized operation, unity, system-level presence, temporal presence} \rangle.$$

Boundary qualification supplies the bearer. V supplies realized operation. N_1 supplies unity. N_2 supplies system-level causal presence. N_3 supplies bounded temporal presence. Under the constitutive hypothesis, this role-complete organization is not merely a cause or correlate of C_0 ; it is proposed as the physical form of C_0 .

Use the electrical analogy carefully. An engineer sees more than ink because symbols have learned component semantics, component laws compose, inputs and operating conditions are declared, and counterfactual faults have localized consequences. The diagram alone neither powers the circuit nor validates the component laws. Likewise, an N^* diagram neither experiences nor proves the identity. The actual system must realize the dynamics, and independent evidence must support the constitutive hypothesis.

Define schematic legibility:

$$\text{Legible}(M, o / K) \Leftrightarrow D \wedge C \wedge F \wedge E.$$

- **Decomposable semantics (D):** elements have stable role meanings not defined by repeating the target outcome.
- **Compositional readout (C):** a trained reader can derive an intermediate organization profile and apply an explicit rule.
- **Counterfactual fault localization (F):** licensed changes support differentiated predictions about which aspect fails or changes.
- **Evidential independence (E):** the readout procedure is not counted as evidence that its own constitutive rule is true.

The fault map is central. Boundary failure makes the inference unlicensed. Viability failure removes the operating regime. Selective N_1 failure removes unified content for that candidate. Selective N_2 failure leaves content causally isolated. Selective N_3 failure leaves it temporally evanescent. A change in G_c under an intact presence gate alters predicted character rather than presence. These are differentiated failures, not one generic low score.

Teach the three strength levels:

Position	Claim	Explanatory status
Correlational marker	N^* covaries with consciousness evidence	Useful prediction; organization may be incidental
Black-box law	$V \wedge N^*$ reliably guarantees C_0	Strong law; the transition remains opaque
Constitutive schematic	C_0 is the role-complete organization	The trained reader reconstructs presence and its failure map

The strong reading is an a posteriori identity claim, not a dictionary definition. Its opponents can understand it perfectly and deny that the identity is true. Structural intelligibility is not confirmation.

Address the zombie objection. A physical duplicate without experience would defeat sufficiency if independently established. Merely imagining the duplicate or noting the absence of first-person access does not provide that evidence. The substantive dispute is whether the complete organization is constitutive.

Address external observation. Distinguish observing that another system instantiates C_0 , inferring features of its token character q_S , and undergoing q_S from that system's first-person position. A scientific observer need not become the other subject. If the identity is true and supported, observing the realized organization is observing consciousness under a public causal-dynamical description, even though first-person acquaintance remains indexed to the bearer.

Address functionalism. The proposal is a physicalist information-dynamical functionalism, not unrestricted input-output equivalence. Static diagrams, behavioral lookup tables, and merely isomorphic software descriptions do not qualify. Effective causal realization, correct grain, boundary, time, and viability do real work.

End with teachability as a standard. Independent investigators should learn the readout rules, agree on held-out cases under the same frozen specification, predict localized failures, and transport the role profile across materially different implementations when causal organization is preserved. If only the author can "see" the identity, it is not schematic legibility.

Worked teaching sequence: from profile to conditional readout

Keep three objects distinct:

- the printed or digital schematic is a **representation**;

- the running causal network is the **realization**;
- the boundary-qualified candidate system is the possible **bearer**.

Take one held-out profile: boundary unique and stable; V positive; N_1 positive with estimator agreement; N_2 positive after report and executive removal; N_3 positive within the frozen dwell interval. Translate each result into a role vector:

$$\rho(S, c) = (\text{bearer}, \text{operation}, \text{unity}, \text{system-level presence}, \text{temporal presence}).$$

At the operational level, the reader has established only that the frozen profile passes. Under the constitutive hypothesis, the reader then infers C_0 . The phrase **under the constitutive hypothesis** must remain visible. If N_2 alone fails, the schematic predicts loss of system-level causal presence, not generic system failure. If the boundary changes, confident fault localization is unavailable because the bearer itself is unstable.

Treat D, C, F , and E as a conjunctive minimum for legibility:

Condition	Competence question	Possible partial failure
Decomposable semantics D	Can readers state distinct roles without repeating "consciousness"?	Labels are stable but overlap or remain outcome-defined
Compositional readout C	Can readers reconstruct the whole profile from the parts?	Parts are clear but no rule combines them
Fault localization F	Do different perturbations predict different failures?	Many interventions collapse the whole profile or produce coupled faults
Evidential independence E	Was the schematic validated without using its own output as truth labels?	Readout is reproducible but circularly confirmed

A model may score well on one condition and fail another. Graded inter-reader competence can be measured even though the minimum standard is conjunctive.

Use a three-stage teachability protocol. First train readers on component semantics and worked examples. Second test blind readout, correct abstention, calibration, and fault localization on unseen cases. Third test transfer to a materially different implementation whose role structure is preserved. High agreement establishes that the readout is teachable; it does not establish that the identity is true.

Finally, revisit the three explanatory levels with the same dataset. A **correlational marker** reliably covaries with the target labels. A **black-box law** predicts occurrence across held-out cases but does not reveal why different failures matter. A **constitutive schematic** adds stable role semantics, compositional reconstruction, counterfactual fault structure, and an identity claim. Each level is an achievement. None gains the warrant of the next merely by being renamed.

Display plan

Draw:

$$S_{N^*} \rightarrow \rho_{C_0} \rightarrow C_0$$

Label the first arrow **mechanistic interpretation** and the second **exposed identity hypothesis**. Underline that only independent evidence bears on the truth of the second arrow.

Slide sequence

- The remaining explanatory lacuna
- What an expert reads in a schematic
- The N^* schematic
- The role vector

5. Decomposable semantics
6. Compositional readout
7. Counterfactual fault localization
8. Evidential independence
9. Marker, law, or constitution?
10. Observation versus undergoing
11. Limits of the analogy
12. Teachability and transport

Collaborative exercise: Blind schematic-readout trial

Purpose. Students test schematic legibility as a teachable competence rather than an author's claim that the model is intuitively clear.

Materials. Supply a one-page component-law sheet, two training cases, and four unseen schematics: a positive profile, a boundary failure, a single-component fault, and a coupled fault in which an N_3 intervention also reduces N_1 . Include one profile whose validation labels were generated from its own component scores.

Roles and procedure. A training designer teaches only the frozen meanings of bearer, operation, unity, system-level presence, and temporal presence. Blind readers A and B independently reconstruct the role vector, assign an outcome, localize the fault, and note which step depends on the constitutive identity rather than operational evidence. The adjudicator compares the forms, classifies disagreements, and applies D , C , F , and E .

Readers may abstain when boundary or coupling prevents fault localization. Correct abstention counts as competence. The printed page is a representation; the running system is the realization; the boundary-qualified candidate is the possible bearer. No diagram becomes conscious by depicting the conditions.

Deliverable. Submit two independent Worksheet 6 forms and an adjudication table with agreement, disagreement source, calibration, fault specificity, and evidence-independence checks. Calculate simple agreement but do not interpret it as confirmation of the identity.

Debrief. Compare the one-number classifier, black-box law, and constitutive schematic. Reliable prediction may establish a useful marker; compositional roles and fault maps add legibility; independent evidence still bears on whether the identity is true.

Collaborative discussion: Constitutive-identity tribunal

Assign identity theorist, correlationist, conceivability critic, and evidential referee. Give the tribunal a system that passes every frozen component test and supports reliable fault predictions. The group must issue a tiered verdict about:

1. marker status;
2. lawlike predictive status;
3. warrant for a constitutive identity;
4. the remaining explanatory or perspectival gap.

The referee marks every slide from intelligibility to confirmation. Ask whether two trained readers can agree on the schematic while rationally disagreeing about the identity, and distinguish observing that the realized organization occurs from undergoing the bearer's token experience.

Common misconceptions

- **"The schematic proves consciousness visually."** A representation is not a realization, and readability is not confirmation.

- **“Role language smuggles phenomenality into the premises.”** The roles are operationalized without a consciousness verdict; the identity supplies the phenomenal claim.
- **“First-person privacy makes all third-person evidence defective.”** It limits the mode of access, not necessarily the possibility of warranted attribution.
- **“The circuit analogy claims equal scientific maturity.”** It describes a target epistemic form, not current evidential parity.

Exit ticket

State the difference between a black-box law and a constitutive schematic, and name one way the latter could fail empirically even if it remains intelligible.

Readings

Required: Stilwell, *Reading Consciousness from the Schematic*.

Recommended: Machamer, Darden, and Craver (2000); Craver (2007); Levine (1983); Chalmers (1995).

Session 11. Biological and Clinical Applications

Learning objectives

Students should be able to apply the full inference order to variable human states, distinguish target-specific from whole-system conclusions, and identify the major validity risks in report-limited clinical and experimental cases.

Instructor preparation

Assign the worked-case sections of the core, boundary, availability, indeterminacy, and ablation papers. Divide students into clinical, sleep/anesthesia, perceptual, and split-system teams.

Timed plan

	Time	Teaching move
	0-15 min	Longitudinal demonstration: one patient across three windows
	15-30 min	Target-specific results, negative license, and repeated sampling
	30-58 min	Evolving multidisciplinary case conference
	58-75 min	False-positive and false-negative council
	75-94 min	Sleep, anesthesia, rivalry, and covert-command applications
	94-104 min	Split-system horizons and test-performance synthesis
	104-110 min	Exit ticket

Lecture notes

Require the same order for every case:

1. specify the target C ;
2. declare the candidate system, grain, regime, and interval;
3. audit boundary status;
4. assess V independently;
5. estimate N_1 , N_2 , and N_3 for the same target;
6. apply conflict and uncertainty rules;
7. state the model output; and
8. separate that output from clinical or ethical action.

Use ordinary waking perception only as a calibration case. It should not define consciousness by report. For a dominant percept in a healthy adult, a stable boundary, viable regime, integration, availability, and recurrence are expected. The scientific advantage is abundant anchor evidence; the danger is overfitting every operationalization to report-capable adults and assuming transport.

REM dreaming. Concurrent motor report is absent and external interaction is reduced. The candidate can still satisfy an internally defined N_2 if dream contents causally modulate diverse internal recipients. Retrospective reports are temporally displaced and memory-dependent. A positive result requires dream-state criteria for availability and recurrence, not waking executive criteria imposed by default.

Deep NREM sleep. The model allows sparse, episodic, content-specific, or indeterminate presence. A single negative awakening report cannot establish dreamlessness because memory and timing can fail. Repeated state sampling, perturbational measures, and carefully timed awakening anchors are needed. Different intervals in one night may receive different outputs.

General anesthesia. Agents should not be pooled as one mechanism. Some primarily alter thalamocortical organization, some recurrent connectivity, and some neuromodulatory regimes. The model predicts loss of C_0 when the bundle collapses, but a valid test must distinguish viability failure, component failure, and unlicensed measurement. Recovery and dreaming-like anesthesia experiences are important near-miss cases.

Locked-in syndrome. Motor report failure is not a consciousness failure. A stable boundary and viable N^* bundle can support a positive prediction when speech and movement are absent. The case is a direct reason for report-independent N_2 .

Unresponsive wakefulness syndrome and cognitive motor dissociation. Behavioral non-response has many possible causes. Task-based fMRI or EEG can reveal command following in some behaviorally unresponsive patients. A positive command-following result supports at least some awareness-relevant capacity but does not automatically reveal continuity or richness. A negative task result is licensed only when hearing, comprehension, arousal, attention, memory, and motor-independent response generation had a fair opportunity to operate. Repeated assessment matters because states fluctuate.

Minimally conscious state. The framework expects time- and content-indexed variability. A single negative window should not be generalized. Report the distribution of positive, negative, indeterminate, and unlicensed windows along with causes.

Blindsight. Accurate forced choice in a blind field does not by itself establish visual C_0 . The theory predicts no conscious blind-field content if the relevant integrated, available, recurrent bundle fails, while allowing unconscious information to guide behavior. The important comparison is not performance versus no performance but the causal component profile for the content.

Binocular rivalry. Physical stimulation can remain stable while perceptual content alternates. Involuntary markers can time the dominant content without requiring trial-by-trial report. The model predicts that content-specific N^* tracks the dominant percept. Eye movement is an anchor or timing aid, not a constituent. Physical-alternation controls and report/no-report contrasts separate perceptual, introspective, and motor activity.

Split brain. The case first tests boundaries, not the count of reports. Evaluate left, right, and residual joint candidates. If each side becomes autonomous, role-complete, stable, and pairwise separable, multiple candidate bearers are supported. If shared subcortical recurrence preserves a joint candidate or results vary by task and horizon, report a larger, layered, or boundary-indeterminate result. Do not compare pre- and post-disconnection component scores as one persistent subject unless boundary identity is preserved.

Use the following summary matrix as hypotheses, not diagnoses:

Case	Initial component expectation	Proper caution
Waking dominant percept	$V +, N^* +$	Baseline anchors do not define the theory by report
REM dream	$V +$ under dream regime; N_1, N_3 plausible; N_2 internal	Retrospective report and memory are displaced
Deep NREM	fluctuating, sparse, or unresolved	One negative awakening report is weak
Adequate anesthesia	multiple component collapse likely	Agent, dose, timing, and covert experience vary
Locked-in syndrome	$V +$ and N^* may remain	Motor silence is not N_2 failure

Case	Initial component expectation	Proper caution
Unresponsive wakefulness	variable and often unresolved	Negative license and repeated assessment are essential
Blindsight	content-specific bundle may be incomplete	Forced choice is access-like evidence, not decisive presence
Binocular rivalry	bundle should track dominant content	Involuntary markers time content but do not constitute it
Split brain	one, multiple, layered, or unresolved candidates	Anatomy alone does not count subjects

Worked teaching sequence: one patient across three windows

Use a fictional longitudinal case and make clear that it is a teaching device, not clinical advice.

Window	Key observations	Scientific output	Cause profile and next step	Separate action note
Day 2, low arousal	No command response; hearing not established; heavy artifact; physiology unstable	Unlicensed	Auxiliary route, data quality, and regime fail; stabilize and use a nonauditory, low-demand protocol	Preserve low-burden supportive options; do not treat the negative as absence
Day 9, fluctuating arousal	Valid sensory route; EEG N_2 surrogate positive, imaging surrogate negative; boundary stable	Indeterminate	Surrogate discordance s dominates; repeat in matched state and adjudicate routes	Continue reversible assessment and communication opportunities
Day 21, target-specific window	Repeated content-specific imagery modulation; V, N_1, N_2, N_3 pass with route controls; no motor output	Positive for the tested command-related content under the model	Track stability, broaden contents, and assess false-positive controls	Offer an appropriate communication confirmation pathway

The Day 21 result does not establish continuous rich awareness across the day. A target-specific positive and a broader indeterminate state estimate can coexist. Conversely, the Day 2 negative behavior never entered a licensed negative region.

Use the case to teach $\{e, s, m, b, t\}$. Day 2 is dominated by invalid auxiliary conditions and data quality, so a scientific tuple should mark the assessment unlicensed rather than allocate ordinary evidential uncertainty. Day 9 has valid but discordant evidence, so s is high and the remedy is surrogate adjudication. A model-dependent disagreement at Day 21 would raise m even if every individual theory were internally decisive.

Negative-license conditions should be spoken aloud:

1. the target and candidate are appropriate;
2. the state had a fair detection opportunity;
3. sensory, comprehension, attention, memory, and response auxiliaries required by the task were available;
4. the measure is sensitive in this population and regime;
5. quality and artifact controls pass;
6. the result falls in a validated rule-out region; and
7. alternative causes of non-detection are controlled.

Repeated assessment matters because both state and test quality fluctuate. Report coverage, indeterminate rate, unlicensed rate, and selective risk rather than a single accuracy number. A tool that classifies only stable, easy windows can be useful but should not be described as universally decisive.

For split systems, cross candidate with horizon and task. A left subsystem may be separable for one lateralized task over seconds, while a larger subcortically coordinated candidate remains plausible over longer regulation. This does not allow the analyst to choose whichever boundary yields a preferred subject count. Candidate families and mappings must be prospective, and the output may legitimately be nested or indeterminate.

Display plan

Maintain one fixed case worksheet with rows for target, boundary, V , N_1 , N_2 , N_3 , conflicts, output, cause profile, and action note. Complete it in public for one case before team work.

Slide sequence

1. The fixed inference order
2. Waking as calibration, not definition
3. REM dreaming
4. Deep NREM
5. Anesthesia by mechanism and window
6. Locked-in syndrome
7. Disorders of consciousness
8. Blindsight
9. Binocular rivalry
10. Split-brain boundaries
11. Case matrix

Collaborative exercise: Evolving multidisciplinary case conference

Purpose. Students integrate boundary, component, evidential, temporal, and ethical analysis without letting clinical urgency distort the scientific label.

Materials. Use a three-envelope dossier. Envelope 1 contains bedside behavior, sensory screening, arousal fluctuation, and basic physiology. Envelope 2 adds a technically questionable command-following result and a negative assessment administered during low arousal. Envelope 3 adds repeated no-report, perturbational, and route-control evidence. The data should permit a target-specific positive while broader ongoing status remains unresolved.

Roles and procedure. A boundary specialist freezes candidate and horizon. A component-measure specialist completes V , N_1 , N_2 , N_3 with intervals. A clinical-methods lead audits sensory access, comprehension, timing, task demands, and repeated sampling. An epistemic auditor assigns the four-way outcome and cause profile. An ethics and decision liaison writes a separate recommendation from declared losses and reversibility.

After each envelope, teams must commit to a six-place assessment tuple, identify the dominant uncertainty cause, and request exactly one next measurement. They record whether the new envelope changed the system estimate, the inference license, or only the practical decision. Earlier verdicts may be revised, but not erased.

Deliverable. Submit a progressive Worksheet 1 for all three windows, a cause-profile trajectory, a repeated-assessment plan, and a separate action note. Include one target-specific conclusion that coexists with a broader indeterminate result.

Debrief. Contrast an invalid negative window, a valid but discordant window, and a licensed positive window. More precision cannot repair a task performed through a blocked route; repeated sampling can help with genuine fluctuation. No clinical treatment advice should be inferred from the classroom dossier.

Collaborative discussion: False-positive and false-negative council

Assign patient advocate, treating clinician, test developer, epistemic conservative, and institutional ethics chair. Give the council one reversible communication trial and one irreversible clinical decision. Members rank harms of false attribution, false denial, delay, burden, and resource diversion while preserving the same scientific status.

The deliverable is a two-scenario decision matrix that identifies which differences arise from values and losses rather than evidence. The council also addresses whether retrospective dream reports can calibrate online measures and how split-system classifications may change with task and horizon without implying arbitrary metaphysics.

Common misconceptions

- **“Wakefulness is phenomenal presence.”** Arousal is a regime variable, not the target.
- **“Command following measures the full richness of experience.”** It supports a narrower capacity and depends on auxiliary demands.
- **“No-report paradigms are confound-free.”** Involuntary markers have their own false positives and target assumptions.
- **“Split brain means two subjects by anatomy.”** Multiplicity requires causal autonomy, role completeness, and separability.

Exit ticket

Choose one clinical or experimental case and identify the single most likely route to a false negative. State the control required for a licensed negative inference.

Readings

Required: Worked-case sections from all six empirical companion papers.

Recommended: Bodien et al. (2024); Wannez et al. (2017); Frassle et al. (2014); Tsuchiya et al. (2015).

Session 12. Nonhuman Animals, Organoids, and Artificial Systems

Learning objectives

Students should be able to explain constrained substrate independence, conduct a transport audit, avoid both biological chauvinism and syntactic overattribution, and formulate a precautionary action without changing the scientific status.

Instructor preparation

Assign the nonhuman, organoid, and AI applications in the core, boundary, availability, indeterminacy, and ablation papers. Ask students to choose one target system and diagram its actual running architecture or organismic causal organization.

Timed plan

	Time	Teaching move
	0-16 min	Transport as construct, causal-role, calibration, and boundary preservation
	16-32 min	Animals, development, organoids, and artificial viability
	32-60 min	Collaborative cross-substrate transport jigsaw
	60-75 min	Discussion: Constituent, interface, support, or environment?
	75-94 min	Horizon-sensitive hybrid and distributed-system cases
	94-104 min	Proportional precaution without epistemic inflation
	104-110 min	Exit ticket

Lecture notes

The theory is substrate-independent in form but not permissive in application. A cross-substrate mapping must preserve the relevant causal-dynamical roles, not merely a diagram or input-output function. At minimum it must preserve directionality, effective coupling, temporal grain, recurrent organization, functional recipient structure, dynamic parameter ranges, and substrate-appropriate viability within preregistered tolerances.

This position avoids two symmetrical errors. **Neural chauvinism** denies consciousness to any non-neural realization without asking whether the causal role is preserved. **Empty functionalism** grants consciousness to any behavioral or abstract isomorph. CØ / N* requires actual realized organization under a validated mapping.

Teach the three transport bridges:

1. **Construct preservation:** does the measure still detect the same functional or dynamical property?
2. **Causal-role preservation:** does that property play the same theoretically relevant role in the new system?
3. **Calibration preservation:** do thresholds, error rates, and uncertainty rules remain defensible?

Failure of any bridge produces transport indeterminacy or an unlicensed application. Running the same code or formula on a new substrate does not carry the old interpretation.

Nonhuman animals and infants. Recipient classes, viability indicators, intervention burdens, and anchors must be species- and stage-appropriate. Failure to learn a human-designed task can reflect ecology, sensory differences, motivation, or training demand. Use homologous causal roles, flexible behavior with known sensory dependencies, motivational tradeoffs, learning and generalization, perturbational predictions, and converging physiological evidence. A mammalian neural marker can be informative in another species only with an explicit bridge.

Organoids. Neural activity, oscillation, complexity, and perturbational response can be measured, but their interpretation was developed in organisms with differentiated anatomy, sensory history, vascular and regulatory support, and behavior. The first question is the candidate boundary: tissue alone, tissue-electrode loop, tissue plus controller, or the full apparatus. If a digital controller supplies indispensable recurrence or memory and generic replacement fails, a hybrid boundary may be appropriate. If the controller imposes the trajectory, the assembly may fail autonomy. None of these boundary results is itself a consciousness verdict.

Artificial systems. Inspect the running architecture, not the product label. A base model, prompt, context window, retrieval service, persistent memory, scheduler, tools, evaluators, other models, and human operators can form different candidates at different horizons. A stateless model call may have complex internal transformation but little recurrent continuity. An agent loop can add persistence and availability while remaining externally scripted. A multi-agent service may contain several autonomous candidates, a larger workspace-like candidate, or no role-complete candidate.

The proper tests include state resets, memory substitution, scheduler ablation, message delay, route interruption, service removal, human-in-the-loop removal, latent or token interventions, and matched-controller replacement. Logging services, mirrored replicas, idle subscribers, and fluent language do not automatically add integration or functional diversity.

Use artificial systems first as construct laboratories. They allow precise criterion, mechanism, and measurement ablations and can reveal whether proposed N_i measures behave as intended. This validates the measurement architecture, not consciousness. Phenomenal interpretation still requires independent anchors and transport discipline.

Ethical decisions remain separate. An indeterminate animal, organoid, or AI assessment can justify documentation, monitoring, harm minimization, scaling limits, or independent review when potential harm is serious and irreversible. The practical action reflects evidence plus a declared loss structure. It must not be reported as scientific proof of consciousness.

Worked teaching sequence: what exactly must transport?

Represent transport as a declared mapping T from a source-domain realization to a target-domain realization. The claim is not that identical numbers mean identical constructs. The mapping must preserve, within tolerance:

1. **construct semantics:** what the measure is intended to estimate;
2. **causal role:** how the measured organization makes a difference inside the candidate;
3. **calibration:** how scores, uncertainty, rule-in, and rule-out regions relate to independent anchors; and
4. **boundary and regime:** which system and operating conditions the inference concerns.

Transport is a license condition, not a fifth consciousness outcome. Record two levels:

Transport and target evidence	Within-theory four-way result	Field-level qualifier
Mapping is licensed and target evidence clears the rule-in region	Positive under the declared theory	Add model-indeterminate if serious theories interpret the preserved role differently
Mapping is licensed and target evidence clears a validated rule-out region	Negative under the declared theory	Add model-indeterminate if rival theories do not license the same negative
Mapping is licensed but valid target evidence straddles or credible surrogates conflict	Indeterminate	State whether the unresolved cause is evidence, model, boundary, or calibration
Construct, causal role, calibration, boundary, or validity fails	Unlicensed	No within-theory positive or negative; theory disagreement may be secondary

A theory-conditional positive or negative and field-level model indeterminacy can coexist. Conversely, adequate transport only permits interpretation; the target evidence must still satisfy the ordinary rule-in or rule-out criteria.

For artificial viability, compare several proposals rather than defining the answer by intuition.

Candidate ✓ criterion	What it captures	Likely false inclusion or exclusion
Uptime	continued execution	a passive script or externally maintained process
Endogenous fault recovery	restoration after bounded disruption	a narrow retry service with no broader organization
Operating-variable regulation	maintenance of memory, timing, resources, or state integrity	a managed platform whose regulation is entirely external
Functional-organization maintenance	preservation of the causal role profile	systems that maintain roles only because an operator continually reconstructs them
Goal-preserving adaptation	flexible recovery of task organization	sophisticated control without phenomenal relevance

The theory needs a prospectively justified, substrate-appropriate envelope. Biology is not required by definition, but neither is mere execution enough.

Use a horizon-sensitive artificial system. Over one model call, the base model and current context may be the only fast causal unit; retrieval, persistent memory, scheduler, and human oversight are interfaces or supports. Over a day-long agent process, state resets, memory substitution, scheduler ablation, and matched service replacement may show that some of these become constitutive. If a human continually chooses every goal and repairs every failure, the larger candidate may fail autonomy even though the product appears coherent.

For animals, distinguish mechanistic homology from causal-role analogy. A mammalian marker may transport through conserved circuitry; a different architecture may transport only after its causal role is independently demonstrated. Failure on a human-designed task can reflect sensory ecology, motivation, or motor incompatibility.

For organoids, separate four achievements: a brain-like measure is observed; the intended construct is implemented; a stable tissue or hybrid candidate is identified; and phenomenal interpretation is calibrated. Success at the first does not entail the fourth. Tissue-only, tissue-controller, and controller-dominated candidates may receive different boundary results under removal and replacement tests.

Proportional precaution considers severity, reversibility, alternatives, monitoring burden, and current evidence. It can justify stricter safeguards while transport remains indeterminate or unlicensed. It cannot serve as a hidden positive anchor.

Display plan

Draw a transport bridge with three load-bearing supports: **construct**, **causal role**, and **calibration**. Remove one support and label the result **unlicensed transport**, not negative.

Slide sequence

1. Substrate independence with teeth
2. Neural chauvinism and empty functionalism
3. Three transport bridges
4. Species-appropriate recipient classes
5. Developmental and ecological demands
6. Organoid boundary alternatives
7. Hybrid constitution versus imposed control
8. Actual AI running architectures
9. Service and state perturbations
10. Construct validation versus consciousness evidence

11. Precaution without relabeling

Collaborative exercise: Cross-substrate transport jigsaw

Purpose. Students distinguish numerical resemblance from construct, causal-role, calibration, and boundary preservation across domains.

Materials. Give every group one human-validated measure and one target dossier: octopus, infant, organoid-controller hybrid, or tool-using artificial agent. Each dossier includes apparently favorable similarities and at least one ecological, causal, or calibration mismatch.

Roles and procedure. The construct specialist asks whether the target variable means the same thing. The causal-role specialist identifies homologous or analogous difference-making. The calibration specialist audits thresholds, error rates, and anchor relations. The boundary specialist evaluates tissue-only, hybrid, service-level, or organism-level candidates across two horizons. The ethics and action specialist applies proportional precaution without changing the scientific result.

After twelve minutes in substrate teams, students meet same-role specialists from other substrates and compare failure modes. They then return to revise the original ledger. Every team records three separate fields: transport license, the within-theory four-way result, and any field-level model-indeterminacy qualifier.

Deliverable. Submit a transport ledger naming source construct, target implementation, mapping, tolerance, causal-role evidence, calibration evidence, boundary status, transport license, within-theory result, field-level qualifier, falsifying observation, smallest next study, and separate action note.

Debrief. Compare cases with the same numerical score but different causal-role evidence. A brain-like value in an organoid or a human-like benchmark in an AI may demonstrate an interesting construct while leaving phenomenal interpretation unlicensed.

Collaborative discussion: Constituent, interface, support, or environment?

Assign system architect, interventionist, boundary theorist, skeptical reviewer, and recorder. Ask whether an external persistent-memory service belongs to an artificial candidate. Reveal removal, matched-replacement, latency-perturbation, and horizon-change results one at a time.

The group submits a boundary verdict for a short and long horizon plus the intervention that most strongly bears on constitution. It must also address whether artificial viability requires mere uptime, endogenous fault recovery, maintenance of functional organization, regulation of operating variables, or goal-preserving adaptation. Close by distinguishing a lower precaution threshold from a lower scientific rule-in threshold.

Common misconceptions

- **“Substrate-independent means software equivalence is enough.”** The mapping must preserve realized causal dynamics and operating ranges.
- **“Biological markers absent in AI prove no consciousness.”** An out-of-domain biological test may be unlicensed rather than negative.
- **“A brain-like number in an organoid has the same meaning.”** Measurement and causal-role transport must be established.
- **“Precaution is evidence.”** It is a policy response to evidence and asymmetric risk.

Exit ticket

Name one feature that must be preserved for cross-substrate mapping and one reason that applying a human-calibrated test to a new substrate could be unlicensed rather than negative.

Readings

Required: Nonhuman, organoid, and AI application sections across the paper corpus.

Recommended: Butlin et al. (2023); Birch et al. (2020); Kagan et al. (2022); Smirnova et al. (2023).

Session 13. Differentiation, Validation, and the Strongest Objections

Learning objectives

Students should be able to state what is genuinely distinctive about CØ / N*, organize evidence into validation layers, compare the model fairly with neighboring theories, and identify observations that would require revision rather than reinterpretation.

Instructor preparation

Assign the theory-comparison and objections sections of all seven papers. Ask each student to prepare the strongest version of one objection, not a rhetorical caricature.

Timed plan

	Time	Teaching move
	0-16 min	Distinctiveness as risky contrast rather than package novelty
	16-30 min	Validation ladder, dependencies, and maturity levels
	30-58 min	Collaborative versioned rival-prediction laboratory
	58-73 min	Research-program health board
	73-94 min	Steelmanned objections and progressive versus degenerating revision
	94-104 min	Comparative synthesis across theory families
	104-110 min	Exit ticket

Lecture notes

Begin with the obvious challenge: the components are not individually novel. The distinctive proposal lies in their **division of labor and joint empirical discipline**. The theory says:

1. the target is minimal phenomenal presence rather than report or self-reflection;
2. one frozen candidate must realize all roles in one window;
3. integration, availability, and recurrence are jointly required and individually insufficient;
4. viability is independently assessed and cannot be retrofitted;
5. availability is causal, functionally diverse, and report-independent;
6. unresolved credible conflict blocks a positive verdict;
7. the biconditional is vulnerable in both directions;
8. every conjunct competes with leave-one-out and grouped rivals on diagnostic cases;
9. character is a separately validated second layer; and
10. the intended identity has a compositional fault map rather than an unexplained label.

This is more than “IIT + GNWT + RPT” because source theories do not jointly force these restrictions. IIT does not ordinarily require separate report-independent broadcast and bounded recurrence. GNWT does not ordinarily require synergy plus a separate character bridge. RPT does not ordinarily require functionally diverse availability. None by itself supplies the boundary-first autonomy procedure, four-outcome precedence policy, diagnostic-set audit, or schematic-legibility standard.

Present a validation ladder. Higher rungs presuppose lower ones:

1. **Conceptual clarity:** targets, roles, scope, and exclusions are explicit.
2. **Construct separability:** measures can distinguish integration, availability, recurrence, and viability in synthetic and benchmark systems.
3. **Measurement validity:** surrogates track their intended roles rather than activity, report, arousal, or one latent factor.
4. **Boundary validity:** candidate systems are independently individuated and stable.
5. **Bundle discrimination:** $V \wedge N^*$ predicts anchor patterns across states and contents.
6. **Component necessity:** full models beat proper submodels on adequate diagnostic coverage.
7. **Causal intervention:** selective perturbation and rescue alter independent anchors as preregistered.
8. **Transport:** roles and thresholds generalize without favorable retuning.
9. **Character bridge:** causal geometry predicts held-out phenomenal relations and deformations.
10. **Schematic competence:** independent investigators can read, fault-localize, and transport the organization.

Stress that success at one rung does not automatically establish the next. A synthetic network can validate construct separability without consciousness. A state-level classifier can support bundle discrimination without component necessity. A character bridge can fail while presence remains supported.

Use a discriminating comparison matrix:

Case	CØ / N*	IIT-style integration sufficiency	GNWT-style access emphasis	RPT-style recurrence sufficiency
High integration, local and unstable	No C_0 prediction	Some versions may favor consciousness	No global access	No stable recurrence
Broad report-free reach, low integration	No C_0 prediction	Integration requirement fails	Availability may favor consciousness	Recurrence status decides
Stable local loop, low reach	No C_0 prediction	Depends on integration	No broad access	Some versions may favor consciousness
Motor report loss with intact bundle	C_0 predicted	Depends on causal structure	Report-focused implementations risk false negative	May preserve consciousness
Rich unconscious geometry	No ordinary character attribution without gate	IIT may treat causal structure as phenomenal	Content access may be absent	Recurrence may or may not occur
Full bundle with strong negative anchors	Sufficiency threat	Theory-specific response	Theory-specific response	Theory-specific response

The matrix must be tailored to precise versions of rival theories; it is a teaching device, not a substitute for reading them.

Steelman the major objections.

Hybrid without novelty. Reply with the exposed conjunction and methodological package, then admit that distinctiveness must yield predictions no source theory already forces.

Access smuggling. Reply that N_2 is internal, counterfactual, content-specific causal availability, and must pass without report or executive classes. Admit that the objection succeeds if every credible implementation requires those classes.

Viability escape hatch. Reply with independent preregistration and versioned rules. Admit that repeated bespoke revisions damage transport credibility.

Threshold arbitrariness. Reply that explicit, stress-tested thresholds are scientifically preferable to hidden ones. Admit that brittle classifications should remain fragile or indeterminate.

Correlation rather than constitution. Reply that interventions, route tests, ablations, and rescue matter more than passive decoding. Admit that even causal relevance does not by itself prove metaphysical identity.

Hard problem and explanatory gap. Reply that the program offers a mid-level constitutive identity and structural legibility, not a derivation of why any physical organization is accompanied by experience. The metaphysical objection remains live.

Embodiment and affect underplayed. Reply that boundary, V , content dynamics, and the character profile can include bodily and affective organization. Admit that future evidence may require an organismic or affective conjunct.

Conjuncts are causally inseparable. Reply that the audit includes a causally coupled outcome. If inseparability is robust, the theory should revise its factorization rather than advertise separately identified necessity.

No consciousness oracle. Reply that held-out convergent anchors permit comparative empirical tests without pretending to direct observation. Admit that the results remain anchor-relative and cannot close the other-minds gap.

End with revision rules. A healthy research program specifies what changes after failure: remove a redundant conjunct; combine coupled components; narrow a transport claim; revise an invalid measure; replace the sharp biconditional with a probabilistic rule if threshold instability is systematic; add a conjunct if repeated diagnostic cases demand it; or reject the bundle if licensed counterexamples persist.

Worked teaching sequence: from family labels to risky comparisons

Require a version card before comparing theories. A card states exact explanandum, bearer, proposed necessary and sufficient conditions, role of report, temporal commitment, source passage, and prediction for the current case. This prevents “IIT says integration” or “GNWT says report” from standing in for version-specific analysis.

Use the following comparison protocol:

1. Match targets. If one theory explains access and another minimal presence, either construct a bridge or stop the direct comparison.
2. Freeze versions and candidate boundaries.
3. Identify an ordinary agreement case; agreement does not differentiate.
4. Construct a near-miss on which commitments diverge.
5. State prospective observations, validity conditions, and revision rules for both theories.
6. Score the result without retuning either side.

The sixteen-family landscape remains relevant, but not every family makes a case-level neuroscientific prediction. Higher-order, predictive, attention-schema, embodied, illusionist, panpsychist, and dualist views may disagree at the level of explanandum, constitution, evidence, or metaphysics. Say which level is under comparison rather than forcing all positions into one experimental table.

Render validation as a branching dependency structure rather than one ladder:

Branch	Required foundation	Achievement	What branch failure leaves intact
Bundle attribution	semantic separability, measurement validity, stable boundary and regime	licensed $V \wedge N^*$ classification and diagnostic prediction	the philosophical target and untested operationalizations
Component necessity	valid bundle measures, diagnostic coverage, independent anchors	incremental support; selective intervention and rescue can add causal support	other components and the bundle's source-domain possibility
Transport	a validated source-domain claim plus construct, causal-role, calibration, and boundary mapping	warranted scope extension to a new domain	the source-domain result

Branch	Required foundation	Achievement	What branch failure leaves intact
Character bridge	positive presence gate plus independently estimated empirical quality structure	supported mapping from causal geometry to character relations	positive presence if the bridge fails
Schematic competence	stable role semantics and independent test cases	teachable readout, calibrated abstention, and fault prediction	empirical truth of the constitutive identity

Constitutive-identity support is an integrative judgment drawing on several branches, not a final rung reached only after universal transport. Failure propagates selectively. Poor construct separability blocks a component-necessity claim; it need not refute the philosophical target. A failed character bridge leaves a positive presence gate intact. Failed transport narrows domain but does not automatically refute the source-domain result.

Use a maturity rubric for every rung:

Level	Meaning
Not yet attempted	A criterion or protocol exists only as a proposal
Construct demonstration	The measure behaves as intended in synthetic or tightly controlled cases
Single-domain support	Frozen predictions succeed in one relevant domain
Replicated support	Independent teams reproduce the diagnostic result
Transported support	Role and calibration survive a materially different domain

Do not confuse existence of a detailed protocol with evidence that the theory has climbed the ladder.

Turn objections into argument maps. Record premises, hidden assumption, strongest reply, residual pressure, and an adjudicating result. For the access-smuggling objection, for example, the hidden question is whether report-independent causal availability is a weak constitutive role or simply access under another name. The decisive work is a case where the criteria diverge, not a terminological vote.

Finally, audit revisions. Progressive revision is predeclared or at least principled, narrows scope honestly, and creates sharper future predictions. Degenerating revision redraws the boundary after results, moves thresholds without independent calibration, adds bespoke exceptions, or never risks a unique failed prediction. Flexibility is scientifically healthy only when it increases future constraint.

Display plan

Draw the validation ladder. On the side write: **No upward inference without the lower license.**

Slide sequence

1. What is genuinely distinctive?
2. More than a hybrid list
3. The ten-rung validation ladder
4. Comparative near-miss profiles
5. Hybrid-without-novelty objection
6. Access and report objection
7. Viability and threshold objections
8. Correlation / constitution objection
9. Hard problem and embodiment objections
10. Coupling and oracle objections

11. Revision rules
12. What would make the program progressive?

Collaborative exercise: Versioned rival-prediction laboratory

Purpose. Students compare exact theory versions on matched targets and convert advertised distinctiveness into risky prediction rather than package novelty.

Materials. Prepare version cards for CØ / N*, IIT, GNWT, RPT, one higher-order theory, one predictive or active-inference account, an attention-schema view, an embodied or enactive account, illusionism, and a fundamentalist view. Each card states explanandum, bearer, necessary and sufficient commitments, report role, temporal requirement, and a source passage. Also provide four near-miss profiles: integrated but isolated, broadly available but reducible, recurrent but local, and full-bundle positive with strong negative anchors.

Roles and procedure. A CØ / N* advocate records its prediction before discussion. An assigned-rival advocate does the same for one exact version. A neutral comparison chair rejects target-mismatched contrasts. A source and evidence auditor checks that claims follow from the version card and identifies the evidence needed to adjudicate. Rotate rivals after one profile.

Teams identify ordinary agreement cases, genuine diagnostic disagreement cells, and one observation that discriminates without defining either theory as true. “The rival depends” is not a finished answer; the relevant version must be specified or the comparison returned as indeterminate.

Deliverable. Submit Worksheet 2 plus a 200-word adversarial-test abstract containing target, candidates, rival predictions, decisive observation, licenses, and revisions. The rival advocate certifies whether the proposed test is fair.

Debrief. Ask whether the conjunction produced a unique prediction or merely reorganized familiar measurements. Reject comparisons in which GNWT targets access while CØ / N* targets presence unless the bridge between those targets is itself part of the test.

Collaborative discussion: Research-program health board

Assign program proponent, rival theorist, empirical methodologist, and philosophy-of-science chair. Give groups four failure scenarios: a redundant conjunct, repeated transport failure, unstable thresholds, and licensed negative anchors in a full-bundle case. For each, choose measure repair, scope restriction, factorization change, probabilistic reformulation, addition or removal of a conjunct, or program rejection.

The deliverable is a public failure-to-revision rule with one recorded dissent. A progressive response narrows or changes the theory in a way that creates sharper future predictions. A degenerating response shifts boundaries or thresholds after inspection, multiplies exceptions, or leaves every future result compatible. Finish by ranking which present vulnerability is greatest and what unique success would most improve the program’s standing.

Common misconceptions

- **“A theory is novel only if every component is new.”** New constraint architecture and predictions can be theoretically substantive.
- **“Falsifiability requires a consciousness oracle.”** Comparative risky predictions can be tested against independent anchor structures, though not with metaphysical certainty.
- **“Every objection has been answered.”** Several replies explicitly preserve residual philosophical and empirical pressure.
- **“A component failure destroys every paper.”** The program is modular; some failures motivate revision rather than total collapse.

Exit ticket

State the strongest differentiating claim of $C\emptyset / N^*$ and one empirical result that would force a nontrivial revision of that claim.

Readings

Required: Theory-comparison and objection sections across all seven papers.

Recommended: Seth and Bayne (2022); Mudrik et al. (2025); Doerig et al. (2021); Cogitate Consortium (2025).

Session 14. Capstone Adversarial Research Design

Learning objectives

Students should be able to integrate the complete research workflow, design a study with genuine rival predictions, define project-appropriate favorable, disfavoring, indeterminate, and unlicensed outcomes in advance, and defend a revision rule.

Instructor preparation

Launch the capstone at the end of Session 9 and distribute Appendix C then. Assign teams with deliberately mixed theoretical preferences. Use Sessions 10-13 for the first four staged checkpoints and require the complete mock-defense packet before Session 14. In a compressed course, require Checkpoints 1, 3, and 5 and label the others recommended rather than pretending that five reviews fit into two weeks.

Timed plan

	Time	Teaching move
	0-10 min	Recap: the complete inference architecture
	10-22 min	Weak versus strong adversarial design example
	22-64 min	Collaborative adversarial preregistration studio
	64-88 min	Oral defenses before a skeptical panel
	88-103 min	Outcome-and-revision tribunal and disposition logs
	103-110 min	Individual confidence commitments and final synthesis

Lecture notes

The complete workflow is:

target $\rightarrow$ candidate boundary $\rightarrow V \rightarrow (N_1, N_2, N_3) \rightarrow$ conflict and validity gates $\rightarrow$ presence output $\rightarrow$ component aud

A capstone project should not attempt every element. It should select one tractable point where a result would alter the model. Good projects include:

- an N_2 diagnostic case that preserves N_1 and N_3 ;
- a feedforward long-timescale adversary for N_3 ;
- a boundary intervention in a hybrid biological-digital system;
- a grouped ablation that tests whether N_1 and N_3 are jointly redundant;
- a transport audit of one surrogate across species or substrate;
- a geometry-changing intervention that differentiates causal-trajectory and stimulus-only models; or
- a reproducibility study of schematic readout across independent analysts.

Require every team to state:

1. the exact target and unit of analysis;
2. the full and rival claims;
3. the cases on which the claims disagree;
4. the boundary, grain, regime, and interval;
5. the discovery, calibration, and final-test split;

6. component estimators and uncertainty bounds;
7. independent anchors and their validation genealogy;
8. intervention selectivity and retained-component equivalence margins;
9. rule-in, rule-out, conflict, indeterminate, and license-failure conditions;
10. sensitivity and transport analyses;
11. the result each team member expects; and
12. the revision that follows from every declared outcome.

A good adversarial question has three properties. First, rival predictions differ before the result. Second, both sides consider the test fair. Third, the outcome cannot be absorbed by moving the boundary, target, threshold, viability criterion, or anchor set after inspection.

Model one example: a recurrent visual content is stabilized locally and remains integrated, but a reversible route intervention prevents modulation of diverse recipients while preserving source-content identity, stimulus representation, measurement quality, and the retained components. The full model predicts loss or degradation of independently constructed presence-relevant anchors when N_2 fails. The N_2 -ablated rival predicts that those held-out anchors remain stable. Measures derived from the interrupted recipient routes cannot also serve as outcome anchors. The study must verify that N_1 , N_3 , V , and boundary are stable; report and motor pathways cannot be decisive anchors; and the intervention cannot create a new regime.

The skeptical panel should ask:

- Could the boundary change?
- Is the target narrower than the conclusion?
- Does the intervention selectively move the intended component?
- Is any measure used twice?
- Is there adequate diagnostic coverage?
- Could common input, arousal, task demand, report, or sensor leakage explain the result?
- What output occurs if thresholds straddle?
- What observation would make the team abandon or revise its favored interpretation?

Close by returning to the course stance. A mature understanding of a new theory is not the ability to repeat its slogans. It is the ability to expose its commitments, preserve its inferential limits, and design the fair test its proponents and critics should both want.

Worked teaching sequence: from an idea to an executable test

Require every capstone to state a minimum meaningful effect, equivalence margins for retained components, expected diagnostic coverage, multiplicity plan, missing-data rule, and specification neighborhood. These need not involve advanced power calculations. They must show how the same null result could mean redundancy, inadequate coverage, measurement failure, or genuine equivalence under different prospective conditions.

Require a causal diagram with nodes for the component intervention, retained components, boundary, viability, anchors, report or task paths, common causes, and selection into the diagnostic set. Ask four questions:

1. Can the intervention reach anchors through a route other than the target component?
2. Can a common cause move both the component estimate and anchors?
3. Does selection into the diagnostic set create collider bias?
4. Can the intervention change the bearer, regime, or content itself?

Adversarial collaboration needs governance, not only good will. Proponent and rival should have symmetric input into prediction statements, controls, exclusion rules, and result language. Maintain a disagreement log. Use blind or scripted analysis where feasible. Freeze stopping rules and declare how unresolved disagreement will be reported.

Contrast two designs.

Feature	Weak design	Stronger design
Target	"Consciousness during anesthesia"	Presence of a specified auditory content in a declared window and candidate
Rival	Unspecified "other theories"	Full model versus a recurrence-omitting, duration-matched rival that retains V, N_1, N_2
Cases	Waking versus deep anesthesia, all variables move together	Enriched cases with retained V, N_1, N_2 and varied N_3
Evidence	Same recurrent score defines the component and outcome	Frozen component measure and independent held-out anchor vector
Intervention	Global anesthetic dose	Selective recurrent-feedback interruption with V, N_1, N_2 equivalence, content-fidelity checks, and pathway-specific rescue
Null	"No effect; more research needed"	Redundancy, insufficient coverage, indeterminate interval, or unlicensed selectivity distinguished prospectively
Revision	Threshold may be adjusted	Outcome-to-revision table frozen before test

The stronger design can still fail. Its virtue is that different failures mean different things.

Reproducibility begins before data. Require a variable dictionary, analysis skeleton or pseudocode, simulated-data recovery check, file and version plan, and a statement of which decisions are confirmatory. A preregistration too vague to implement is not complete.

Interpret results granularly. A theory challenge, component-operationalization failure, intervention-selectivity failure, anchor-validity failure, boundary reversal, and transport failure are not interchangeable. Each has a different next commitment. The final oral defense should reward teams that expose this structure even when their proposed study remains difficult.

Display plan

Keep a visible checklist with the twelve required fields. Place a red mark beside any element that a team attempts to choose after seeing the critical result.

Slide sequence

1. Complete inference architecture
2. Choose one tractable pressure point
3. Full and rival predictions
4. Diagnostic cases
5. Freeze the specification
6. Split the evidence
7. Protect anchor independence
8. Define four outcomes
9. Sensitivity and transport

10. Revision rules
11. Skeptical-panel questions

Collaborative exercise: Adversarial preregistration studio

Purpose. Teams integrate the course into an implementable study whose diagnostic contrast and revision rules are accepted as fair by both a proponent and a serious rival.

Minimum viable capstone. Require one exact target, one primary rival, one diagnostic contrast, one intervention or decisive observation, one primary anchor outcome, one major negative control, and one outcome-to-revision table. A study of “consciousness in general” is too broad.

Team roles. The principal theorist reconstructs the full model. The assigned-rival representative has symmetric authority over rival predictions and controls. The causal-design lead freezes the candidate and draws intervention, common-cause, report, anchor, and selection paths. The statistics and reproducibility lead specifies meaningful effects, equivalence margins, diagnostic coverage, multiplicity, missingness, sensitivity, data structure, and analysis skeleton. The independence and licensing auditor controls data partitions, dual use, outcome precedence, and transport.

Forty-two-minute sequence. During minutes 0-8, state rival predictions and the diagnostic cell. During 8-18, draw the causal diagram and freeze candidate, grain, regime, and interval. During 18-28, specify measures, anchors, selectivity, negative controls, and rescue. During 28-35, complete the outcome-to-revision table. During 35-42, exchange with a red-team group. The reviewers must identify one circularity route, one selectivity or validity failure, and one unclassified result; the original team visibly revises one challenged field.

Deliverable. Submit Appendix C, a one-page causal diagram, variable dictionary, analysis skeleton or pseudocode, simulated-data recovery plan, versioned specification, disagreement log, and red-team disposition log. The rival representative signs a fairness statement or records the unresolved objection.

Debrief. Panelists classify criticisms as fatal, repairable before data collection, or limitation to be reported. Teams may not answer by redefining the target. A null with adequate equivalence and coverage differs from a null caused by sparse diagnostic cases or invalid measurement.

Collaborative discussion: Outcome-and-revision tribunal

After the oral defenses, give each team one result card: positive effect with very low diagnostic coverage; null with adequate equivalence; apparent target effect accompanied by a boundary change; anchor conflict; or transport failure. Assign project proponent, rival representative, statistician, reproducibility officer, and tribunal chair.

The tribunal submits three statements: **result**, **inference licensed**, and **next commitment**. It must identify whether the result bears on the theory, a component operationalization, intervention selectivity, anchor validity, boundary stability, or transport. End with a private sentence from every student:

The theory would gain my confidence if ____ under license ; it would lose my confidence if ____ under license ____.

Collect these before plenary discussion. Genuine understanding consists in exposed commitments, not agreement with the theory or the instructor.

Common misconceptions

- **“A capstone must test consciousness directly.”** A rigorous construct, boundary, transport, or component audit can make a decisive contribution.
- **“More measures guarantee independence.”** Measures can share sensors, preprocessing, task demands, and theoretical assumptions.
- **“An indeterminate result means the study failed.”** A preregistered cause-specific indeterminate result can identify the next discriminating experiment.
- **“Revision rules signal weakness.”** They distinguish a scientific program from a protected doctrine.

Final exit ticket

Write the one result you think would most increase confidence in $C\emptyset$ / N^* and the one result you think would most decrease it. For each, state the inferential license required before it counts.

Readings

Required: Review all protocols and preregistration appendices in the seven-paper corpus.

Recommended: Melloni et al. (2023); Cogitate Consortium (2025); Yaron et al. (2022).

Part III. Instructor Argument Bank

This section gives compact, classroom-ready arguments for differentiating and evaluating the theory. Each argument includes the claim, its rationale, the evidence needed, and the principal residual pressure. These are arguments for a research program, not declarations that the empirical burden has already been met.

How to Use the Argument Bank Collaboratively

Treat each argument as a testable structure, not a speech for the theory. Use groups of four with rotating roles:

- **Claim reconstructor:** states premises, conclusion, scope, and excluded claim without advocacy.
- **Rival advocate:** supplies the strongest theory or simpler model that explains the ordinary agreement cases.
- **License auditor:** identifies boundary, validity, independence, negative-license, and transport conditions.
- **Synthesis chair:** records the diagnostic case, prospective outcomes, required revisions, and residual pressure.

Thirty-minute argument clinic

1. Every student privately records an initial confidence estimate and one reason.
2. The reconstructor presents the argument in premise-conclusion form.
3. The rival advocate supplies the strongest alternative and one ordinary case on which both accounts agree.
4. The license auditor identifies the case on which the accounts genuinely differ and the evidence required to interpret it.
5. The group writes prospective comparative support, comparative challenge, indeterminate evidence, and unlicensed inference conditions. These argument-clinic labels are not substitutes for a case-level consciousness classification.
6. The reconstructor and rival advocate exchange positions and improve one another's cases.
7. Every student records a revised confidence estimate and the consideration that most affected it.

The required product is one page containing the exact target, argument, strongest rival, diagnostic case, evidence licenses, prospective outcomes, theory revision, and residual pressure. Consensus is not required. Unresolved disagreement must be recorded rather than averaged away.

Argument 1. Explanandum discipline

Claim. CØ / N* improves theory comparison by fixing minimal phenomenal presence as its initial target and refusing to identify it with report, attention, intelligence, wakefulness, self-consciousness, or detailed character.

Rationale. Many apparent disputes compare theories of different explananda. A target-first framework prevents report-related or metacognitive mechanisms from being credited as necessary for minimal presence without a separate argument.

Validation route. Show that component measures track presence-relevant contrasts after report, executive demand, and overt behavior are controlled, and that richer capacities can dissociate from the minimal bundle.

Residual pressure. The target may be too thin to satisfy those who regard ownership, affect, or qualitative structure as constitutive of any experience.

Argument 2. Conjunctive differentiation

Claim. Integration, availability, and recurrence solve different false-positive problems, so a conjunction can outperform single-factor theories.

Rationale. Integration blocks heaps, availability blocks isolated contents, and recurrence blocks transients. Each condition alone permits a characteristic near miss.

Validation route. Enrich for single-failure diagnostic cases and compare the full model with every leave-one-out and scientifically motivated grouped rival on independent anchors.

Residual pressure. The conditions may be manifestations of one deeper process or too causally coupled to warrant separate necessity claims.

Argument 3. Boundary-before-measurement non-circularity

Claim. Any theory using system-level quantities requires an independently motivated candidate system before those quantities can support a consciousness verdict.

Rationale. Integration, availability, recurrence, complexity, and causal power all change with the selected node set, grain, and horizon. Choosing the boundary that maximizes the final score lets evidence select its own bearer.

Validation route. Demonstrate that autonomy-first boundaries recover stable causal units, predict held-out perturbations, and outperform anatomical, product-label, random, and score-maximizing baselines.

Residual pressure. The autonomy measure, proposal family, and inclusion-minimality rule may miss a genuinely larger or cross-scale subject.

Argument 4. Report-independent availability

Claim. System-wide availability can be operationalized as a causal disposition without making report or executive access constitutive.

Rationale. Content-specific interventions can reveal timely modulation of functionally diverse recipients even when no output is produced. Diffusion, common input, arousal, and copied messages can be excluded by selectivity, route, and diversity controls.

Validation route. Show that N_2 remains stable in no-report, motor-dissociation, posterior-dominant, and nonexecutive conditions while unconscious diffusion and message duplication fail.

Residual pressure. If all reliable recipient effects require memory, decision, executive control, or report, the proposed separation from access consciousness fails.

Argument 5. Epistemic discipline increases rather than weakens falsifiability

Claim. Positive, negative, indeterminate, and unlicensed outputs make empirical claims more accountable by preventing missing or invalid evidence from being converted into favorable verdicts.

Rationale. A sharp theory can coexist with a limited test. Validity and quality gates, a preregistered conflict rule, and protected gray zone prevent cherry-picking and false negative inflation.

Validation route. Report coverage, selective risk, indeterminate rate, unlicensed rate, cause profiles, and calibration over repeated assessments. Show that abstention improves conditional error without hiding systematic failures.

Residual pressure. Indeterminacy can become a rescue category unless thresholds, applicability, conflict rules, and maximum acceptable unresolved rates are set prospectively.

Argument 6. The necessity audit prevents syntactic self-confirmation

Claim. A component earns its place only when the full and ablated models differ on valid diagnostic cases and the full model predicts independent held-out anchors better.

Rationale. A term is logically necessary inside a written conjunction but may be empirically redundant. Diagonal datasets cannot distinguish components. Labels generated by N^* cannot validate N^* .

Validation route. Establish diagnostic coverage, preserve anchor independence, verify selective mechanism changes, use matched model flexibility, test grouped ablations, and transport without favorable retuning.

Residual pressure. Anchor evidence remains theory-laden and diagnostic conditioning can create selection bias.

Argument 7. Modular extension to phenomenal character

Claim. Separating presence from character allows the theory to explain phenomenal differentiation without letting representational geometry manufacture consciousness.

Rationale. A presence gate and an independently tested geometry-to-quality bridge answer different questions and can fail independently.

Validation route. Fit the bridge on independent anchors, predict held-out relations and intervention-induced deformations, beat nuisance models, and report unbroken automorphisms and transport limits.

Residual pressure. Relational geometry may not exhaust intrinsic, semantic, affective, bodily, or historical aspects of character.

Argument 8. Schematic legibility supplies an explanatory standard

Claim. If the constitutive identity is true, the organization can become readable through stable component semantics, compositional inference, localized counterfactual faults, and independent validation.

Rationale. A theory should make a trained reader able to reconstruct why a state is classified, which role fails under a perturbation, and where the inference becomes unlicensed.

Validation route. Test inter-rater reproducibility, fault prediction, role-preserving transport, estimator convergence, and compression from detailed causal data to a discriminating role profile.

Residual pressure. Structural intelligibility does not by itself bridge the metaphysical explanatory gap or prove the identity.

Argument 9. Constrained substrate independence avoids opposite extremes

Claim. The model permits multiple realization while rejecting both material chauvinism and consciousness from abstract similarity.

Rationale. A qualifying implementation must realize directionality, effective coupling, temporal grain, dynamic ranges, role structure, and substrate-appropriate viability.

Validation route. Establish construct, causal-role, and calibration preservation across substrates; use intervention and replacement tests rather than behavioral imitation alone.

Residual pressure. Nonbiological viability and cross-substrate phenomenal anchors remain underdeveloped.

The validation program at a glance

Phase	Scientific aim	Success criterion	What success does not yet establish
A. Synthetic construct separation	Test whether V, N_1, N_2, N_3 can vary and be measured separately	Recovery of known Boolean and continuous profiles under controls	Consciousness of the synthetic system
B. Near-miss atlas	Find natural and clinical diagnostic cases	Valid single-failure and grouped-failure coverage with complete audit records	Causal necessity without intervention
C. Selective perturbation and rescue	Change one realization while preserving retained roles	Licensed selectivity, predicted anchor change, and component-specific rescue	Metaphysical identity from one domain
D. Held-out transport	Test generality across subjects, tasks, modalities, and substrates	Stable prediction without favorable retuning	Universal substrate independence
E. Adversarial adjudication	Compare full and rival models under shared rules	Prospective outcome categories and independent implementation	Finality of the theory landscape

Argument-to-Exercise Validation Crosswalk

Argument	Collaborative task	Required artifact	Validation or differentiation payoff	Prohibited overreach
1. Explanandum discipline	Target-sorting controversy	Matched-target grid and corrected overclaim	Prevents report, access, character, and presence from being compared as coextensive	Target clarity does not validate the identity
2. Conjunctive differentiation	Single-failure design studio	Near-miss profile and full-versus-ablated predictions	Finds cases that distinguish the conjunction from single-factor and leave-one-out rivals	Logical differentiation is not empirical necessity
3. Boundary-first non-circularity	Frozen-boundary hearing	Candidate set, proposal rule, replacement test, and reversal analysis	Tests whether verdicts survive independently chosen boundaries	A stable candidate is not thereby conscious
4. Report-independent availability	Causal-routing laboratory	Recipient ledger, route controls, and reduced-family results	Differentiates causal availability from report, diffusion, and executive uptake	An N_2 pass is not sufficient for C_0
5. Four-outcome discipline	Evidence tribunal	Outcome, cause profile, next study, and separate action note	Tests calibrated rule-in, rule-out, abstention, and invalidity	Indeterminate is not partly conscious
6. Necessity audit	Diagnostic-cell review	Coverage, independence check, selectivity audit, and repaired design	Distinguishes genuine component tests from diagonal or self-labeled studies	Better anchor prediction does not prove metaphysical identity
7. Character extension	Bridge-breaking workshop	Presence result, bridge result, deformation prediction, and surviving symmetries	Separates presence from character and causal from stimulus geometry	A failed bridge does not negate presence
8. Schematic legibility	Blind fault-localization trial	Inter-reader role reconstruction and fault map	Tests teachability, composition, and counterfactual specificity	Intelligibility is not confirmation

Argument	Collaborative task	Required artifact	Validation or differentiation payoff	Prohibited overreach
9. Constrained substrate independence	Transport jigsaw	Construct, causal-role, calibration, and boundary ledger	Differentiates realized organization from behavior or graph similarity	Failed transport licenses neither positive nor negative attribution

Grade the diagnostic contrast and inferential discipline, not whether a group ultimately favors CØ / N*. Every artifact should state both what its evidence would establish and what it would leave open.

Quick objection-and-reply matrix

Objection	Best available reply	Residual pressure to acknowledge
The theory is just IIT + GNWT + RPT	The novelty is the exposed conjunction, distinct roles, viability, boundary order, conflict policy, and component audit	It must produce discriminating predictions not already forced by the source theories
N_2 is access consciousness in disguise	Internal causal modulation and reduced-family tests exclude report and executive necessity	Implementation may still depend on access-like consumers
V is unfalsifiable	Freeze indicators and thresholds before the verdict and require concrete interpretability consequences	Repeated bespoke exceptions would weaken the theory
Thresholds are arbitrary	Make thresholds explicit, calibrated, cross-validated, and stress-tested	Brittle threshold dependence remains a real weakness
The model confuses correlates with constitution	Prioritize perturbation, route interruption, selective ablation, and rescue	Causal relevance still underdetermines metaphysical identity
It ignores qualia	It brackets presence first and adds a separate causal-geometry bridge for character	The bridge remains incomplete and symmetry-limited
It overattributes consciousness to AI	Mere behavior and graph isomorphism fail without causal realization, viability, boundary, and transport	Artificial-system viability and anchors remain contested
There is no oracle for C_0	Use independent convergent anchors and rival prediction, not self-generated labels	Conclusions remain anchor-relative
Consciousness may be graded	Keep continuous measures, thresholded scientific outputs, and a possible probabilistic extension	A sharp biconditional may ultimately need revision
Embodiment and affect are constitutive	Let boundary, V , dynamics, and character include organismic roles, and add a conjunct if evidence requires it	The current theory may understate life and value
The hard problem remains	The theory is a mid-level constitutive and empirical program, not a derivation of why experience exists	Philosophical dissatisfaction is not dissolved
Indeterminacy protects the model	Preregister all license and conflict rules and report unresolved rates as test properties	Excessive unresolved output would reduce usefulness

Reusable Graduate-Seminar Discussion Formats

Structured academic controversy

Two pairs prepare opposing positions, present them, then exchange positions and improve the argument they initially opposed. The four students produce a joint statement containing agreed premises, the genuine disagreement, a discriminating case, and remaining uncertainty.

Progressive-disclosure case conference

Give teams evidence in four rounds: target and context; boundary and viability; component measures; and validity, conflict, and anchors. Require a provisional verdict after each round. Students record what changed their verdict and whether the change concerned the system or only the license to infer.

Component jigsaw

Initial expert groups study boundary, V / N_1 , N_2 / N_3 , or evidence licensing. New mixed groups receive one expert from each area and must produce a complete classification. The design makes every student responsible for information that the group cannot omit.

Fishbowl adjudication

An inner group publicly adjudicates a difficult case while an outer group records target drift, illicit negatives, boundary changes, circular validation, and ignored rival predictions. Groups exchange places. The observer sheet, not rhetorical victory, supplies the assessment evidence.

Preregistration gallery walk

Teams display only target, rivals, diagnostic region, outcome rules, and revision commitments. Visitors leave one **fatal unless repaired** note and one **discriminating improvement** note. Teams file a disposition log explaining which comments they accepted, rejected, or converted into sensitivity analyses.

Part IV. Reproducible Course Materials

Appendix A. Corpus-to-Course Coverage Audit

This matrix lets the instructor verify that the course incorporates the conceptual contributions of the full paper corpus rather than relying only on the central formula.

Source	Concept or commitment	Primary treatment
Core paper	Minimal phenomenal presence distinguished from report, attention, and C_1	Sessions 1-2
Core paper	Necessary, sufficient, conjunction, and biconditional logic	Session 2; Problem Set 1
Core paper	Causal graph; structural, functional, and effective connectivity	Session 2
Core paper	Temporal grain and whole-brain / subcortical scope	Sessions 2, 5, 6
Core paper	N_1 integration or synergy and candidate surrogates	Session 4
Core paper	N_2 availability separated from report and fixed anatomy	Sessions 2 and 5
Core paper	N_3 bounded recurrent dwell and metastability	Session 6
Core paper	Independent viability and rescue-clause guardrail	Session 4
Core paper	Surrogate-conflict rule, preregistration, and robustness	Sessions 2 and 7
Core paper	Sufficiency, necessity, and exclusion-event falsifiers	Sessions 2 and 13
Core paper	Theory landscape and constrained substrate independence	Sessions 1, 12, 13
Core paper	Deliberate limitation to presence and initial quality-space proposal	Sessions 1 and 9
Boundary paper	Boundary relativity and boundary reversal	Session 3
Boundary paper	Physical, causal, and subject boundaries	Session 3
Boundary paper	Prohibition on retrospective score-maximizing boundaries	Session 3
Boundary paper	Endogenous capacity, inbound dependence, and autonomy ratio	Session 3
Boundary paper	Candidate reliability versus boundary stability	Session 3; Appendix D worksheet
Boundary paper	Role completeness and confidence-bound decisions	Sessions 3 and 7
Boundary paper	Inclusion-minimality and cross-scale hierarchy	Session 3
Boundary paper	Pairwise separability for multiple candidates	Sessions 3 and 11

Source	Concept or commitment	Primary treatment
Boundary paper	Constituent, interface, enabling support, and replacement test	Sessions 3 and 12
Boundary paper	Five boundary statuses	Sessions 3 and 7
Boundary paper	Split brain, brain-body, coupled agents, organoid-computer, distributed AI	Sessions 11-12
Availability paper	Activation / availability, availability / report, broadcast / diffusion	Session 5
Availability paper	Globality / extent, diversity / node count, posterior-subcortical / executive	Session 5
Availability paper	Independent recipient classes from causal repertoires	Session 5
Availability paper	Preparation burden and context robustness	Session 5
Availability paper	Content fidelity, class-native effects, and latency	Session 5
Availability paper	Selectivity and route-interruption evidence	Session 5
Availability paper	Coverage, class count, and diversity evenness	Session 5; Problem Set 2
Availability paper	Report- and executive-excluded recomputation	Session 5
Availability paper	Positive, negative, indeterminate, and unlicensed N_2 outputs	Sessions 5 and 7
Indeterminacy paper	Assessment tuple and validity envelope	Session 7; Appendix D worksheet
Indeterminacy paper	Four-outcome precedence rule	Session 7; Problem Set 2
Indeterminacy paper	Positive and Negative License Principles	Sessions 7 and 11
Indeterminacy paper	Five-part cause profile	Session 7; Appendix D worksheet
Indeterminacy paper	Surrogate-discordance rule and model-output sets	Session 7
Indeterminacy paper	Boundary and transport indeterminacy	Sessions 3, 7, 12
Indeterminacy paper	Coverage, indeterminate rate, unlicensed rate, selective risk	Session 7
Indeterminacy paper	Scientific status separated from practical action	Sessions 7 and 12
Ablation paper	Logical, incremental empirical, and causal necessity	Session 8
Ablation paper	Diagnostic-set theorem and approximate diagnostic region	Session 8; Problem Set 2
Ablation paper	Diagnostic coverage and underidentification	Session 8
Ablation paper	Hypothesis-not-definition and two evidence streams	Session 8
Ablation paper	Development, calibration, and test splits; no dual use	Sessions 8 and 14
Ablation paper	Held-out anchor risk and Δ_k	Session 8
Ablation paper	Criterion, mechanism, and measurement ablation	Session 8

Source	Concept or commitment	Primary treatment
Ablation paper	Selectivity license and rescue	Sessions 4, 6, 8
Ablation paper	Indispensable, redundant, coupled, indeterminate, unlicensed	Session 8
Ablation paper	Grouped ablation and weak versus strong minimality	Session 8
Ablation paper	Synthetic, near-miss, perturbation, transport, adversarial phases	Sessions 8, 13, 14
Character paper	Presence gate versus character map and two-layer error matrix	Session 9
Character paper	Perturbation-derived causal-trajectory geometry G_c	Session 9
Character paper	Independently anchored quality space	Session 9
Character paper	Held-out bridge F_Φ , nuisance controls, and stability	Session 9
Character paper	Geometry-changing intervention and deformation prediction	Session 9; Capstone options
Character paper	Character as an automorphism equivalence class	Session 9
Character paper	Modality, location, intensity, valence, temporal path	Sessions 6 and 9
Character paper	Color, audition, olfaction, pain, temporality, cross-modal scenes	Session 9; optional presentations
Character paper	Supported, refuted, indeterminate, unlicensed bridge outcomes	Session 9
Schematic paper	Black-box versus structurally legible necessity	Session 10
Schematic paper	Bearer, operation, unity, system-level presence, temporal presence role vector	Session 10
Schematic paper	Decomposable semantics, compositional readout, fault localization, independence	Session 10
Schematic paper	Correlational marker, black-box law, constitutive schematic	Session 10
Schematic paper	Observation of occurrence versus inference of character versus undergoing	Session 10
Schematic paper	Limits: representation, confirmation, theory-ladenness, scope, presence-character split	Session 10
Schematic paper	Role-preserving transport, fault prediction, convergence, compression, teachability	Sessions 10 and 13

Appendix B. Assessment Bank and Answer Guides

Problem Set 1. Concepts and arguments

1. Define minimal phenomenal presence and distinguish it from access consciousness and phenomenal character.
2. Translate $V \wedge (N_1 \wedge N_2 \wedge N_3) \Leftrightarrow C_0$ into two directional claims. Give a threat to each.
3. Explain why a boundary reversal is scientifically important even when neither candidate boundary is yet known to be correct.
4. Give one example each of activation without availability and availability without current activation.
5. Explain why the absence of a positive marker is not automatically a negative consciousness result.
6. State the diagnostic profile for auditing N_3 and explain why ordinary waking-versus-anesthesia data may not identify N_3 necessity.
7. Distinguish criterion, mechanism, and measurement ablation.
8. Explain why a high-performing neural manifold is not automatically a phenomenal quality space.
9. Apply D , C , F , and E to a one-number consciousness classifier.
10. State the strongest version of the “hybrid without novelty” objection and the best reply available in the paper corpus.
11. Explain how scientific indeterminacy can justify precaution without becoming a positive attribution.
12. Name one empirical result that should cause removal, combination, or addition of a conjunct.

Answer guide for Problem Set 1

1. C_0 is the minimal what-it-is-like fact. Access concerns usability for reasoning or control; character concerns what the admitted experience is like.
2. Sufficiency: the viable bundle guarantees C_0 ; threatened by licensed $V \wedge N^* \wedge \neg C_0$. Necessity: C_0 requires the viable bundle; threatened by licensed $C_0 \wedge \neg (V \wedge N^*)$.
3. It shows that the consciousness result depends on what was counted as the system and therefore requires independent individuation and sensitivity reporting.
4. Activation without availability: common-drive or arousal recruitment. Availability without activation: an idle recipient shown by nearby, low-burden intervention to be selectively reachable.
5. A negative requires applicability, sensitivity, adequate quality, satisfied auxiliary demands, a rule-out region, and exclusion of alternative causes of non-detection.
6. $V=1, N_1=1, N_2=1, N_3=0$ with uncertainty bounds clearing all gates. If all components move together across waking and anesthesia, full and ablated models agree on every case.
7. Criterion removes a rule term; mechanism changes a proposed physical realization; measurement removes an estimator or data route.
8. It may encode stimulus or task relations, occur unconsciously, or fit only correlationally. The bridge requires the presence gate, independent phenomenal anchors, held-out prediction, and intervention-sensitive deformation.
9. Stable semantics may be absent, the intermediate role profile is absent, faults are not localized, and validation may be circular. A label alone is not a schematic.
10. Objection: all components are borrowed. Reply: the exposed conjunction and its boundary, viability, report-independent, conflict, ablation, and revision rules make a distinct empirical architecture. Residual pressure: it must deliver unique predictions.

11. Scientific status describes evidential warrant; action also depends on stakes, reversibility, and error costs. Protective action does not alter D .
12. Examples: stable equivalence of a leave-one-out rival supports removal; inseparable components support combination; repeated conscious diagnostic cases with one condition absent support deletion or replacement; a new reliably necessary factor supports addition.

Problem Set 2. Formal classification practicum

Case A: Four-outcome decision

A test is applicable and well executed. Two credible surrogates disagree beyond the preregistered tolerance. The validated aggregate score would otherwise exceed τ_+ . What is the outcome and why?

Case B: Boundary status

Two disjoint candidates both clear autonomy and role thresholds. Both are stable across specifications. Interrupting either candidate's specific coupling to the other leaves the other's role profile intact; pairwise separability is .89 with $\kappa = .80$. What boundary status is supported? What does it not establish?

Case C: N_2 aggregation

Five independently calibrated recipient classes have frozen weights (.25, .20, .20, .20, .15). Classes 1, 2, and 4 pass every effect, latency, selectivity, route, and context gate. Class 3 is effect-indeterminate. Class 5 fails route dependence. Let $\alpha = .60$ and $m = 3$. The three qualifying effects are sufficiently even to pass η . Report and executive classes are not among the qualifying three. Is N_2 positive, negative, or indeterminate?

Case D: Diagnostic coverage

A study contains 500 cases. For N_1 , only four cases have the upper N_1 bound below threshold while lower bounds for V, N_2, N_3 clear their gates. The full model predicts their anchor patterns slightly better than the ablated model, but the confidence interval spans the minimum meaningful effect. What component status is warranted?

Case E: Mechanism selectivity

An intervention lowers N_3 by the preregistered amount but also lowers N_2 beyond its equivalence margin and changes the selected boundary. Independent anchors decline. What may be inferred?

Case F: Character bridge

Presence is positive. The bridge predicts baseline similarity well but fails held-out geometry-changing interventions, while the intervention and phenomenal-anchor validity checks pass. What is the bridge outcome and what remains intact?

Answer guide for Problem Set 2

- A. Indeterminate. Credible conflict has precedence over scalar aggregation.
- B. Multiple candidate bearers. The result identifies multiple stable candidates for a later consciousness test; it does not prove that either is conscious.
- C. Positive on the stated gates. Qualifying weight is .65, class count is three, diversity passes, and reduced-family requirements are satisfied. The unresolved class remains reported.
- D. Indeterminate and coverage-limited. Very low diagnostic coverage and an interval spanning the effect margin do not establish indispensability.
- E. The intended selective N_3 mechanism ablation is unlicensed; the result may reveal coupling or boundary reorganization. Anchor decline cannot be assigned specifically to N_3 .

F. Refuted for the specified bridge and domain. Positive presence remains intact; another character mapping may still succeed.

Problem Set 3. Essay and oral-exam prompts

1. Is CØ / N* a genuinely new theory or a disciplined synthesis? Defend a position using at least four commitments not entailed by simply listing IIT, GNWT, and RPT.
2. Does report-independent availability escape the phenomenal/access distinction, or does it relocate access inside the minimal condition?
3. Evaluate inclusion-minimality as a subject criterion. Compare it with IIT-style exclusion and an embodied alternative.
4. Can independent anchors test a consciousness theory without becoming a second consciousness theory?
5. Does the phenomenal-character extension answer the inverted-spectrum objection or formalize its remaining force?
6. Does schematic legibility explain consciousness, or only make a functional identity easier to understand?
7. Design one licensed negative test in a nonreporting population and defend every negative-license condition.
8. Should the sharp biconditional be replaced with a probabilistic model? State what empirical pattern would justify the revision.

Analytic-writing rubric

Criterion	Excellent	Competent	Needs revision
Target clarity	Exact consciousness target and scope are maintained throughout	Target is stated but occasionally drifts	"Consciousness" is used without qualification
Theory accuracy	Roles, order, and outcome rules are reconstructed correctly	Minor omissions do not change the argument	Major components or licenses are conflated
Rival fairness	Strong, version-specific rival and objection	Rival is recognizable but simplified	Strawman or target mismatch
Evidence discipline	Boundary, validity, anchors, confounds, and transport are addressed	Most licenses are addressed	Evidence is equated with the target
Falsifiability	Concrete result and revision rule are stated	General counterevidence is named	No outcome could change the view
Argument quality	Premises, inference, residual pressure, and conclusion are explicit	Coherent argument with limited residual analysis	Assertion or summary without inference

Collaborative work: accountability and assessment

Collaborative work should combine a shared product with evidence of individual preparation, role performance, and revision. Do not grade verbal speed, confidence, social dominance, or agreement with the instructor.

Recommended allocation:

- **60% shared scientific product:** accuracy, rival fairness, diagnostic contrast, evidential licensing, and revision rule;
- **25% individual accountability:** pre-class position note, assigned-role record, and brief individual defense or post-discussion revision; and
- **15% collaborative process:** evidence sharing, response to criticism, equitable participation, and documentation of dissent.

Dimension	Strong	Adequate	Needs revision
Epistemic contribution	Introduces a relevant distinction, datum, model, or repair and connects it to the conclusion	Completes the assigned role accurately	Offers assertions without usable support
Rival charity	Improves the strongest rival and gives it a fair diagnostic prediction	Represents the rival recognizably	Uses a weak or target-mismatched rival
License vigilance	Identifies validity, boundary, independence, or transport failures before verdict	Notices the principal limitation	Treats non-detection or invalid evidence as negative
Responsive revision	Changes a claim or design for a stated reason and records the consequence	Acknowledges criticism	Absorbs criticism without changing anything
Shared reasoning	Invites missing evidence, makes reasoning visible, and preserves dissent	Participates reliably	Dominates, withdraws, or conceals disagreement

Individual process form:

1. State your assigned role and one concrete contribution traceable in the final artifact.
2. Identify one teammate contribution that changed or sharpened your view.
3. Record one disagreement the group resolved and how.
4. Record one disagreement that remains and why averaging it would mislead.
5. State one claim in the group product you would defend differently on your own.

Peer reports should be used only when behavior-specific and corroborated by the group's record. The instructor should intervene when role assignments consistently reproduce status hierarchies or when one student becomes the permanent skeptic, reporter, or technical specialist.

Appendix C. Capstone Preregistration and Rubric

Capstone title and team

Study title:

Team members and declared prior expectations:

Planned registration, data, and code repository:

Collaboration charter and rotating roles

Before designing the study, each member privately records expected outcomes for the full model and its rivals. The team then declares:

- role rotation at each checkpoint;
- how dissent will be preserved;
- how deadlocks will be handled;
- responsibility for version control, citations, analysis, and meeting records;
- the procedure for missed work and redistribution; and
- which design decisions require unanimous agreement because they affect fairness between rivals.

Recommended roles are **target and boundary steward**, **construct and measurement lead**, **rival-model advocate**, **license and anti-circularity auditor**, and, when appropriate, **clinical or welfare liaison**. The rival advocate must be able to certify that the final test is fair to the rival before the protocol is frozen. Failure to certify does not give the rival a veto without reasons; the unresolved objection is recorded for panel adjudication.

1. Exact target

- Project type: attribution / character bridge / component ablation / adversarial model comparison / construct, boundary, or transport audit
- Target form of consciousness:
- Candidate content or state:
- Population or substrate:
- Claim deliberately excluded from the study:

2. Full model and rivals

State the full CØ / N* prediction and at least one serious rival. The rival may be a leave-one-out conjunction, grouped ablation, probabilistic version, source theory, nuisance model, or nonconjunctive theory.

Model	Prospective prediction	Expected anchor pattern	Revision if disfavored
Full model			
Rival 1			
Rival 2, if used			

3. Diagnostic region

- Exact cases on which the models disagree:
- Expected diagnostic coverage and how it will be estimated:

- Continuous diagnostic margin and uncertainty rule:
- Off-manifold or sampling concerns:

4. Candidate-system specification

- Observational universe Ω and known omissions:
- Proposed candidates and discovery-only generator:
- Grain g , interval Δ , and operating regime R :
- Autonomy estimators, thresholds, and uncertainty bounds:
- Inclusion-minimality, separability, and stability rules:
- Constituent, interface, and enabling-support tests:

5. Component operationalization

Component	Estimator family	Threshold and uncertainty	Main confound	Negative control
V				
N_1				
N_2				
N_3				

For N_2 , additionally state recipient-class calibration, content and matched control, admissible contexts, preparation burden, effect and latency thresholds, selectivity and route tests, class weights, diversity rules, and report/executive exclusions.

6. Intervention and selectivity

- Target mechanism or criterion:
- Matched sham, energy, timing, and nuisance controls:
- Minimum intended component change:
- Equivalence margins for every retained component:
- Boundary and viability preservation checks:
- Rescue or recovery condition:
- Conditions that make the intervention unlicensed:

7. Independent anchors

- Anchor vector and target relation of each element:
- Validation genealogy and known auxiliary demands:
- Prohibited dual uses:
- Leave-family-out and shared-channel sensitivity analysis:
- Preregistered translation from each model to expected anchor patterns:

8. Evidence separation

Partition	Permitted use	Prohibited use
Development	Choose candidate estimators and interventions	Final model adjudication
Calibration	Freeze thresholds, bridge rules, and uncertainty	Outcome-driven tuning
Final test	Score frozen predictions once	Any redesign except labeled exploration

9. Outcome rules

Define each outcome before critical data access.

Use vocabulary appropriate to the project:

- attribution: **positive, negative, indeterminate, unlicensed;**
- character bridge: **supported, refuted, indeterminate, unlicensed;**
- ablation: **indispensable, redundant, causally coupled, indeterminate, unlicensed;**
- adversarial model comparison: **relative support for the full model, relative support for the rival, equivalence or tie, indeterminate, unlicensed;** and
- construct, boundary, or transport audit: the exact status vocabulary defined by that protocol.

Project-appropriate outcome	Exact rule	Scientific interpretation	Next study
Primary model-favorable or rule-in result			
Primary rival-favorable, refuting, redundant, or rule-out result			
Equivalence, tie, or coupled result, if applicable			
Indeterminate			
Unlicensed			

10. Robustness and transport

- Reasonable specification neighborhood:
- Boundary, grain, estimator, threshold, and anchor variants:
- Multiplicity control:
- Transport target and three bridge requirements:
- Minimum result stability:
- Maximum acceptable unresolved-output rate:

11. Practical action note

If the study bears on clinical or welfare decisions, state possible actions, error costs, reversibility, and precaution independently of the scientific classification.

12. Stopping and revision commitments

- Sample-size and stopping rule:

- Exclusion and missingness policy:
- Model change after each possible result:
- Result that would cause abandonment of the favored interpretation:
- Exploratory analyses permitted after confirmatory completion:

Staged cross-team review

Recommended sequence: form teams and complete the collaboration charter after Session 9; Checkpoint 1 after Session 10; Checkpoint 2 after Session 11; Checkpoint 3 after Session 12; Checkpoint 4 after Session 13; and Checkpoint 5 before the Session 14 defense. A compressed version retains Checkpoints 1, 3, and 5.

Checkpoint	Team submission	Reviewing team's required response
1. Target and rivals	Exact target, exclusions, strongest rival	One target drift and one stronger rival
2. Diagnostic region	Disagreement set, expected coverage, boundary	One case where models still agree and one possible reversal
3. Evidence architecture	Measures, anchors, splits, intervention	One circularity route and one selectivity failure
4. Outcome freeze	Four outcomes and revision rules	One unclassified result and one commitment still movable
5. Mock defense	Complete preregistration	One fatal-unless-repaired judgment and one valuable-even-if-null judgment

Every team submits a disposition log. Rejected criticism must receive a substantive reason; accepted criticism must produce a visible protocol change. Recommended capstone grade composition is 70% team preregistration under the rubric, 20% individual oral defense with questions assigned unpredictably, and 10% collaboration charter, role records, dissent log, and peer-review disposition.

Capstone rubric

Dimension	Points	Full-credit standard
Target and scope	10	Exact explanandum, system, content, regime, and excluded claims
Rival quality	10	Strong rivals with genuinely different prospective predictions
Diagnostic identification	12	Correct disagreement set, coverage plan, and uncertainty margin
Boundary discipline	10	Independent proposal, stability, minimality, and separability addressed
Construct validity	12	Component measures and confounds are explicitly validated
Anti-circularity	12	Evidence splits, anchor independence, and no dual use are enforceable
Intervention selectivity	10	Retained components, identity, viability, and rescue are protected
Outcome discipline	10	Project-appropriate outcomes, indeterminate, unlicensed, and special statuses are prospective
Robustness and transport	8	Reasonable alternatives and transfer limits are declared
Revision commitments	6	Every declared result has a substantive consequence for the model or protocol

Individual oral defense criteria (20% of the overall capstone grade): accurate reconstruction without teammate prompting; ability to defend the rival fairly; explanation of one validity or independence risk; interpretation of an unanticipated result; and a clear personal revision commitment.

Collaboration-record criteria (10%): completed charter and role rotations; behavior-specific contribution record; preserved dissent; timely cross-team reviews; and a disposition log in which accepted criticism changes the protocol and rejected criticism receives a substantive reason.

Appendix D. Reusable Instructor Worksheets

Worksheet 1. Complete case-classification record

Field	Entry
Target form of consciousness	
Candidate system, boundary status, and omitted variables	
Grain, regime, and interval	
Independent viability result	
N_1 result and surrogate agreement	
N_2 result and reduced-family checks	
N_3 result and temporal interpretation	
Assessment applicability and data quality	
Final outcome	
Cause profile $\{e, s, m, b, t\}$	
Most informative next study	
Separate practical-action note	

Worksheet 2. Theory-comparison grid

Question	CØ / N*	Rival theory
Exact explanandum		
Candidate bearer		
Necessary condition(s)		
Claimed sufficient condition		
Evidence treated as constitutive		
Report and access role		
Phenomenal-character account		
Substrate commitment		
Unique diagnostic case		
Direct falsifier		

Worksheet 3. Boundary audit

Step	Required record
Observational universe	Included variables, known omissions, interfaces
Specification	Grain, horizon, regime, intervention family

Step	Required record
Candidate proposal	Discovery-only method and candidate family
Autonomy	J_{self}, J_{in} , ratio, uncertainty, reliability
Role completeness	V, N_1, N_2, N_3 for every frozen candidate
Minimality	Qualifying proper subsets
Separability	Reciprocal interruption and preservation
Replacement	Constituent, interface, or support
Stability	Cross-specification mapping and result
Boundary status	Unique, multiple, nested, none, or indeterminate

Worksheet 4. Component-necessity audit

Question	Entry
Component under audit	
Full and ablated model	
Exact diagnostic profile	
Diagnostic coverage	
Component estimator and threshold	
Independent anchor vector	
Development / calibration / test split	
Dual-use check	
Mechanism selectivity, if claimed	
Full versus rival held-out risk	
Specification stability	
Status: indispensable, redundant, coupled, indeterminate, unlicensed	

Worksheet 5. Phenomenal-character bridge

Field	Entry
Positive presence-gate evidence	
Content family and causal trajectories	
Geometry type, distance, transitions, occupancy	
Independent phenomenal anchors	
Train / validation / test split	
Null and nuisance models	
Held-out metric and transition performance	
Geometry-changing intervention	

Field	Entry
Deformation prediction and observed change	
Surviving automorphisms	
Transport license	
Bridge outcome	

Worksheet 6. Schematic-legibility check

Condition	Evidence
Decomposable semantics	Roles are stable and do not repeat the outcome label
Compositional readout	Intermediate organization profile can be reconstructed
Counterfactual fault localization	Different changes yield different predicted failures
Evidential independence	Readout is not used to validate its own identity claim
Teachability	Independent readers agree on held-out cases
Transport	The role profile survives materially different realizations when appropriate

Worksheet 7. Collaborative argument map

Field	Entry
Exact target and scope	
Argument reconstructed as premises and conclusion	
Strongest rival explanation	
Ordinary agreement case	
Diagnostic disagreement case	
Full-model prediction	
Rival prediction	
Required validity and independence conditions	
Project-appropriate outcome vocabulary and exact rules	
Revision required by each licensed result	
Residual conceptual or metaphysical pressure	
Individual pre- and post-discussion confidence	

Worksheet 8. Progressive evidence record

Evidence round	Provisional output	What changed?	System change or license change?	Next discriminating evidence
Target and context				
Boundary and viability				
Components				

Evidence round	Provisional output	What changed?	System change or license change?	Next discriminating evidence
Validity, conflict, and anchors				
Final				

Worksheet 9. Peer-review disposition log

Criticism	Source role or team	Fatal, strengthening, or optional	Accepted?	Protocol change or reason for rejection
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Appendix E. Collaborative Case Packets

These packets are designed for progressive disclosure. Print each numbered round as a separate envelope; when a round contains several cards, cut those cards apart and distribute them among team members only after the envelope is opened. Do not hand out later rounds at the start. Students initially report the evidence on their current card without interpretation. After all cards in a round are recorded, the team updates Worksheet 8 and commits to a provisional output before the instructor releases the next envelope. After Round 4, teams complete Worksheet 1 for the final case record and Worksheet 7 when an argument or rival comparison is central. Release the instructor key only after the final written commitment.

For every packet, teams must state:

1. the exact target;
2. the candidate bearer;
3. the strongest licensed output;
4. the strongest forbidden inference;
5. the principal rival prediction;
6. the most informative next study; and
7. any separate practical action.

Packet 1. The silent patient

Teaching aims: explanandum discipline, report-independent evidence, four outcomes, negative license, and science-action separation.

Round 1 - context card. Patient P is behaviorally unresponsive after traumatic brain injury. Eye opening fluctuates. No reliable motor output has been observed. The team is asked about phenomenal presence during a specified thirty-second auditory trial, not about continuous awareness, prognosis, or quality of life.

Round 2 - route and boundary cards. Auditory brainstem responses are present. In the high-arousal imagery window, an independent cortical speech-discrimination check passes; the route is not reverified in the later low-arousal window. A discovery-only connectivity analysis nominates a thalamocortical candidate that remains stable across two grains. Physiological viability is adequate in high-arousal windows and poor in low-arousal windows.

Round 3 - component cards. During one high-arousal window, an imagery classifier exceeds its healthy-control threshold on two artifact-free, cross-validated runs out of ten. Its preregistered task-level positive rule is at least two qualifying runs with familywise false-positive probability below .05. N_1 and N_3 surrogates are above threshold but have wide intervals. A content-specific N_2 perturbation was not performed. During a later window, an EEG assessment is negative but contaminated by muscle artifact and administered after arousal declines.

Round 4 - validity and action cards. The imagery classifier has good specificity but modest sensitivity in this injury class. Hearing demand was satisfied for the high-arousal imagery window but not for the later negative EEG window. A low-burden visual communication trial is available. Missing genuine command following carries a high cost; an unnecessary trial carries a low cost.

Instructor key. Classify three targets separately. The covert-command-following assessment is **positive** under its preregistered narrow task rule. The later EEG assessment is **unlicensed** because quality and state opportunity fail. The full CØ / N* assessment for phenomenal presence is **indeterminate** because N_2 is unassessed and N_1 / N_3 intervals remain wide; continuous phenomenal richness is not tested. The low-burden visual trial can be justified separately by asymmetric loss. The forbidden inference is “positive command following establishes the complete bundle,” “no behavior means no experience,” or “precaution proves consciousness.” The smallest next study confirms a sensory route and repeats a content-specific, report-independent component assessment in a stable high-arousal window.

Packet 2. Selective route interruption

Teaching aims: N_2 operationalization, single-failure profiles, mechanism selectivity, rescue, and incremental necessity.

Round 1 - baseline card. In a calibrated recurrent visual preparation, one content passes a stable boundary test, V , N_1 , N_2 , and N_3 . A held-out presence-relevant anchor vector was independently validated before the experiment and shares neither sensors, derived features, nor recipient-route measures with the N_2 test.

Round 2 - intervention card. A reversible pathway intervention reduces content-specific modulation of sensory-comparison, orienting, and value-updating recipients. The intervention leaves source decoding, local activity, energy, and initial feedforward response within equivalence margins.

Round 3 - retention cards. Held-out measures show N_1 and N_3 remain above threshold; V and the candidate boundary remain stable. Report and motor routes were absent from the paradigm. Route mediation and matched-content controls pass.

Round 4 - anchor and rescue cards. The independently validated held-out anchor vector declines in the predicted direction. Restoring the pathway rescues both N_2 and the anchors without globally increasing every component. The N_2 -ablated rival predicts stable anchors.

Instructor key. This is meaningful evidence for the incremental and mechanism-level contribution of N_2 in the tested domain, assuming selectivity and anchor independence withstand audit. Because the anchor vector shares no recipient, sensor, or intervention channel with the N_2 measure, it does not merely redescribe downstream availability. The result does not by itself prove the full biconditional, metaphysical identity, or N_2 necessity across substrates. The strongest rival may invoke an unmeasured common cause or hidden change in content fidelity. The next study uses a second intervention route and independent replication to test measurement-route invariance.

Packet 3. The adaptive dyad

Teaching aims: boundary, inclusion-minimality, separability, replacement, group minds, and horizon dependence.

Round 1 - context card. Agents A and B solve problems through a shared memory service. Their conversation produces sophisticated joint behavior. The candidate family is $\{A, B, A \cup B, A \cup B \cup M\}$ at a one-minute and one-day horizon.

Round 2 - autonomy cards. Each agent maintains endogenous state and role completeness when communication is briefly interrupted. The joint dyad is highly coordinated, but much inbound structure comes from the shared memory service. At the one-day horizon, both agents use that memory to resume unfinished tasks.

Round 3 - replacement cards. Replacing the shared cache with a generic store preserving latency and content leaves each agent's role profile intact. Removing agent A leaves B intact and vice versa. Interrupting the specific A-B channel does not collapse either candidate, although joint performance declines.

Round 4 - minimality and separability cards. A and B each qualify and are proper subsets of the larger candidates at the same short-horizon specification. Pairwise separability exceeds the frozen threshold. The long-horizon identity mapping remains uncertain because both agents repeatedly write to and reconstruct state from the shared memory.

Instructor key. At the short horizon, multiple candidate bearers are supported; the larger dyad loses under inclusion-minimality. This does not establish that either candidate is conscious. At the long horizon, memory may be a constituent or interface depending on replacement and identity-continuity evidence, so the boundary may be nested or indeterminate. Joint intelligence and shared storage are not sufficient for one subject. The next study uses matched memory substitution and state-reset interventions across the longer horizon.

Packet 4. The behavioral duplicate

Teaching aims: constrained functionalism, recurrence, causal realization, artificial viability, and transport.

Round 1 - behavior card. System R and System L produce identical responses across a large benchmark and identical verbal self-descriptions. R is a recurrent agent with persistent internal state. L is an explicitly acyclic feedforward transducer. The benchmark harness supplies L with an address encoding the full prior input history; that history oracle is outside the declared L candidate and has no writable state inside L.

Round 2 - causal cards. Perturbing an internal R state changes later processing through recurrent paths; matched rescue restores the prior trajectory. L contains no state carried between steps; changing an intermediate feedforward value affects only the current pass, while changing the external history address changes output. Both run on reliable hardware. The oracle's large inbound influence creates a separate autonomy and boundary concern for L.

Round 3 - component cards. R has candidate evidence for integration, diverse causal availability, and bounded recurrence. L has output equivalence but no demonstrated irreducible joint event or recurrent stabilization inside the declared candidate. Whether reliable platform support counts as V for either system has not been calibrated.

Round 4 - transport cards. All available phenomenal anchors were validated in humans or animals. No agreed artificial-domain calibration bridge exists. Rival behavioral functionalism treats the systems alike; CØ / N* distinguishes their realized organization.

Instructor key. The case differentiates CØ / N* from unrestricted input-output functionalism: L does not pass merely by behavioral duplication. R is a stronger component candidate, but phenomenal attribution remains transport-limited and viability-underdetermined. The forbidden conclusions are “R is conscious” and “L is known not to be conscious.” The next studies validate artificial-domain component constructs and compare role-preserving perturbations without using self-report as the sole anchor.

Packet 5. The hue-warp intervention

Teaching aims: presence-character separation, independent anchors, deformation prediction, automorphisms, and bridge refutation.

Round 1 - presence card. In a human no-report color paradigm, boundary, V , N_1 , N_2 , and N_3 pass for all tested color contents. Presence status is frozen before character analysis.

Round 2 - baseline geometry cards. Perturbation-derived V4-like causal geometry predicts baseline low-demand similarity and discrimination relations. The bridge is trained on three colors and correctly places a fourth held-out color. Stimulus wavelength geometry is included as a nuisance model.

Round 3 - intervention card. A validated adaptation manipulation produces the preregistered contraction in causal geometry. Attention, task difficulty, retinal input, and response mapping controls pass.

Round 4 - phenomenal anchor card. Independently collected low-demand similarity, confusion, and transition measures show no corresponding contraction, with intervals excluding the predicted deformation. The measures remain reliable and were not used to tune the intervention.

Instructor key. The frozen geometry-to-quality bridge is refuted for this domain and manipulation. Positive presence remains intact, as does the fact that the experiences have character. A different mapping, additional latent factor, or nongeometric structure may be required. The prohibited inference is “no phenomenal experience occurred.” The next study tests an alternative causal geometry or a richer quality structure with the same independence firewall.

Packet 6. The consciousness dashboard

Teaching aims: circular validation, schematic legibility, calibration, coverage, and model comparison.

Round 1 - performance card. A dashboard outputs a consciousness score from 0 to 100 and achieves 94% accuracy on a held-out dataset. Its developers claim the score makes consciousness directly observable.

Round 2 - label card. Training and test labels were generated by thresholding the same integration, availability, and recurrence measures that feed the dashboard. Clinical diagnoses influenced thresholds during development.

Round 3 - legibility cards. The dashboard offers no stable intermediate role vector. Two different faults can yield the same score. It cannot distinguish boundary failure from component failure or predict which function should change after a selective perturbation.

Round 4 - abstention card. The dashboard forces every case into positive or negative. It achieves its high accuracy mainly in ordinary waking and deeply anesthetized cases; near-miss and transport cases are rare.

Instructor key. The accuracy is circular evidence for the target because the labels repeat the inputs. The tool also fails schematic legibility, diagnostic coverage, and outcome discipline. It may summarize its own component rule reliably, but it has not independently validated consciousness. Redesign requires independent anchor genealogy, frozen outcome rules, component-level readout, fault predictions, diagnostic-case enrichment, and explicit indeterminate and unlicensed outputs. The smallest useful next study is blind fault localization and held-out prediction on near-miss cases with labels independent of the dashboard inputs.

Appendix F. Glossary and Notation

Term or symbol	Teaching definition	Guardrail
C_0 or $C\emptyset$	Minimal phenomenal presence	Not report, intelligence, wakefulness, or full character
C_1	Enriched reflective or metacognitive consciousness	Not built into the minimal gate
N^*	$N_1 \wedge N_2 \wedge N_3$	A defined bundle; its identity with consciousness is a hypothesis
V	Independently assessed capacity to realize the relevant dynamics	Not biological life alone and not a post hoc rescue clause
N_1	Integration or synergy	Not activity, correlation, or complexity in general
N_2	Counterfactual system-wide availability	Not report, diffusion, anatomical extent, or node count
N_3	Bounded recurrent stability	Not mere lateness, persistence, or memory
S	Boundary-qualified candidate system	Must be frozen before theory-specific scoring
c	Target content or content-bearing state	Results can be content-relative
g	Spatial, functional, and temporal grain	Cross-grain comparisons need a mapping
R	Operating regime	State changes can alter validity and boundaries
Δ	Evaluation horizon or interval	Presence is windowed rather than instantaneous
Effective connectivity	Directed causal influence among modeled variables	Not anatomy or statistical dependence alone
Synergy	Joint information or causal contribution irreducible to parts	Estimator families require construct validation
Recipient class	Functionally distinct group defined by causal response repertoire	Not an anatomical label or parcel count
Preparation burden Γ	Cost of placing a recipient in a nearby test context	Prevents installing new access machinery
Content selectivity	Effect beyond energy-matched and content-destroying controls	Broad response alone is diffusion-compatible
Route dependence	Loss of a content-specific effect under matched route interruption	Redundant routes require adequate family coverage
C_A	Weighted coverage across qualifying recipient classes	Weights cannot simply reward node count
K_A	Number of qualifying recipient classes	Prevents one class from passing alone
H_A	Evenness of effect across qualifying classes	Prevents token threshold crossings around one dominant class
N_2^{-R}	Availability recalculated without report classes	Report remains optional evidence
N_2^{-E}	Availability recalculated without executive classes	Executive uptake cannot be the sole pass route

Term or symbol	Teaching definition	Guardrail
Validity envelope V	Domain in which an evidence-target interpretation is established	A formula does not carry its meaning automatically
Positive	Evidence licenses attribution of the exact target	Not certainty or complete explanation
Negative	Valid, sensitive, adequate evidence licenses rule-out	Not failure to cross a positive threshold
Indeterminate	Valid evidence does not discriminate enough	Not half conscious and not invalid
Unlicensed	Interpretation conditions failed	Meta-output, not evidence of absence
Cause profile	Evidential, surrogate, model, boundary, and transport limitations	Not a consciousness score
Surrogate conflict	Credible measures exceed preregistered disagreement tolerance	Must be checked before aggregation
Boundary reversal	Result changes when the candidate boundary changes	Shows that individuation is part of the claim
J_{self}	Candidate's endogenous causal capacity	Blocks inert independence
J_{in}	Inbound external imposition on organization	Does not penalize outbound agency
A_{θ}	Ratio of endogenous capacity to endogenous plus inbound dependence	Eligibility for candidacy, not consciousness amount
Inclusion-minimality	No qualifying proper subsystem at the same specification	Not fewest nodes or highest score
Separability	Each candidate preserves organization under reciprocal channel interruption	Low separability does not prove one larger subject
Constituent	Specific dynamics are required after generic support replacement	More than remote causal necessity
Interface	Bounded input or output channel compatible with autonomous role completeness	Causal contact alone does not make it internal
Enabling support	Generic replacement restores the role profile	Time-horizon dependent
Diagnostic set D_k	Cases where X_k fails and every retained component passes	Only region where full and leave-one-out conjunctions differ
Diagnostic coverage Γ_k	Target-distribution mass in the diagnostic region	A rare corner limits generality
Criterion ablation	Remove a term from the decision rule	Tests incremental prediction, not mechanism
Mechanism ablation	Intervene on a proposed realization	Requires selectivity and preserved identity
Measurement ablation	Remove an estimator or data route	Tests robustness, not physical necessity
Anchor vector Y	Independent convergent evidence used for model comparison	Not an infallible consciousness oracle
Causally coupled	Components cannot be varied separately within the tested intervention family	Motivates composite modeling or better intervention
Weak minimality	No single component can be removed	Can miss grouped redundancy
Strong minimality	Full model beats every preregistered proper submodel	Still relative to anchors, domain, and rival class

Term or symbol	Teaching definition	Guardrail
G_c	Content's neighborhood in a causal-trajectory geometry	Not a snapshot or arbitrary latent embedding
Q	Independently anchored empirical quality-space structure	Need not be Euclidean or globally commensurable
F_ϕ	Frozen bridge from causal-geometry features to quality structure	High in-sample correlation is insufficient
Q_ϕ	Symmetry-qualified phenomenal-character map	Applied only after a positive presence gate
$Aut(Q)$	Transformations preserving every measured quality relation	Records residual inversion and label underdetermination
Π_c	Temporal path features of an admitted character	Distinct from N_3 's presence gate
Schematic legibility	Publicly teachable compositional and counterfactual readout	Intelligibility is not proof
Role vector ρ_{C_0}	Bearer, operation, unity, system-level presence, temporal presence	The intermediate explanatory profile
Transport	Preservation of construct, causal role, and calibration across domains	Surface similarity is insufficient

Appendix G. Primary Corpus and Selected Contextual Readings

Primary CØ / N* corpus

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- Stilwell, P. (2026). *Where is the conscious subject? A dynamical criterion for system boundaries* [Manuscript].
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- Stilwell, P. (2026). *Reading consciousness from the schematic: Structural legibility in the N model** [Manuscript].

Project site: <https://philstilwell.github.io/C0/>

Selected contextual readings

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Closing instructor note

The deepest pedagogical virtue of this corpus is not the compactness of $V \wedge N^* \Leftrightarrow C_0$. It is the insistence that every inference earn its target, bearer, evidence, scope, and revision rule. If students finish the course able to preserve those distinctions while designing a study that could surprise both proponents and critics, the manual has done its job.